DOT POINT

Answers

Science Press

Notes

1.1.1.1

Completed sentences
All matter is made up of particles which are in continuous motion.
The particles of matter are always moving except at absolute zero.
Particles in matter are held together by forces which vary in strength.
The particles in solids cannot move freely because they are held together by strong forces.
Particles in solids simply vibrate in their fixed positions.
In liquids particles can roll over one another.
The forces holding particles together in liquids are weaker than those holding particles together in solids.
Particles in gases are free to move in any direction so that gases always fill their containers.
Particles in gases are not held together.
Particles of matter are far too small to be seen even under a microscope.
We deduce the behaviour of the particles of matter by studying the properties of matter.
We use the kinetic theory of matter to explain and predict the behaviour of matter.

1.1.2.1 D
1.1.2.2 C
1.1.2.3 B
1.1.2.4 D
1.1.2.5 B
1.1.2.6 A
1.1.2.7 A
1.1.2.8 C
1.1.2.9 A
1.1.2.10 C
1.1.2.11

	Fact	Kinetic theory explanation
(a)	Air can be compressed.	Air is a gas and its particles are not close together, therefore they can be pushed closer together.
(b)	Liquids diffuse into each other.	Particles in a liquid are free to roll over one another and as they do so they gradually mix or diffuse together.
(c)	Cooking odours spread throughout the house.	Gas particles from the cooking food are free to move and diffuse relatively quickly through the house.
(d)	You can dry yourself with a towel.	Water particles are not held together by strong forces and they bond to the particles in the towel with stronger forces.
(e)	Liquids do not fill their containers.	The forces between the liquid particles hold them together and prevent them from breaking free from each other (evaporating) to fill the container.
(f)	Glass objects shatter when dropped onto a hard floor.	The forces involved in the collision between the glass and the floor are stronger than the forces holding glass particles together.
(g)	Concrete footpaths have black rubber strips every few metres.	These are soft expansion joints so that in hot weather, when the concrete blocks expand they squash the rubber rather than pushing against each other and breaking.
(h)	Gases expand quickly when heated.	Gas particles are not held together and are free to move when the gas is heated. They move faster and collide with each other more violently and take up more space if it is available (if not, the pressure increases).
(i)	Solids have a constant shape.	Particles in a solid are held together by relatively strong forces.
(j)	Liquids expand when heated.	Liquid particles roll over one another faster when heated and need more room to move about.

1.1.2.11

	Fact	Kinetic theory explanation
(k)	If a soccer ball gets a hole in it, the air leaks out.	The high pressure inside the ball (more particles per unit volume than outside) results in more collisions between particles and their random movement in all directions results in some particles moving out of the ball through the hole.
(l)	Large bridges sit on, and are free to move on rollers at one end.	In hot weather, particles within the bridge material move faster, collide more violently and push each other further apart – the bridge expands. Without the freedom to move over the rollers it would buckle.
(m)	Your breath is 'foggy' on a very cold day.	Warm breath contains water particles which slow down rapidly when they meet the cold air and forces between them clump them together to form tiny drops of water vapour.
(n)	A balloon expands on a hot day.	On a hot day the particles of air inside the balloon are moving faster (as are the ones outside the balloon) but have nowhere to spread out. So they collide with each other and the walls of the balloon more violently and push the balloon material outwards – the balloon expands.

1.1.3.1

	Completed sentences
(a)	Temperature is a measure of how fast the particles of matter are moving.
(b)	Temperature is a measure of the average kinetic energy of the particles of matter.
(c)	The higher the temperature, the faster the particles are moving.
(d)	When matter is heated its particles absorb energy and move faster.
(e)	When matter is cooled its particles lose energy and slow down.
(f)	Gas particles have more energy due to their state than liquid particles at the same temperature.
(g)	Similarly, liquid particles have more energy than solid particles at the same temperature.
(h)	In both cases (f) and (g), this is due to the extra energy they have because the particles move more freely.

1.1.3.2 Our theory is that temperature is a measure of the average kinetic energy of the particles of matter and kinetic energy depends on mass and speed (squared). The particles will always have mass, so the kinetic energy can only be zero if their motion is zero. Because motion cannot be less than zero, this means there is a temperature (absolute zero) which is the lowest possible temperature. Absolute zero is 0 K (kelvin) = –173.14°C.

1.1.3.3
(a) D = E faster than B faster than C = A. The hottest (both at 100°C) have particles moving equally fast, and the others are in order of decreasing temperature.

(b) Taking temperature as a measure of average kinetic energy, then the order for kinetic energy will be the same as the order for temperature and rate of movement, so, answer is same as (a) = D = E more than B more than C = A.

(c) D. Being a vapour, the stream has more degrees of freedom of movement than the particles in water at the same temperature, so the steam particles will have the most total energy.

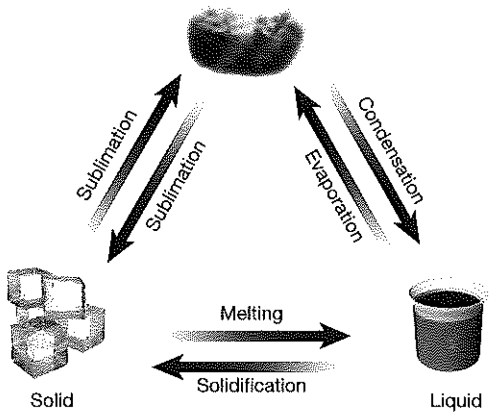
(d) The particles in the ice at 0°C will have least total energy as their freedom of movement is less than that of water at the same temperature.

(e) Steam particles move more freely than water particles and are not held to other steam particles, so they can penetrate pores in our skin much more effectively and give a deeper burn.

1.1.4.1

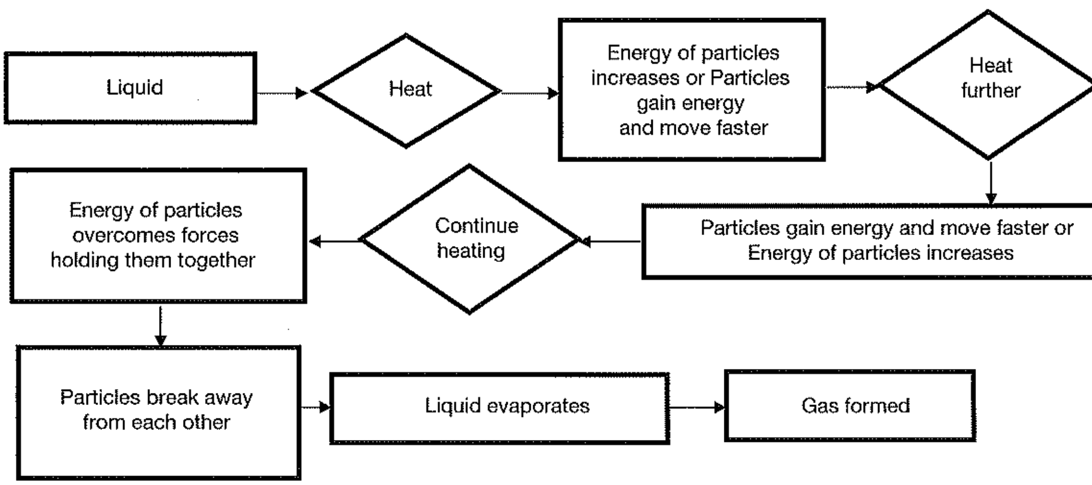
	Completed sentences
(a)	As a gas (or any matter) is cooled its particles move more slowly.
(b)	As the particles lose energy the forces between them can start to hold them together.
(c)	Eventually, the strength of the forces between the particles is strong enough to hold them close together.
(d)	They start to form a lattice but are still able to roll over one another.
(e)	As more energy is removed from the system, more particles move into the lattice positions.
(f)	Eventually the system loses sufficient energy for all particles to become bonded into the lattice.
(g)	The gas has condensed and become a liquid.

1.1.4.2



1.1.4.3
(a) Change of state is liquid to gas (evaporation).
(b) The average *KE* of the particles does not change (except during any heating process causing a rise in temperature) because the question indicates that the process is occurring at a constant temperature.
(c) The forces between the particles are overcome by the energy of their motion as they change from the liquid to the gaseous state.
(d) In the liquid state (diagram A).
(e) Again, given that the temperature remains constant, the rate of movement of the particles will also remain constant. However, the way in which they move changes from rolling over one another to moving freely.
(f) A = liquid
B = liquid in the process of evaporating
C = gas

1.1.4.4



1.2.1.1
(a) Latent energy is the energy involved in changing the state of matter.
(b) Latent energy is used to overcome bonds between particles of matter (evaporation and melting) or released as bonds between particles form (solidification or condensation).

1.2.1.2

Label	Correct position(s) for label	Label	Correct position(s) for label
Boiling	E	Solid	A
Gas	C	System gains kinetic energy	F, H, J
Latent heat of fusion	K	System gains potential energy	G, I
Latent heat of vaporisation	L	Temperature of system increases	M, O, Q
Liquid	B	Temperature of system remains constant	N, P
Melting	D		

1.2.1.3 (a) Energy added is used to overcome forces between particles when liquids are evaporating or solids are melting, so provided the energy supply is not excessive, the temperature of the substance will not change while this process is occurring. Similarly, as gases condense or liquids solidify, the freedom of movement of the particles decreases so energy is released even though temperature does not decrease.

(b) During condensation, the degrees of freedom of movement of the particles decreases and this energy is released – therefore the process is exothermic.

1.2.1.4 (a)

Section	State or states of matter present
C	Solid only
D	Solid and liquid
E	Liquid only
F	Liquid and gas
G	Gas only

(b)

Section	Change occurring
1 to 2	Solid is becoming hotter
2 to 3	Solid is melting
3 to 4	Liquid is becoming hotter
4 to 5	Liquid is evaporating
5 to 6	Gas is becoming hotter

(c) A = boiling point

B = melting point

(d) 2 to 3 = latent heat of fusion

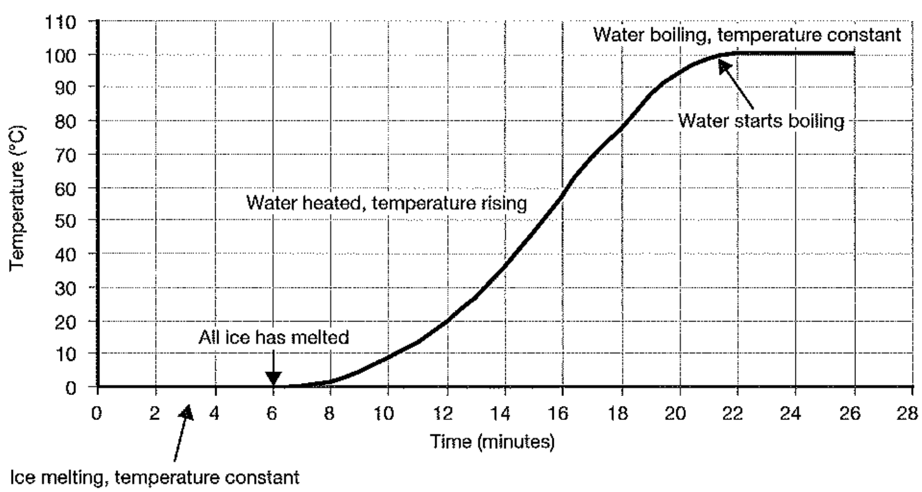
4 to 5 = latent heat of vaporisation

(e)

Diagram	Represents	Refers to graph section:
V	Gas	G
W	Liquid	E
X	Liquid changing to gas	F
Y	Solid	C
Z	Solid changing to liquid	D

(f) W and Z both represent liquid, by showing the particles further apart. In reality, they are as close together as in solids, but roll over one another rather than vibrate.

1.2.2.1



1.2.2.2 See graph.

1.2.2.3 The energy is being absorbed by the ice to melt it.

1.2.2.4 The energy is being absorbed to evaporate the water.

1.2.2.5 (a) Latent heat of fusion, L_f is the energy required to changer the state of a substance from solid to liquid.

(b) The temperature of the ice does not change while it is melting despite a steady heat energy input. This illustrates the latent heat of fusion.

(c) The energy is being taken in by the particles of ice to overcome the forces holding them in fixed positions, and allowing them to break away from the solid lattice and roll over one another.

1.2.2.6 (a) Latent heat of vaporisation, L_v is the energy required to change the state of a substance from liquid to gas. Also referred to as the specific heat of vaporisation.

(b) The temperature of the water does not change while it is boiling despite a steady heat energy input. This illustrates the latent heat of vaporisation.

(c) The energy is being taken in by the particles of water to overcome the forces holding them together, and allowing them to break away from each other and escape into the environment (evaporate).

1.2.3.1 (a) The energy is taken in by the particles of the solid to overcome the forces holding them in fixed positions, and allowing them to break away from the solid lattice and roll over one another.

(b) The energy is being taken in by the particles of the liquid to overcome the forces holding them together, and allowing them to break away from each other and escape into the environment (evaporate).

(c) Energy is released from the particles of the liquid as they slow down (matter becomes cooler). Eventually the motion of the liquid particles is not sufficient to overcome the forces between them and they become locked into fixed (solid) positions.

(d) As the gas particles cool and slow down, interparticle forces can pull them closer together. Eventually, the kinetic energy of the particles will not be sufficient to keep them free from each other and they will join together to form the liquid form of the matter.

1.2.3.2 (a) $\frac{X}{4}$

(b) $\frac{X}{4}$

(c) For nitrogen, 0.6 kg. For alcohol, 0.15 kg.

1.2.3.3 210 kJ kg^{-1}

1.2.3.4 (a) 161 kJ

(b) From $P = \frac{W}{t}$, $t = \frac{W}{P} = \frac{161}{12} = 13.4$ s

1.2.3.5 41.625 kJ

1.2.3.6 (a) 6.125 kJ

(b) 10.2 J s^{-1}

(c) 33.5 kJ

1.2.3.7 (a) About 24°C

(b) About 62°C

(c) $\frac{500}{240} = 2.08$ A

(d) Total energy = $Pt = 500 \times 36 \times 60 = 1.08 \times 10^6$ joules

(e) Latent heat of fusion is greater because it takes more time to evaporate the liquid than to melt the solid with the same rate of energy supply.

(f) Energy to melt = $500 \times 3.5 \times 60 = 105\,000$ J for 45 g

Therefore for 1 kg = latent heat of fusion = 2333.3 kJ kg^{-1}

(g) Energy to boil = $500 \times 5.0 \times 60 = 150\,000$ J for 45 g

Therefore for 1 kg = latent heat of fusion = 3333.3 kJ kg^{-1}

(h) Heat energy added between time 11 and 25 = $500 \times 14 \times 60 = 420\,000$ J = $mc\Delta T$

Therefore c = 333.3 kJ kg^{-1} $°C^{-1}$

1.2.4.1 (a) Different substances are made up of different combinations of atoms.

(b) Different atoms have different masses.

(c) Kinetic energy of a particle is due to its mass and the rate at which it moves ($E_K = \frac{1}{2}mv^2$).

(d) The temperature of matter is a measure of the average kinetic energy of its particles.

(e) Particles with greater mass will require more energy to increase their kinetic energy than the same amount as particles with smaller masses.

(f) So raising the temperature of 10 kg of substance A by T°C will require a different amount of energy compared to increasing the temperature of 10 kg of substance B by the same amount.

(g) In addition, raising the temperature of 10 kg of a substance by T°C will require more energy than raising the temperature of 5 kg of the same material by the same amount.

1.2.4.2 The amount of energy required to increase or decrease the temperature of a substance depends on its mass.

1.2.4.3 (a) Different substances are made up of different combinations of atoms.

(b) Different atoms and molecules within substances are held together by forces that vary in strength.

(c) Kinetic energy of a particle is due to its mass and the rate at which it moves ($E_K = \frac{1}{2}mv^2$).

(d) The temperature of matter is a measure of the average kinetic energy of its particles.

(e) Particles held together with larger forces will require more energy to increase their kinetic energy the same amount as particles which are held together with weaker force.

1.2.4.4 The amount of energy required to increase or decrease the temperature of a substance depends on the strength of the forces holding its particles together.

1.2.4.5 The amount of energy required to increase or decrease the temperature of matter also depends on how far we wish to raise or lower that temperature.

1.2.4.6 The specific heat of a substance is the amount of energy required to raise the temperature of 1.0 g of a material by 1°C.

1.2.4.7 (a) Ammonia, alcohol, water. This is the increasing order of their latent heats of vaporisation.

(b) Gold – it has the lowest latent heat of fusion of the three metals.

(c) Gold – it has the lowest specific heat.

1.2.4.8 (a) 33.5 kJ

(b) 16.1 kJ

(c) 72.25 kJ

(d) 83.25 kJ

1.2.4.9 (a) 41.38 kJ

(b) 12.870 kJ

(c) 89.100 kJ

(d) 75.9×10^6 kJ

1.2.4.10 (a) Energy to melt 75 g ice = mL_f = 24.975 kJ

Energy to heat ice water to 100°C = $mc\Delta T = 0.075 \times 4.18 \times 100 = 31.35$ kJ

Energy to boil the water = $mL_v = 0.075 \times 2260 = 169.5$ kJ

Therefore total energy = 225.825 kJ

(b) Energy to melt 375 g ice = mL_f = 124.875 kJ

Energy to heat ice water to 100°C = $mc\Delta T = 0.375 \times 4.18 \times 100 = 156.75$ kJ

Energy to boil the water = $mL_v = 0.375 \times 2260 = 847.5$ kJ

Therefore total energy = 1129.125 kJ

(c) Energy to condense vapour = $mL_v = 0.5 \times 2260 = 1130$ kJ (removed)

Energy to cool water to 0°C = $mc\Delta T = 0.5 \times 4.18 \times 100 = 209$ kJ (removed)

Energy to freeze 500 g water = mL_f = 166.5 kJ (removed)

Therefore total energy = 1505.5 kJ (removed)

1.3.1.1 (a) Metals and metal alloys allow heat energy to transfer through them.

(b) For this reason they are known as heat conductors.

(c) Non-metallic substances do not allow heat energy to pass through them.

(d) These substances are called heat insulators.

(e) Conduction of heat occurs because any applied energy causes the particles of the metal to vibrate more violently.

(f) This increased vibration causes these particles to bump into the particles next to them and cause these to vibrate more as well.

(g) Gradually the heat energy is passed through the metal from particle to particle.

(h) In addition to resulting in conduction of heat, the increased vibrations cause the metal to expand.

(i) This happens because the particles need more space in which to vibrate.

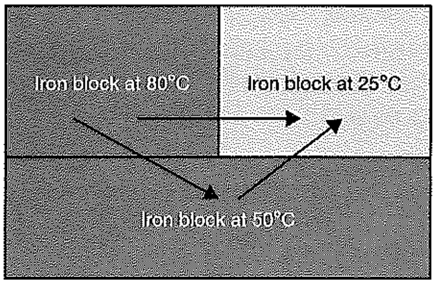
(j) Air is one of the best insulators.

(k) Many substances that contain a lot of air, such as cork, polystyrene foam, fibreglass and wool, are therefore good insulators.

1.3.1.2 The person on the left must be holding a metal rod because the heat from the fire has conducted down its length to burn her hand. The person on the right must be holding a non-conducting rod and she seems unaffected by heat transfer.

1.3.1.3 Most likely X because the diagram implies heat energy transfer by conduction along the rod – the particles are moving more violently further from the heat source in the second diagram.

1.3.1.4 (a)



(b) They will all end up at the same temperature.

(c) Assuming that size indicates relative masses, because the block are all the same material, we could estimate the final temperature as:

Top left block has (say) 40 units of energy (half the size of the bottom block × temperature)

Top right block has 12.5 units of energy (same logic)

Bottom block has 50 units of energy

This gives a total of 102.5 units of energy to be averaged over the total mass = 102.5 ÷ 2 = 51.25

So estimate the final temperature to be 51.25°C

1.3.2.1 (a) As water at the bottom of a beaker is heated, it expands.

(b) As the water expands it becomes less dense.

(c) The less dense water then 'floats' to the top of the beaker.

(d) At the top of the beaker this water cools down as it is further from the heat source.

(e) More rising hot water from the bottom pushes the cooling water to the sides.

(f) The cooling water, now more dense, sinks to the bottom of the beaker to take the place of more rising hot water.

(g) This occurs because the particles in the water are not held together tightly but can flow over one another.

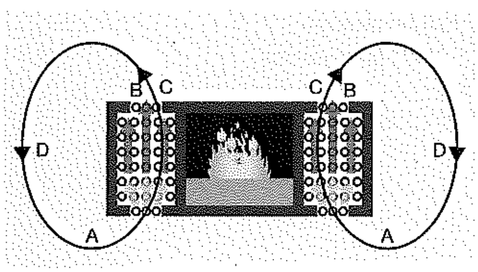
(h) The transfer of heat energy by convection occurs in gases and liquids.

(i) Gases and liquids are collectively known as fluids because they both flow, unlike solids which maintain a constant shape.

(j) This is also because their particles can move more freely than the particles in solids.

1.3.2.2 The cat. In a convection current, warm air rises and cooler air, being denser, falls. The cat is in the warmer position.

1.3.2.3



Note: The hottest air will be as it comes out of the convection heater not at the top of the convection cycles.

1.3.2.4 Particles in gases are free to move and can therefore move about much more than particles in liquids which are free only to roll over one another. Gases can therefore transfer heat energy by convection faster and more efficiently than particles in liquids. Particles in solids are held in fixed positions and are not free to move other than to vibrate in those fixed positions.

1.3.2.5 Convection will play very little role in heating the water because the water at the top of the pool is being heated, and that is where it will stay – convection currents will not form. Convection works best when the heat source is at the bottom of the matter involved in the convection currents.

1.3.3.1 (a) Heat energy can also travel through fluids and space by a process known as radiation.

(b) Radiation does not involve the particles of matter.

(c) It is because of this that heat energy can be transferred through space or a vacuum.

(d) Heat radiation occurs through electromagnetic waves, also known as infra-red rays.

(e) When infra-red radiation hits a surface it is partly absorbed and partly reflected.

(f) It is the energy absorbed by a substance that causes its temperature to rise.

(g) Dull, dark surfaces absorb heat energy more rapidly than bright, shiny surfaces.

(h) Bright, shiny surfaces reflect heat energy more than dull, dark surfaces.

(i) Bright, shiny surfaces radiate heat energy more rapidly than dull, black surfaces.

1.3.3.2 (a) Heat radiation refers to the transfer of heat energy outwards from a hot source via electromagnetic radiation (infra-red rays).

(b) At the side, you only receive heat energy by radiation, and very little heat, if any, reaches you at the side. Above the flame heat energy reaches you via radiation and convection currents. Because air is a good insulator little heat is conducted in any direction.

(c) Its colour and whether is has a bright, shiny surface or a dull surface.

Note that in the following calculations, the conversion of masses from g to kg has not been done because the units appear in all parts of each question and simply cancel. It is simpler to leave the mass units as grams.

1.4.1.1 Energy lost by water and calorimeter = energy gained by ice and ice water

$(m \times c \times \Delta T)_{\text{water}} + (m \times c \times \Delta T)_{\text{calorimeter}} = (m \times \Delta H_{\text{fusion}})_{\text{ice}} + (m \times c \times \Delta T)_{\text{cold water}}$

Because we don't know the change in temperature, let the final temperature = T

Therefore, temperature change for water = $80 - T$ (it gets cooler)

And temperature change for cold water = $T - 0$ (it gets warmer)

Substituting $250 \times 4.18 \times (80 - T) + 50 \times 0.39 \times (80 - T) = 40 \times 333 + 40 \times 4.18 \times (T - 0)$

From this $83600 - 1045T + 1560 - 19.5T = 13320 + 167.2T$

$83600 + 1560 - 13320 = 1045T + 167.2T + 19.5T$

$71840 = 1231.7T$

$T = 58.3°\text{C}$

1.4.1.2 (a) Energy lost by hot water = energy gained by colder water

$(m \times c \times \Delta T)_{\text{hot water}} = (m \times c \times \Delta T)_{\text{cold water}}$

Because we don't know the change in temperature, let the final temperature = T

Therefore, temperature change for hot water = $80 - T$ (it gets cooler)

And temperature change for cold water = $T - 10$ (it gets warmer)

Substituting $50 \times 4.18 \times (80 - T) = 200 \times 4.18 \times (T - 10)$ (conversion of mass to kg cancels)

$16720 - 209T = 836T - 8360$

$25080 = 1045T$

Therefore final temperature, $T = 24°\text{C}$

(b) Energy gained by water + energy gained by cup = energy lost by hot water

$(m \times c \times \Delta T)_{\text{cold water}} + (m \times c \times \Delta T)_{\text{glass}} = (m \times c \times \Delta T)_{\text{hot water}}$

Because we don't know the change in temperature, let the final temperature = T

Therefore, temperature change for cold water and glass = $T - 10$ (it gets warmer)

And temperature change for hot water = $80 - T$ (it gets cooler)

Substituting $200 \times 4.18 \times (T - 10) + 250 \times 0.84 \times (T - 10) = 50 \times 4.18 \times (80 - T)$

$836T - 8360 + 210T - 2100 = 16720 - 209T$

$1255T = 27180$

So, final temperature, $T = 21.7°\text{C}$

1.4.1.3 Energy lost by ammonia = energy gained by ice and ice water

$(m \times c \times \Delta T)_{\text{ammonia}} = (m \times \Delta H_{\text{fusion}})_{\text{ice}} + (m \times c \times \Delta T)_{\text{cold water}}$

The final temperature = 15°C

Therefore, temperature change for ice water = $15 - 0 = 15$ (it gets warmer)

And temperature change for ammonia = $24 - 15 = 9$ (it gets cooler)

Substituting $m \times 2.09 \times 9 = 130 \times 333 + 130 \times 4.18 \times 15$

From this $18.81m = 43290 + 8151$

$18.81m = 51441$

Therefore final amount, $m = 2735$ g

1.4.1.4 (a) Energy lost by warm water = energy gained by cold water

$(m \times c \times \Delta T)_{\text{water}} = (m \times c \times \Delta T)_{\text{cold water}}$

Because we don't know the change in temperature, let the final temperature = T

Therefore, temperature change for warm water = $75 - T$ (it gets cooler)

And temperature change for cool water = $T - 15$ (it gets warmer)

Substituting $50 \times 4.18 \times (75 - T) = 40 \times 4.18 \times (T - 15)$

$15675 - 209T = 167.2T - 2508$

$18183 = 376.2T$

Therefore final temperature, $T = 48.3°\text{C}$

(b) Energy gained by cold water + energy gained by beaker = energy lost by hot water

$(m \times c \times \Delta T)_{\text{cold water}} + (m \times c \times \Delta T)_{\text{beaker}} = (m \times c \times \Delta T)_{\text{hot water}}$

Because we don't know the change in temperature, let the final temperature = T

Therefore, temperature change for cold water = $T - 15$ (it gets warmer)

Temperature change for beaker = $T - 20$

And temperature change for warm water = $75 - T$ (it gets cooler)

Substituting $40 \times 4.18 \times (T - 15) + 150 \times 0.39 \times (T - 20) = 50 \times 4.18 \times (75 - T)$

$167.2T - 2508 + 58.5T - 1170 = 15675 - 209T$

$434.7T = 19353$

So, final temperature, $T = 44.5°\text{C}$

1.4.1.5 Energy lost by hot metal = energy gained by colder metal

$(m \times c \times \Delta T)_{\text{hot metal}} = (m \times c \times \Delta T)_{\text{cold metal}}$

Because we don't know the change in temperature, let the final temperature = T

Therefore, temperature change for hot metal = $125 - T$ (it gets cooler)

Temperature change for cool metal = $T - 15$

Substituting $3 \times 0.39 \times (125 - T) = 2 \times 0.45 \times (T - 15)$

$146.25 - 1.17T = 0.9T - 13.5$

Therefore $159.75 = 2.07T$

Therefore $T = 77.2°\text{C}$

1.4.1.6 Energy lost by copper = energy gained by water

$(m \times c \times \Delta T)_{\text{copper}} = (m \times c \times \Delta T)_{\text{water}}$

Because we don't know the change in temperature, let the final temperature = T

Therefore, temperature change for copper = $90 - T$ (it gets cooler)

Temperature change water = $T - 15$

Substituting $180 \times 0.39 \times (90 - T) = 250 \times 4.18 \times (T - 15)$

$6318 - 70.2T = 1045T - 15675$

Therefore $21993 = 1115.1T$

Therefore $T = 19.7°\text{C}$

1.4.1.7 Energy = $9.200 = (m \times c \times \Delta T)_{\text{metal}}$

$= 0.3 \times c \times 25$ (Note mass in kilograms and energy in kJ)

Therefore $c = 1.2$ kJ kg^{-1} °C^{-1}

1.4.1.8 Energy = $(m \times c \times \Delta T)_{\text{metal}}$

$= 0.09 \times 0.45 \times 160$ (Note mass in kilograms and energy in kJ)

Therefore $E = 6.48$ kJ

1.5.1.1 B

1.5.1.2 D

1.5.1.3 C

1.5.1.4 C

1.5.1.5 B

1.5.1.6 C

1.5.1.7 A

1.5.1.8 A

1.5.1.9 A

1.5.1.10 D

1.5.1.11 D

1.5.1.12 B

1.5.1.13 B

1.5.1.14 B

1.5.1.15 C

1.5.1.16 B

1.5.1.17 1200 kJ or 1.2×10^6 J

1.5.1.18
(a) The work done by the gas against the piston.
(b) 1.8 kJ or 1800 J
(c) 1.8 kJ or 1800 J
(d) As it expands its temperature will fall as the gas will do work on the piston and therefore its internal heat energy will decrease. As it is compressed back to its initial volume, the temperature will rise to its initial level as the piston does work on the gas to compress it.

1.5.1.19
(a) Yes. By the external heating source initially, and then by the gas in pushing the piston up to expand its volume.
(b) The internal thermal energy increases because the temperature of the gas rises.
(c) 450 J decrease
(d) 450 J

1.5.1.20
(a) Work was done on the gas to decrease its volume.
(b) Approximate work done (estimate the area under the curve) = approximately 5.4 J
(c) From $W = mc\Delta T$, change in temperature = 0.036°C

1.5.1.21
(a) 4.05×10^1 J
(b) Temperature of the gas must increase in order for it to do work on the piston to increase its volume. Because the gas will be less dense, it must do more energetic collisions on the piston to maintain the pressure.

1.6.1.1

Contribution to thermodynamics	Correct order of scientists	Scientists (alphabetical order)
In 1698, he developed and patented the first crude steam engine to pump water out of coal mines.	Thomas Savery (1650-1715)	Joseph Black (1728-1799)
Invented the first atmospheric steam engine, first used in 1712 to pump water out of a mine.	Thomas Newcomen (1663-1729)	Nicolas Carnot (1796-1832)
In the early 1820s, hypothesised the existence of an engine which ran on a reversible cycle. His ideas led to the development of the second law of thermodynamics.	Nicolas Carnot (1796-832)	Josiah Gibbs (1839-1903)
Known for his improvements of the steam engine. In 1765, his work improving the Newcomen engine helped bring about the Industrial Revolution.	James Watt (1736-819)	James Joule (1818-1889)
Did experiments on the freezing and melting of liquids that led to the concept of latent heats of fusion and vaporisation. His work led to the concepts of heat capacity or specific heat.	Joseph Black (1728-1799)	Antoine Lavoisier (1743-1794)
Helped explain what happened when substances were burned. In 1783 he designed the calorimeter for measuring the amount of heat given off during combustion or respiration.	Antoine Lavoisier (1743-1794)	James Clerk Maxwell (1831-1879)
Connected mechanical and heat energy. In 1850 he published a calculated value for the *mechanical equivalent of heat* which is defined in an early version of the first law of thermodynamics.	James Joule (1818-1889)	Thomas Newcomen (1663-1729)
Defined the absolute temperature scale in 1847. In 1851 he published ideas leading to the second law of thermodynamics and supported his friend James Joule's mechanical equivalent of heat. Introduced the term 'kinetic energy' in 1856.	William Thomson (Lord Kelvin) (1824-1907)	William John Rankine (1820-1872)
Established relationships between the temperature, pressure and density of gases, and expressions for the latent heat of evaporation of a liquid. Formulated a science of energetics describing dynamics in terms of energy and its transformations rather than force and motion.	William John Rankine (1820-1872)	Thomas Savery (1650-1715)
One of his most important achievements was his contribution to kinetic theory of gases. Calculated the mean free path of molecules and formulated the equations for the kinetic theory of gases.	James Clerk Maxwell (1831-1879)	William Thomson (Lord Kelvin) (1824-1907)
Created statistical mechanics explaining the laws of thermodynamics. He published his ideas in 1875 and 1878 in a work which consisted of about three hundred pages and contains exactly seven hundred numbered mathematical equations.	Josiah Gibbs (1839-1903)	James Watt (1736-1819)

1.6.2.1
(a) Chemical energy of fuel is converted to thermal energy by oxidation with air in the combustion chamber.
(b) Thermal energy raises the temperature and pressure of the gases within the engine cylinder.
(c) High pressure gases expand against the resistance of the engine mechanism.
(d) The expansion of the gases moves the piston up ready for the next cycle of the engine.
(e) Movement of the piston up and down produces a rotational torque which turns the crankshaft.

1.6.2.2

Stroke	What happens during this stroke	Diagram
Intake stroke	Air and vaporised fuel are drawn in as the piston moves downwards.	A
Compression stroke	Fuel vapour and air are compressed as the piston moves upwards. Gas volume in the cylinder reduces, causing a rise in temperature.	B
Combustion stroke	Fuel is ignited by sparks from spark plugs and burns explosively increasing the gas temperature and pressure rapidly and then pushing the piston downwards as the gas expands rapidly and cools as it pushes the piston down.	C
Exhaust stroke	Exhaust is driven out as the piston moves up again.	D

1.6.2.3

Table 1	
Position	**Correct label**
E	Intake valve open
F	Air-fuel mixture
G	Exhaust valve closed
H	Combustion chamber
I	Valves closed
J	Compressed air-fuel mixture
K	Valves closed
L	Spark plug firing
M	Intake valve closed
N	Exhaust valve open
O	Exhaust gases
P	Exhaust gases

1.7.1.1

	Completed sentences
(a)	100% of the energy entering Earth's atmosphere comes from the Sun.
(b)	About 50% of the energy from the Sun is absorbed by the Earth's surface, the land and oceans.
(c)	About 30% of the Sun's energy is directly reflected back to space by the atmosphere.
(d)	About 20% of the Sun's energy is absorbed by the atmosphere.
(e)	Most of the energy re-emitted from the surface does not go directly out to space.
(f)	This re-emitted energy is reabsorbed by clouds and gases in the atmosphere.
(g)	Some of the re-emitted energy gets redistributed by convection.
(h)	More re-emitted energy is released into the atmosphere through condensation.
(i)	The majority of the energy re-emitted from the surface is absorbed by the greenhouse gases.
(j)	These gases constantly emit the Sun's energy back into the atmosphere and keep the Earth a habitable temperature.
(k)	Eventually, most of the energy makes its way back out to space and Earth's energy balance is fairly well maintained.
(l)	The energy that doesn't make its way out is responsible for global warming.
(m)	Earth's energy imbalance is the difference between the amount of solar energy absorbed by Earth and the amount of energy the planet radiates to space as heat.
(n)	Earth's energy imbalance is the single most crucial measure of the status of Earth's climate and it defines expectations for future climate change.

1.7.1.2 If the energy was originally in the atmosphere, why could it not radiate into space without having to go through a cycle of evaporating water which then condenses to put it back into the atmosphere. The idea is a little circular in that it really does not offer a viable strategy for reducing surface energy.

1.7.1.3 (a) Climate change models are based on information which is possibly not accurate.

Climate change scientists make the data fit the outcomes they want it to show.

(b) The statement has no reference as to its origin and therefore it is impossible to assess its credibility. However, the ideas it puts forwards are commonly stated criticisms of those against the idea of global warming, which makes the ideas themselves warrant consideration. One should not make any conclusion nor form a firm opinion based on a simple statement like this. Much more research would be needed before this could be done.

1.7.2.1 (a) There is no difference between these terms.

(b) Personal choice by the person drawing the diagram. There is no scientific reason for the difference.

(c) Author's choice is diagram Y because it shows usages in the past (as does Z, but only until the present) and predicts usage into the future (as does X, but only for one data point).

(d) Solid bars represent energy actually used (by whatever measurement strategy was employed) while the dotted bars represent predictions into the future.

(e) Diagram Y because it shows a 50 year plot of data from which it is much easier to draw in a predictive trend line.

(f) All three indicate significant increases in energy use over time, which in turn predicts that there will eventually be shortages of fossil fuels on which most of our energy use is based. They suggest that we need to find alternate sources for energy if it is to be sustainable.

(g) The term refers to viable sources of energy being available for eternity.

(h) They will have to be renewable sources currently in existence, or newly developed renewable sources. The fossil fuels are going to be depleted at some stage in the future.

2.1.1.1 (a) Models change to suit observations that are made but cannot be explained by the existing model.

(b) As technology and knowledge grows, new observations are made and changes are needed.

(c) They all have a small central nucleus.

2.1.1.2 (a) Rutherford (1909).

(b) This was his famous alpha particle scattering experiment and the observation that led to the proposal of the existence of a small, dense nucleus was that a few alpha particles were reflected straight or almost straight back.

(c) Any four of:

It did not explain atomic spectra.

It did not explain the differing intensities of spectral lines.

It did not explain the effect of a magnetic field on spectral lines (the Zeeman effect).

It did not explain the anomalous Zeeman effect.

It did not explain electrical discharge in a discharge tube.

It did not explain the photoelectric effect.

2.1.1.3 (a) They were attempting to explain atomic spectra

(b) A = absorption spectrum

B = emission spectrum

C = continuous spectrum

(c) A = Light passes through a gas and photons, representing energy transitions of the atoms of that gas, are absorbed – the energy values in the spectrum appear as black lines because it does not reach the photographic plate.

B = Photons emitted as electrons in excited atoms return to lower energy levels and produce the emission spectrum lines – the rest of the spectrum is black because only the energy values associated with the particular electron transmission in the particular atom produce spectral lines.

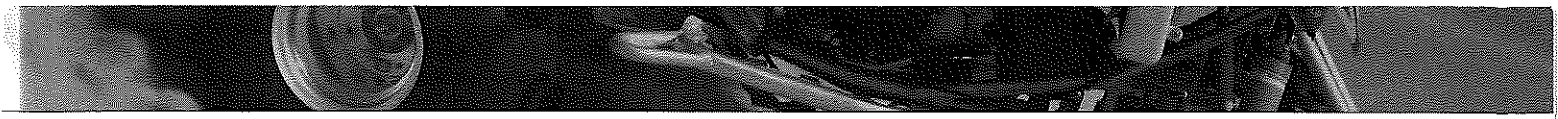
C = All spectral lines appear in the spectrum; our eyes do not have the resolution to see them as separate lines – the spectrum appears continuous because the source (e.g. the Sun) is emitting all (visible) frequencies.

(d) They are spectra of the same element. The frequencies of radiation emitted as excited electrons fall back to lower energy levels and produce the emission spectrum are the same energy frequencies absorbed from a beam of radiation which contains all frequencies. This energy excites those electrons into higher energy levels.

2.1.2.1 (a) The atomic number of an atom of an element is equal to the number of protons in its nucleus

(b) The mass number of an atom of an element is equal to the sum of the number of protons and neutrons in its nucleus.

(c) The atomic mass of an element is the average mass of its atoms when all component isotopes are considered.



2.1.2.2 (a) A = protons

B = electrons

C = nucleus

D = neutrons

Key
- O Neutron
- ● Proton
- ◎ Electron

(b) 5

(c) 11

(d) Boron

(e) It has one less proton in its nucleus and also one less neutron than the next atom which is carbon.

2.1.2.3 (a) A = atomic number

B = atomic symbol

C = name (iron)

D = mass number (this is often stated as the atomic mass in periodic tables in which case it will not be an integral number)

(b) 26

(c) 56

2.1.2.4 A

2.1.2.5 D

2.1.2.6 B

2.1.2.7 A

2.1.2.8 D

2.2.1.1 (a) If any nucleus is to stay together and not burst apart due to the electrostatic repulsions there must be another force. We call this force the strong nuclear force.

(b) A = average distance between nucleons

B = electrostatic force of repulsion between protons

C = strong nuclear force

(c) Holds all nuclear particles together, whether charged or uncharged.

Is much stronger than electrostatic forces.

Is attractive over only a very small distance (about 10^{-15} m).

Becomes repulsive at small distances (less than the diameter of a nucleon).

2.2.1.2 The strong nuclear force in particular requires ideal distances between the protons. If the distance between protons is too small then the strong nuclear force is repulsive. If the average separation of protons is too large, then the strong nuclear force is too weak to hold them together. The neutrons help create these ideal distances. If there are too few neutrons, or too many neutrons, the nucleus becomes unstable. Large nuclei have many neutrons which increase the distance between protons thereby making the atoms unstable.

2.3.1.1 (a) Isotopes are atoms of the same element which have different numbers of neutrons in their nucleus. In other words, the atoms have the same atomic number but different mass numbers.

(b) The atomic numbers of the atoms of the isotopes of a particular element wil be identical.

(c) The mass numbers will be different due to different numbers of neutrons in the nuclei.

(d) Mass number will always be an integral number, being the simple sum of the number of protons and neutrons in nuclei. The mass number however, reflects the average mass of the nuclei with the percentages of component isotopes considered.

2.3.1.2

Isotope	Number of protons	Number of neutrons	Number of electrons	Atomic number	Mass number
Lithium-6	3	3	3	3	6
Lithium-7	3	4	3	3	7
Lithium-8	3	5	3	3	8

2.3.1.3

Isotope	Number of protons	Number of neutrons	Number of electrons	Atomic number	Mass number
A	6	6	6	6	12
B	6	7	6	6	13
C	6	8	6	6	14

2.3.1.4 (a) Regardless of which particle represents the protons, the atoms have different numbers of protons, so they are from different elements.

(b) The number of protons and electrons should be the same, so atom Y has 1 electron short. An additional electron needs to be added.

2.3.2.1 (a) The decaying nucleus emits an alpha particle which is a helium nucleus, $({}^{4}_{2}He)^{2+}$. The mass number of the nuclide decreases by 4 and the atomic number by 2.

(b) The nucleus emits an electron and an electron antineutrino. The atomic number increases by 1 as a neutron decays, but the mass number stays the same because the neutron decay also results in the formation of an additional proton.

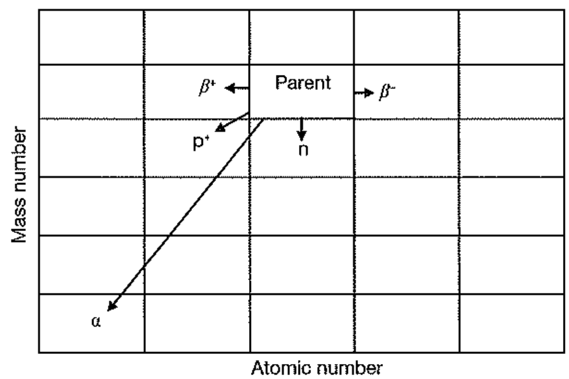
(c) A neutron is ejected from the nucleus. Atomic number does not change, mass number decreases by 1.

(d) The nucleus emits a positron (a positively charged electron) and an electron neutrino as a proton decays. Atomic number decreases by 1, mass number stays the same as a neutron is also formed.

(e) A proton is ejected from the nucleus. Both atomic number and mass number decrease by 1.

(f) Note that gamma rays usually accompany alpha and beta decay, but not always. It depends on the particular decay and the amount of energy involved. There is no rule for predicting gamma rays in any particular decay reaction.

2.3.2.2



2.3.2.3 (a)

Decay reaction	General equation for reaction
Alpha	${}^{A}_{Z}X \rightarrow {}^{A-4}_{Z-2}X + {}^{4}_{2}He$
Beta	${}^{A}_{Z}X \rightarrow {}_{Z+1}{}^{A}X + {}^{0}_{-1}e + {}^{0}_{0}\overline{v}$ (Note that beta decay is always accompanied by an antineutrino)
Positron	${}^{A}_{Z}X \rightarrow {}_{Z-1}{}^{A}X + {}^{0}_{+1}e + {}^{0}_{0}v$ (Note that positron decay is always accompanied by a neutrino)
Proton	${}^{A}_{Z}X \rightarrow {}^{A-1}_{Z-1}X + {}^{1}_{1}p$ (or ${}^{1}_{1}H$)
Neutron	${}^{A}_{Z}X \rightarrow {}^{A-1}_{Z}X + {}^{1}_{0}n$

(b) Gamma rays usually accompany alpha and beta decay, but not always. It depends on the particular decay and the amount of energy involved. There is no rule for predicting gamma rays in any particular decay reaction.

2.3.2.4 A

2.3.2.5 D

2.3.2.6 A

2.3.2.7 B

2.3.2.8 C

2.3.2.9 C

2.3.2.10 B

2.3.2.11 B

2.3.3.1 The concept of entropy is that 'things in nature seek their lowest energy state'. In their lowest energy state, things are most stable and less likely to change. Applying this to atoms, stable atoms have low energy states while unstable atoms will try and become stable by getting to a lower energy state. They will typically do this by emitting some form of radioactivity and change in the process.

2.3.3.2 The strong nuclear force in particular requires ideal distances between the protons. Remember, if the distance between protons is too small then the strong nuclear force is repulsive. The neutrons help create these ideal distances. If there are too few neutrons, or too many neutrons, the nucleus becomes unstable.

2.3.3.3 If an atom has more than 82 protons in the nucleus, there is no arrangement of neutrons that can produce more attractive forces than repulsive forces. Therefore, all isotopes of elements beyond lead are radioactive. Their only route to stability is to reduce the overall size of the nucleus.

2.3.3.4 Science still does not completely understand why certain isotopes are more stable than others. There are theories based on observations of stable and unstable isotopes, but no real understanding of why. This is just another, unexplained observation.

2.3.3.5 (a) Binding energy refers to the energy bound up in the strong nuclear force holding the nucleons together in the nucleus.

(b) Elements between mass number 50 and 82 are the most stable, having the highest binding energy per nucleon.

(c) For nuclei with fewer than 50 nucleons, the average distance between them is smaller than ideal, so the strong nuclear force is not as strong. For atoms with atomic mass 82, the number of nucleons is larger than ideal so, on average, they are further apart than ideal, so again, the strong nuclear force is weaker.

(d) No. It simply indicates relative stability, not absolute stability or instability.

2.3.3.6 (a) The line of stability indicates the stable isotopes of the elements.

(b) Above the line of stability, the neutron/proton is higher than that indicated in more stable nuclei, so nuclear decay would be most likely to occur to reduce the number of neutrons and increase the number of protons in order to approach the line of stability – hence beta decay.

(c) Below the line of stability, the neutron/proton is lower than that indicated in more stable nuclei, so nuclear decay would be most likely to occur to reduce the number of neutrons to approach the line of stability – hence positron decay.

(d) All nuclei in this area (mass number greater than 82) are too large to be stable, and would therefore be most likely to decay in a way which reduces their size – i.e. nuclear fission, or by releasing an alpha particle (or both).

(e) The graph is only an indicator of trends, not of absolute information.

2.3.4.1

Radiation	Charge	Mass (amu)	Penetrating power	Ionising power	Path through electric field	Path through magnetic field	Stopped by	What is it?
Alpha	+2	4	Low	High	Deflected towards negative	Circular	20 cm air. Sheet of paper.	Helium nucleus
Beta	−1	$\frac{1}{1836}$	Medium	Medium	Deflected towards negative	Circular in opposite direction	Thin sheet of aluminium.	Electron
Gamma	0	0	High	Low	Not deflected	Not deflected	Sheet of lead.	Electromagnetic radiation

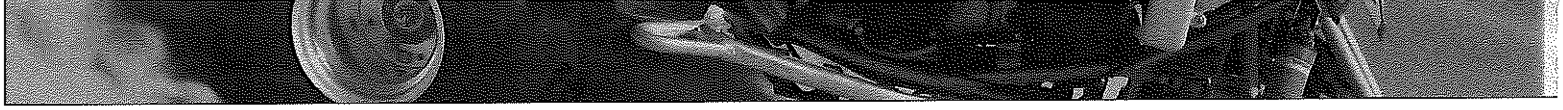
2.3.4.2 (a) Ionising radiation describes radiation which has enough energy to cause atoms to become ionised to ions. During ionising radiation, an electron is ejected off the atom, causing the atom to lose an electron and become ionised.

(b) Particles will penetrate matter further if they are uncharged (and therefore not repelled by the matter atoms) and small. Alpha particles therefore miss out on both counts, while gamma rays have both advantages. Beta particles are much smaller than alphas but are also charged, so they fit into the middle in terms of penetrating power.

(c) The reverse properties apply for ionising power. The greater the mass of the particle, the more chance the particle has to knock electrons out of orbit. The charge on the particle also helps as it can contribute to the force pulling electrons out of orbit.

2.3.4.3

	Completed sentences
(a)	In alpha particle decay, a positively charged particle is emitted from the nucleus.
(b)	This alpha particle consists of two protons and two neutrons.
(c)	Alpha particles are helium nuclei.
(d)	Alpha particles can only travel a few centimetres in air.
(e)	Alpha particles can be stopped by a sheet of paper.
(f)	Alpha particles can be stopped by human skin.
(g)	In beta decay, an electron is emitted from the nucleus.
(h)	Beta particles can penetrate several metres of air.
(i)	Beta particles can penetrate several millimetres of human skin.
(j)	Most beta particles will be stopped by a few centimetres of a light material like aluminium or plastic.
(k)	Gamma decay occurs because the nucleus of an atom is in a very high energy state.
(l)	The nucleus 'falls down' to a lower energy state, emitting a photon, more commonly known as a gamma ray.
(m)	Gamma particles or gamma rays are actually electromagnetic radiation.
(n)	Gamma particles travel at the speed of light.
(o)	Gamma rays can be stopped by lead, steel or concrete.
(p)	Gamma rays can be stopped by several metres of water.



2.3.5.1 (a) $^{238}_{92}U \rightarrow ^{234}_{90}Th + ^{4}_{2}He$

(b) $^{234}_{90}Th \rightarrow ^{234}_{91}Pa + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(c) $^{234}_{91}Pa \rightarrow ^{234}_{92}U + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(d) $^{234}_{92}U \rightarrow ^{230}_{90}Th + ^{4}_{2}He$

(e) $^{230}_{90}Th \rightarrow ^{226}_{88}Ra + ^{4}_{2}He$

(f) $^{226}_{88}Ra \rightarrow ^{222}_{86}Rn + ^{4}_{2}He$

(g) $^{222}_{86}Rn \rightarrow ^{218}_{84}Po + ^{4}_{2}He$

(h) $^{218}_{84}Po \rightarrow ^{214}_{82}Pb + ^{4}_{2}He$

(i) $^{214}_{82}Pb \rightarrow ^{214}_{83}Bi + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(j) $^{214}_{83}Bi \rightarrow ^{214}_{84}Po + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(k) $^{214}_{84}Po \rightarrow ^{210}_{82}Pb + ^{4}_{2}He$

(l) $^{210}_{82}Pb \rightarrow ^{210}_{83}Bi + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(m) $^{210}_{83}Bi \rightarrow ^{210}_{84}Po + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(n) $^{210}_{84}Po \rightarrow ^{206}_{82}Pb + ^{4}_{2}He$

2.3.5.2 (a) $^{210}_{81}Tl \rightarrow ^{210}_{82}Pb + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(b) $^{8}_{4}Be \rightarrow ^{4}_{2}He + ^{4}_{2}He$

(c) $^{14}_{6}C \rightarrow ^{14}_{7}N + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(d) $^{211}_{83}Bi \rightarrow ^{207}_{81}Tl + ^{4}_{2}He$

(e) $^{24}_{11}Na \rightarrow ^{24}_{12}Mg + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(f) $^{137}_{55}Cs \rightarrow ^{137}_{56}Ba + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

(g) $^{24}_{11}Na \rightarrow ^{24}_{12}Mg + ^{0}_{-1}e + ^{0}_{0}\bar{\nu}$

2.3.5.3 D

2.3.5.4 A

2.3.5.5 D

2.3.5.6 C

2.3.5.7 A

2.3.5.8 B

2.3.5.9 A

2.3.5.10 B

2.4.1.1 The half-life of a radioisotope is the time required for one half of the amount of the isotope to degrade into a more stable material.

2.4.1.2 (a) $100\ s^{-1}$

(b) 9.00 pm

(c) 2

2.4.1.3 (a) $200 \div 2 \div 2 \div 2 \div 2 \div 2 = 6.25$

(b) $5 \times 8.04 = 40.2$ days

(c) The thorium would have the greater activity because it has the longer half-life.

(d) For the thorium, 50 days = 2.8 half-lives

Therefore fraction remaining = 0.1436

For the iodine, 50 days = 6.22 half-lives

Therefore fraction remaining = 0.0136

So activity thorium : activity iodine = 10.5 times greater

2.4.1.4 $\frac{2.00 \text{ mg}}{128.0 \text{ mg}} = 0.015625$ of the sample remains

Therefore, from $\left(\frac{1}{2}\right)^n = 0.015625$

$n \log 0.5 = \log 0.015625$

$n = \frac{\log 0.015625}{\log 0.5} = 6$

$\frac{24 \text{ days}}{6 \text{ half-lives}} = 4.00$ days (the length of the half-life)

2.4.1.5 $\frac{5.00}{100.0} = 0.05$ (decimal fraction remaining)

$\left(\frac{1}{2}\right)^n = 0.05$

$n = 4.32$ half-lives

36.0 hours × 4.32 = 155.6 hours

2.4.1.6 Fraction remaining = 0.25

Therefore $\left(\frac{1}{2}\right)^n = 0.25$

$n = 2$

Therefore time = 24.52 years.

2.4.1.7 40.2 days = 5 half-lives

Therefore fraction remaining = 0.03125 $\left(= \left(\frac{1}{2}\right)^n\right)$

0.3125% remaining.

2.4.1.8 Fraction remaining = 0.625

Therefore $\left(\frac{1}{2}\right)^n = 0.625$

$n = 0.68$

Therefore time = 0.68 × 18.72 = 12.7 days

2.4.1.9 Recognise $\frac{1}{4}$ as a fraction associated with 2 half-lives (from $\left(\frac{1}{2}\right)^4 = \frac{1}{4}$)

3.82 days × 2 = 7.64 days

2.4.1.10 In 24 hours, the sample goes from 100% to 95%

$\left(\frac{1}{2}\right)^n = 0.95$

$n \log 0.5 = \log 0.95$

$n = 0.074$

$\frac{24 \text{ hrs}}{0.074} = 324$ hrs = one half-life

2.4.1.11 Fraction remaining = $\frac{(5.20 \times 10^6)}{(5.32 \times 10^9)}$

9.77×10^{-4}

Therefore $\left(\frac{1}{2}\right)^n = 9.77 \times 10^{-4}$

$n = 10$

Therefore time = 301.7 years.

2.4.1.12 99% lost = 1% remaining = 0.01

Therefore $\left(\frac{1}{2}\right)^n = 0.01$

$n = 6.64$

Therefore time = 95 days.

2.4.1.13 (a) 159 days × 24 hrs/day × 60 min/hour = 228960 min

228960 min × (3.4×10^{13} disintegrations per 26 min) = $2.994 \times 10 \times 10^{17}$ total disintegrations in 159 days

$\frac{2.994 \times 10 \times 10^{17}}{3.25 \times 10^{18}} = 0.0921$

9.21% has disintegrated

(b) 0.9079 is the decimal fraction of the substance remaining since 0.0921 has gone away

$\left(\frac{1}{2}\right)^n = 0.9079$

$n \log 0.5 = \log 0.9079$

$n = 0.139$ half-lives

$\frac{159 \text{ days}}{0.139} = 1144$ days

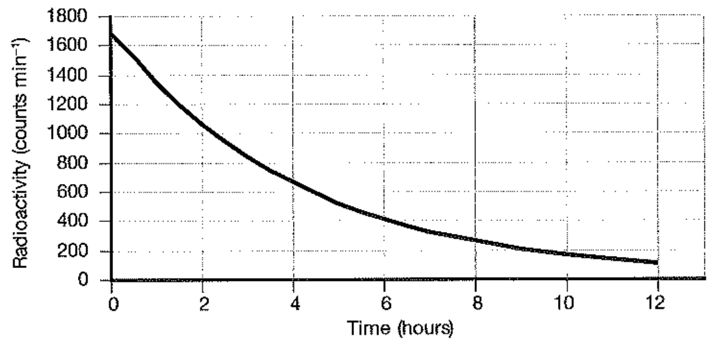
2.4.2.1 (a) 50

(b) The background radiation is always there and it contributes to the overall count. If it is not considered, then the results will be in error by this amount. It would be a 'zero' error and make the result invalid.

(c)

Time (hours)	Total radiation counts recorded (counts min^{-1})	Radiation due to X (counts min^{-1})
0	1730	1680
2	1110	1060
4	715	665
6	465	415
8	320	270
10	215	165
12	160	110

(d)



(e) 3 hours

(f) No. half lives in 20 hours = $\frac{20}{3}$ = 6.67

Fraction remaining after 6.67 half-lives = $\left(\frac{1}{2}\right)^{6.67}$ = 0.00982

Radiation count = 1680 × 0.00982 = 16.5

(g) 36 hours represents 12 half-lives which is beyond a measurable number, so count will simply indicate the background radiation = about 50.

2.5.1.1 (a) It is a technique based on a comparison between the observed abundance of a naturally occurring radioactive isotope and its decay products, using known decay rates.

(b) It is used to date materials like rocks, fossil remains, ancient artefacts, and to determine the historical record within sedimentary rock layers.

(c) It is based on a comparison between the observed abundance of a naturally occurring radioactive isotope and its decay products, using known decay rates.

(d) Rutherford in 1905.

(e) He was trying to determine an age for the Earth.

2.5.1.2 (a) The geological time scale is a chronological list of the species of life on Earth and the sedimentary rock layers on Earth.

(b) Radiometric dating has been used to determine the age of the fossils which in turn determines the age of the rocks the fossils are in, and to determine the age of rocks which do not contain fossils in order to construct the time scale.

2.5.1.3 (a) Carbon-14.

(b) All organisms take in carbon during their lifetimes, plants through photosynthesis, and animals by eating of plants and other animals and through breathing. When an organism dies, it ceases to take in new carbon-14, and the existing isotope decays with a characteristic half-life (5730 years). The proportion of carbon-14 left when the remains of the organism are examined provides an indication of the time elapsed since its death.

2.5.1.4 (a) This represents 11 half-lives, so $\frac{100}{(2^{11})}$ = fraction remaining = 0.0488

(b) The carbon-14 dating limit lies around 58 000 to 62 000 years (10 to 11) half-lives because the amount of C-14 remaining after this time is too small to measure accurately.

2.5.1.5 0.22 per cent of K-40 remains.

$\left(\frac{1}{2}\right)^n = 0.22$

$n \log 0.5 = \log 0.22$

$n = 2.18$

Total elapsed time = (2.18) (1.27×10^9) = 2.77×10^9 yrs = age of the sample

2.5.1.6 Sample is 3.8 parts by mass Ar and 1 part K, so original sample contained 4.8 parts K and zero parts Ar.

Therefore the fraction of K-40 that remains = 0.20833

So $\left(\frac{1}{2}\right)^n = 0.20833$

Therefore $n = 2.263$

Total elapsed time = $(2.263)(1.27 \times 10^9) = 2.87 \times 10^9$ yrs

2.5.1.7 Fraction of C-14 remaining:

$\frac{6.00}{13.6} = 0.4411765$

$\left(\frac{1}{2}\right)^n = 0.4411765$

$n = 1.18057$

Time elapsed = 5730 yr × 1.18057 = 6765 yrs

2.5.1.8 Fraction of C-14 remaining = $\frac{0.109}{0.296} = 0.368243$

$\left(\frac{1}{2}\right)^n = 0.368243$

$n = 1.441269$

5730 yr × 1.441269 = 8258 yrs

2.5.1.9 Fraction of C-14 remaining = 0.300

Half-lives elapsed = $\left(\frac{1}{2}\right)^n = 0.300$

$n = 1.737$

Time elapsed = 5730 yr × 1.737 = 9953 yrs

2.5.1.10 Half-lives elapsed = $\frac{11\,430}{5730} = 1.9947644$

Fraction remaining = $\left(\frac{1}{2}\right)^{1.9947644} = x = 0.25091$

C-14 present in the past = $\left(\frac{0.060 \text{ mg}}{0.25091}\right) = \left(\frac{x}{1}\right)$

$x = 0.239$ mg

2.5.1.11 Half-lives elapsed = $\frac{12\,900 \text{ yr}}{5730 \text{ yr}} = 2.2513$

Fraction remaining = $\left(\frac{1}{2}\right)^{2.2513} = x = 0.21$

Counts remaining = 15.3 × 0.21 = 3.2

2.5.1.12 Fraction remaining = $\frac{9.58}{15.3} = 0.6261438$

$\left(\frac{1}{2}\right)^n = 0.6261438$

$n \log 0.5 = \log 0.6261438$

$n = 0.675434$

Determine number of years:

5730 years × 0.675434 = 3870 years

2.5.1.13 $\frac{10\,000 \text{ yr}}{5730 \text{ yr}} = 1.7452$ half-lives

$\left(\frac{1}{2}\right)^{1.7452} = 0.2983$ = fraction remaining

15.3 times 0.2983 = 4.56 counts min^{-1}

2.6.1.1

(a) Activity is defined as the rate at which the isotope decays. Radioactivity may be thought of as the amount of radiation produced in a given amount of time. The International System (SI) unit for activity is the becquerel (Bq) where 1 becquerel equals one disintegration per second.

(b) Exposure is a measure of the strength of a radiation field at some point in air. This is the measure made by a survey meter. The most commonly used unit of exposure is the roentgen (R).

(c) Absorbed dose is the amount of ionising radiation energy absorbed per kg of matter. Absorbed dose is measured in grays (Gy), where 1 Gy equals 1 joule of energy absorbed per kg. The 'rad' (radiation absorbed dose) is also used. 100 rads equals 1 Gy.

(d) Dose equivalent. α, β and γ radiations affect human tissue differently. Alpha are most damaging because of their high ionising power. Dose equivalent measures dose absorbed per unit time, adjusted for the effect of the radiation on tissue. Dose rates are expressed as sieverts (Sv) per hour, but the rem is commonly used. Rem is an acronym for 'roentgen equivalent in man.' One rem is equivalent to 0.01 SV.

2.6.1.2

(a) Type of radiation involved. All kinds of ionising radiation can produce health effects. The main difference in the ability of alpha and beta particles and gamma and X-rays to cause health effects is the amount of energy they have. Their energy determines how far they can penetrate into tissue and how much energy they are able to transmit directly or indirectly to tissues.

(b) Size of dose received. The higher the dose of radiation, the higher the likelihood of health effects.

(c) Rate the dose is received. Tissue can receive large doses. If this occurs over a number of days or weeks, the results are not as serious as if the dose was received in a matter of minutes.

(d) Part of the body exposed. Extremities such as the hands or feet are able to receive a greater amount of radiation with less resulting damage than blood forming organs housed in the torso.

(e) The age of the individual. As a person ages, cell division slows and the body is less sensitive to the effects of ionising radiation. Once cell division has slowed, the effects of radiation are somewhat less damaging than when cells were rapidly dividing.

(f) Biological differences. Some individuals are more sensitive to the effects of radiation than others. Studies have not been able to conclusively determine the differences.

2.7.1.1 (a) Binding energy is the amount of energy it would take to completely break up a nucleus and separate all the neutrons and protons in it.

(b) The 'binding energy' of a nucleon is of the order of a million times greater than the electrostatic force between electrons and the nucleus.

(c) Theoretically, the greater the binding energy per nucleon, the more stable the nucleus will be.

2.7.1.2 (a) $8.7948 \times 62 = 545.28$ MeV

(b) 8.72×10^{11} J

2.7.1.3 (a) The $^{39}_{19}K$ has the greater total binding energy. From a binding energy graph (see next question) we can see that the binding energy per nucleon for the potassium nucleus is greater than the binding energy per nucleon for phosphorus, and potassium has 39 nucleons to phosphorus's 31, so its total binding energy will be greater.

(b) The $^{39}_{19}K$ will have the greater binding energy per nucleon. (The $^{39}_{19}K$ is higher up on the graph line on the binding energy per nucleon graph – see next question.)

(c) This suggests that the $^{39}_{19}K$ is more stable as each nucleon is held to the other nucleons by stronger forces.

2.7.1.4 (a) Iron. It has the highest binding energy per nucleon.

(b) If we accept the idea of entropy for nuclear reactions, then atoms will react to increase their stability. Atoms with mass numbers less than iron increase stability (binding energy per nucleon) by becoming larger – i.e. nuclear fusion.

(c) Again, accepting the idea of entropy for nuclear reactions, then atoms will react to increase their stability. Atoms with mass numbers more than iron increase stability (binding energy per nucleon) by becoming smaller – i.e. nuclear fission.

2.7.2.1 Nuclear reactions release enormous amounts of energy. When we examine them carefully, we find that the mass of the products is always less than that of the reactants. The difference is called the mass defect.

2.7.2.2 The energy released results from the conversion of the strong nuclear force (the binding energy), which appears as mass in an intact nucleus, into energy as some of the nucleons are separated from others. Because this energy is released during the reaction, the mass equivalence of it is lost by the daughter nuclei.

2.7.2.3 C

2.7.2.4 A

2.7.2.5 B

2.7.2.6 C

2.7.2.7 D

2.7.2.8 A

2.7.2.9 (a) Mass defect $= 235.0439 + 1.0087 - 87.9056 - 135.9072 - (12 \times 1.0087) = 0.1354$ u

$0.1354 \text{ u} \times 931.5 \times 1.6 \times 10^{-13} = 2.02 \times 10^{-11}$ J

(b) Mass defect $= (12.0 + 4.0026) - (15.000109 + 1.007825) = 0.005334$ u

(c) Mass defect $= 7.0160 + 1.007825 - 4.0026 - 4.0026 = 0.018625$ u

(d) Mass defect $= (36 \times 1.007825) + (56 \times 1.0087) - 91.8973 = 0.8716$ u

(e) The mass of the products is greater than the mass of the reactants, so the reaction is endothermic – that is it will not occur spontaneously.

2.8.1.1 (a) Spontaneous transformations are those which will occur without having to initiate the reaction by the input of energy.

(b) Spontaneous transformations are found only in very heavy chemical elements. Because the nuclear binding energy of the elements reaches its limit at about atomic number 82, spontaneous breakdown into smaller nuclei and a few isolated nuclear particles becomes possible at higher atomic numbers.

2.8.1.2 (a) Artificial transformations are those deliberately produced by scientists in high energy particle accelerators where high speed particles are fired at heavy nuclei to break them apart.

(b) They are done so that scientists can use the results to study their atomic structure.

(c) Rutherford's experiments in 1919 where he bombarded nitrogen with alpha particles.

(d) The discovery of the neutron in 1932.

2.8.1.3 (a) The first artificial transmutations were done in 1919 when hydrogen ions and helium nuclei were known but neutrons were not discovered until 1932.

(b) Positively charged particles are pushed away by the positively charged atomic nuclei, since like charges repel. It takes a lot of energy to get these particles to overcome the repulsion and strike the nuclei, and so nuclear reactions were difficult to initiate.

(c) Since they had no charge, but do have significant mass, they were not repelled by nuclei, and were able to hit them easily and transfer energy to split them apart.

(d) Having no charge they cannot be accelerated, so their speed had to be produced from whatever nuclear transformation produced the neutrons themselves. Fortunately, relatively slow moving neutrons can induce fission because of their relatively large mass.

2.8.1.4 (a) ${}^{27}_{13}Al + {}^{4}_{2}He \rightarrow {}^{30}_{15}P + {}^{1}_{0}n$

(b) ${}^{24}_{12}Mg + {}^{4}_{2}He \rightarrow {}^{27}_{14}Si + {}^{1}_{0}n$

(c) ${}^{23}_{11}Na + {}^{4}_{2}He \rightarrow {}^{26}_{13}Al + {}^{1}_{0}n$

2.9.1.1 Neutron induced fission is a reaction in which a heavy nuclide captures a neutron to form a larger, unstable nuclide which then splits into two smaller radioactive nuclides, often with the release of additional neutrons, gamma radiation and energy.

2.9.1.2 It was more massive than electrons, and so carried much more energy and momentum for slower speeds; and uncharged, so it could pass through electron shells unaffected and reach the nucleus.

2.9.1.3 The speed of the neutron caused it to react with target nuclei quite differently, forming quite different products.

2.9.1.4 At high speed it could collide with a nucleus, knock a proton out and decrease the atomic number by one. At slower speeds it spent more time in collision with the nucleus and could be absorbed into it instead of reflecting away. In these cases, the new nucleus was usually unstable and split into two smaller nuclei.

2.9.1.5 In 1939 Lise Meitner and Otto Frisch proposed an explanation for this, coining the term nuclear fission, and included a calculation indicating that a huge release of energy would accompany the artificial fission of uranium.

2.9.1.6 If fission released more neutrons, then the possibility of a self-sustaining chain reaction existed through neutron induced fission.

2.9.1.7 (a) When the N-14 absorbs the neutron it becomes an atom of N-15. Because there is no proton change, it remains a nitrogen atom.

${}^{14}_{7}N + {}^{1}_{0}n \rightarrow {}^{15}_{7}N$

(b) The K-35 absorbs the neutron to become K-36. Again there is no proton change so it remains a potassium atom.

${}^{35}_{19}K + {}^{1}_{0}n \rightarrow {}^{36}_{19}K$ (This is isotope X.)

The K-36 then decays by alpha emission to form S-32. This involves both neutron and proton change, so the daughter atom is a new element (sulfur in this case).

${}^{36}_{19}K \rightarrow {}^{32}_{14}S$ (Isotope Y) $+ {}^{4}_{2}He$

2.9.1.8 (a) ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U$

(b) ${}^{236}_{92}U \rightarrow {}^{92}_{36}Kr + {}^{141}_{56}Ba + 3{}^{1}_{0}n$

(c) The additional neutrons can initiate further neutron capture/fission reactions leading to the production of atomic energy or a nuclear explosion.

2.10.1.1 (a) In a controlled nuclear reaction only one subsequent fission is allowed to occur for each fission initiated.

In an uncontrolled reaction, every neutron produced in each fission initiates further fission reactions resulting in the number of fissions occurring escalating exponentially.

(b) In a controlled reaction the number of fissions occurring remains constant. Energy is produced at a steady rate. In the uncontrolled reaction uncontrollable amounts of energy are released.

(c) Extra neutrons produced during each fission reaction with the energy which makes them able to cause fission must be absorbed by control rods in the nuclear reactor rather than being allowed to hit fuel atoms.

2.10.1.2 (a) 2.5 neutron per fission.

(b) Not all neutrons are produced with the energy needed to initiate further fission reactions.

(c) A sustained nuclear reaction could not be obtained and the reaction would soon cease.

2.10.1.3 (a) (i) ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U$

${}^{236}_{92}U \rightarrow {}^{141}_{56}Ba + {}^{92}_{36}Kr + 3{}^{1}_{0}n$

(ii) ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U$

${}^{236}_{92}U \rightarrow {}^{139}_{57}La + {}^{95}_{42}Mo + 2{}^{1}_{0}n + 7{}^{0}_{-1}e$

(iii) ${}^{235}_{92}U + {}^{1}_{0}n \rightarrow {}^{236}_{92}U$

${}^{236}_{92}U \rightarrow {}^{88}_{38}Sr + {}^{136}_{54}Xe + 12{}^{1}_{0}n$

(iv) $^{235}_{92}U + ^{1}_{0}n \rightarrow ^{236}_{92}U$

$^{236}_{92}U \rightarrow ^{147}_{57}La + ^{87}_{35}Br + 2^{1}_{0}n$

(b) Reactions (i) and (iii) because they both produce more than 2.5 neutrons per fission.

2.10.1.4 (a) Neutron capture by U-235 to form U-236.

(b) Fission of U-236 to form Ba-141 and Kr-92, energy and 3 neutrons.

(c) The chain reaction set up from the bombardment of U-235 with neutrons.

2.10.2.1 (a) Nuclear fission involves a large, unstable nucleus splitting into smaller nuclei either spontaneously, or induced by particle bombardment.

(b) The source of energy in fission is the mass defect involved.

(c) Larger elements (those with mass numbers greater than 56).

(d) Fusion to form even larger nuclei would result in the binding energy per nucleon in the product nucleus being less than that in the reacting nucleus. In fission, the binding energy per nucleon in the daughter products is greater than that in the reacting nucleus – the product nuclei are more stable.

2.10.2.2 (a) $^{90}_{37}Kr$

(b) $^{236}_{92}U \rightarrow ^{143}_{55}Cs + ^{90}_{37}Kr + 3^{1}_{0}n$

(c) Energy.

(d) Experimentation has shown that, on average, 2.5 neutrons released will enable a sustained chain reaction to occur via this fission reaction.

2.11.1.1 (a) Nuclear fusion involves two smaller nuclei merging/combining to form a large nucleus.

(b) The source of energy in fission is the mass defect involved.

(c) Smaller elements (those with mass numbers less than 56).

(d) Fusion of these smaller nuclei to form even larger nuclei results in the binding energy per nucleon in the product nucleus being larger than that in the reacting nucleus. If fission occurred, the binding energy per nucleon in the reactant would be greater than that of the smaller products, so fission is unlikely to occur.

2.11.1.2 Nuclear fusion will occur if the binding energy per nucleon in the product is higher than the binding energies per nucleon in the reactants. The product nucleus is more stable than the reactant nuclei.

2.11.1.3 They are both equally correct. It is just a different way of looking at the same energy. In each case, the energy involved is the energy holding nucleons together in the nucleus.

2.11.2.1 (a)

(a)	Mass defect for the U-235 fission reaction in amu	0.185452 u
(b)	Energy release in MeV	172.74854 MeV
(c)	Mass defect for the U-235 fission reaction in kg	1.2249×10^{-28} kg
(d)	Energy release in joules	1.1024×10^{-11} J
(e)	Mass of one U-235 atom in kg	3.9029×10^{-25} kg
(f)	Energy released per kg of U-235	2.8246×10^{13} J
(g)	Mass defect for the hydrogen fusion reaction in amu	0.0259534 u
(h)	Energy release in MeV	24.17559 MeV
(i)	Mass defect for the hydrogen fusion reaction in kg	4.30956×10^{-29} kg
(j)	Energy release in joules	3.87861×10^{-12} J
(k)	Mass of fusion reactants in kg	6.69033×10^{-27} kg
(l)	Energy released per kg of fusion atoms	5.79733×10^{14} J

(b) The fusion releases 20.5 times as much energy per kg of fuel material.

(c) These values compare the energy released per atom of the fusion and fission, so the mass of fuel is not controlled. The comparisons are not as valid as comparing (f) and (l).

2.12.1.1 The wartime politics worked in Planck's favour as a lot of resources that he would not normally have had access to would have been allocated to scientific programs working towards the war effort. However, in Einstein's case, because he was morally opposed to allowing his ideas to be used for destructive purposes, his facilities were withdrawn and he had to leave Germany. His resources to continue his research would have decreased significantly.

2.12.1.2 Given their country was at war, German society would have applauded Planck and probably declared Einstein unpatriotic, although German Jews, particularly over time, would come to support his actions.

2.12.1.3 The Manhattan project was the beginning of nuclear energy both as a weapons technology and as a source of nuclear energy for peaceful purposes, mainly the production of electricity.

(a) During the 1940s – Only the scientific community really knew what was going on until 1945 and they were excited and concerned about this new field of study with enormous energy potential, but at the same time, enormous destructive potential. After 1945, shock, horror and outcry over the devastation of the bombs and the subsequent radiation sicknesses, which were mostly unexpected; comfort for many because it ended the war and brought soldiers home from hostilities.

(b) Over the subsequent 30 years – Fear over who might have the bomb and who might be developing it; the age of international spies; politically the Cold War; a nuclear 'stand-off'; development of nuclear power stations; lots of cheaper energy; radioisotopes with potential cures for many diseases; nuclear submarines for greater military potential; concern about nuclear accidents; radioactive fallout from nuclear testing.

(c) At the present time – Concern about nuclear accidents and disposal of wastes; concern that small militant nations have nuclear capability – that the nuclear technology is out of control; applications to medical science – cures for cancers, life extended; agriculture – disease-resistant plants, better food storage, longer shelf life, cheaper produce; industry – easier detection of flaws and leaks, cheaper services, more reliable structures; much more control over nuclear development – concern about age of current reactors particularly following Fukushima.

2.12.1.4 The destructive potential of atomic bombs had been grossly underestimated by the theoretical scientists – once tested, many of the scientists working on them campaigned to have the work stopped. Once used, their effects appalled most people. Einstein obviously felt guilty at being significantly responsible for their development.

2.12.2.1 (a) Perhaps the amount is too much to publish as it may generate significantly greater feelings against nuclear energy production. Perhaps it is published, but not circulated beyond those 'who need to know'. Perhaps governments suppress the data so that people and other countries do not know what is happening in their country.

(b) Its level of radioactivity and the half-life of the radioisotopes involved.

(c) Both are needed because the level of radioactivity indicates immediate danger to life while half-life will indicate how long it will be a problem. A low level waste with a long half-life may cause as much damage over time as a very short lived isotope with higher activity.

2.12.2.2 (a) Uranium tailings, also called mill tailings, are waste by-product materials left over from the mining and processing of uranium ore. They are not significantly radioactive. Vast quantities remain at many old mining sites awaiting burial in lined pits.

(b) Exempt waste and very short lived waste. The concentration of radionuclides is so low that they can be considered to have negligible radiological hazard and can be disposed of by normal waste management. Some may have a level of radioactivity that requires them to be stored for up to a few years and it can also be disposed of as regular waste.

(c) Low level waste is generated from hospitals and industry, as well as the nuclear fuel cycle. Low level wastes include paper, rags, tools, clothing, filters, and other materials which contain small amounts of mostly short-lived radioactivity. Some higher activity LLW requires shielding during handling and transport but most LLW is suitable for shallow land burial, ocean depth disposal or incineration. Isolation of LLW, depending on the radionuclides, could be several hundred years.

(d) Intermediate level waste contains higher amounts of radioactivity and in some cases requires shielding. Intermediate level wastes includes resins, chemical sludge and metal reactor nuclear fuel cladding, as well as contaminated materials from reactor decommissioning. It may be solidified in concrete or bitumen for disposal, buried in shallow repositories, or ocean depth disposal, with long-lived isotopes (from fuel and fuel reprocessing) being deposited in underground tunnels in stable geological areas at depths greater than 100 m.

(e) High level waste contains fission products and highly radioactive transuranic elements generated in the reactor cores. HLW accounts for over 95 per cent of the total radioactivity produced in the process of nuclear electricity generation. Storage of HLW in deep geological storage areas (at least several hundred to at least a thousand metes) is considered the most appropriate strategy.

(f) Transuranic waste is waste that is contaminated with alpha-emitting transuranic radioisotopes with half-lives greater than 20 000 years. Because of their long half-lives, It arises mainly from weapons production or decommissioning, and consists of clothing, tools, rags, residues, debris and other items contaminated with small amounts of radioactive elements (mainly plutonium). TRUW is also to be stored in deep geological storage areas.

2.12.3.1 In 1992 the Australian government initiated an Australia-wide survey to site a low level waste storage facility for the storage of the ILW. In 2001 it announced that it would build a facility. The project was abandoned until 2005.

On 15 July 2005 the government announced that three sites on Department of Defence land in the Northern Territory would be investigated to determine their suitability as a site for a co-located LLW and ILW repository. In December 2005 the *Commonwealth Radioactive Waste Management Act 2005* was enacted.

This Act provided for the nomination by an Aboriginal Land Council in the Northern Territory of Aboriginal land for consideration as a potential facility site. A nomination of a site on Ngapa Clan land at Muckaty Station (NT) was accepted by the then government in 2007.

In April 2012, the 2005 Act was repealed and replaced by the National Radioactive Waste Management Act 2012, which requires a facility to be sited on volunteered land. This allows for a Land Council in the Northern Territory to volunteer Aboriginal land on behalf of its Traditional Owners, and if the facility cannot be sited on Aboriginal land nominated by a Land Council in the Northern Territory, a nationwide volunteer process will take place.

Under the Act, the facility will manage radioactive waste generated by Australia's medical, industrial, agricultural and research use of nuclear materials. Currently, this waste is stored at more than 100 sites around Australia, many of which were not built for this purpose.

Australia has accumulated approximately 4000 cubic metres of LLW over fifty years of research, medical and industrial uses of radioactive materials, as well as significant amounts of mill tailings at mine sites and produces an additional 50 cubic metres each year. Over half the volume of this is stored in about 10 000 drums of lightly contaminated soil in about 80 isolated sites in South Australia.

Australia currently has approximately 535 m^3 of long-lived intermediate level radioactive waste (that is, radioactive waste requiring shielding for handling, transport and storage). Most is in storage at ANSTO, with some at selected isolated storage areas with LLW. Australia does not generate any used fuel that would be classified as HLW.

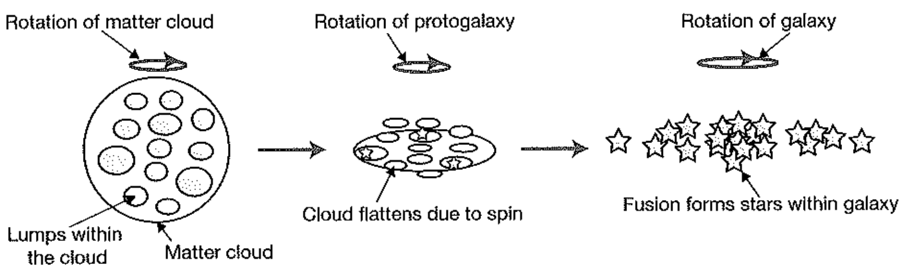
Used fuel from ANSTO's former reactors has been shipped overseas to the UK, France or the USA for reprocessing or storage since 1963. The agreements between these countries and Australia means that a small amount of the reprocessed fuel will be returned to Australia for geological disposal in 2015. The small volumes involved means that Australia can afford to wait and learn from overseas experience in the establishment of such facilities, storing the material in the interim period.

2.13.1.1 The Big Bang event.

2.13.1.2

	Completed sentences
(a)	The Big Bang produced both matter and antimatter with matter in excess.
(b)	The annihilation of some matter/antimatter left a 'lumpy' Universe.
(c)	This consisted of pockets of unevenly distributed matter with enormous empty spaces where annihilation had occurred.
(d)	Gravitational forces drew the matter in the 'lumps' together.
(e)	Individual matter clouds which we call protogalaxies which were also 'lumpy' formed.
(f)	The denser parts exerted greater gravitational attraction on the less dense parts.
(g)	More matter began to accumulate at the more dense parts.
(h)	These clouds of matter were thought to be rotating slowly as a result of the forces associated with the Big Bang event.
(i)	Small, but growing clumps of denser matter started to form within the larger matter cloud.
(j)	As more matter started to be drawn closer into the cloud, its speed of rotation increased.
(k)	This spinning motion caused the cloud to flatten out and form a disc.
(l)	As their size and density increased the 'lumps' of matter collapsed further, becoming compressed and hotter.
(m)	Eventually a core was hot enough to sustain nuclear fusion reactions.
(n)	Stars, composed of hydrogen were born, and galaxies formed.

2.13.1.3



2.13.1.4 Like all accepted theories in science, the Big Bang theory explains more observations that rival theories and makes more accurate predictions of observations that astronomers have been making in modern times.

2.13.2.1 An HR diagram is a plot of the luminosity of stars against their temperature, which also represents their colour ranges and spectral classes.

2.13.2.2 That the stars were arranged in obvious patterns and groups on the diagram.

2.13.2.3 Both colour and spectral class are determined by the temperature of the star.

2.13.2.4 Luminosity is a measure of the total energy output from a star. Brightness is a measure of the energy per square metre reaching Earth.

2.13.2.5 (a) The white dwarfs.

(b) Their energy source is due to the compression of the matter in them by forces of gravitational collapse.

2.13.2.6 (a) Red giants.

(b) Red giants are huge and have enormous surface areas so, despite the fact that each metre of surface area emits only a small amount of energy as seen by us, the total energy emitted, the luminosity, is very high.

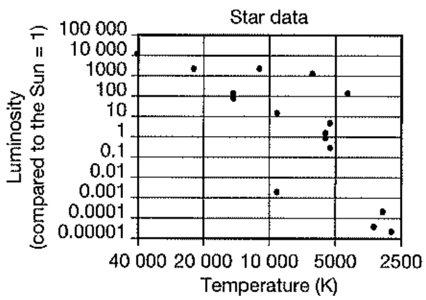
2.13.2.7 The position of a star on the Hertzsprung-Russell diagram indicates its surface temperature, colour, spectral class, approximate mass and luminosity.

2.13.2.8
A Blue supergiants
B Red supergiants
C Red giants
D Main sequence
E White dwarfs
F Red dwarfs

2.13.2.9 The distance of stars relative to Earth makes it difficult to measure brightness since a bright star that is more distant might appear to be as bright as a dimmer star closer to Earth.

2.13.2.10
(a) 10 000 times
(b) White dwarfs are less bright than the Sun.
(c) Red giant or red supergiant.

2.13.3.1



2.13.3.2 Classification of the stars. (*Note:* It is difficult to judge some of these from such a small number of plots because there is not enough data for the main patterns to appear, so you may have any of the answers below, the one in *italics* being the correct classification.) Remember, white dwarfs have low luminosity, but they are still very hot – ranging 6000 K up).

A	The Sun	Main sequence
B	Proxima Centauri	Main sequence
C	Alpha Centauri A	Main sequence
D	Alpha Centauri B	Main sequence
E	Sirius A	Main sequence
F	Sirius B	White dwarf
G	Canopus	Red giant
H	Betelgeuse	Red supergiant

I	Rigel	Blue supergiant
J	Beta Centauri	Red giant, *supergiant*
K	Wolf 349	Main sequence
L	Barnard's Star	Main sequence
M	Arcturus	Red giant
N	Procyon	Main sequence
O	Archenar	Red giant
P	Alderbaran	Red giant

2.13.3.3 Rigel

2.13.3.4 Wolf 359

2.13.3.5 Beta Centauri

2.13.3.6 Luminosity is the total energy emitted each second – this depends not only on its temperature, but also on its size (surface area). Beta Centauri is cooler, but larger than many hotter stars.

2.13.4.1
(a) They all fuse hydrogen to helium as their source of energy.
(b) Main sequence stars occupy the top to bottom diagonal of the HR diagram.
(c) Coolest at the bottom right, hottest at the top left.
(d) Most luminous at the top left, least luminous at the bottom right.
(e) The hottest and largest at the top right because they are fusing their hydrogen at a much faster rate than the cooler stars.

2.13.4.2 A

2.13.4.3 D

2.13.4.4 B

2.13.4.5 C

2.13.4.6 B

2.13.4.7 A

2.13.4.8 D

2.13.4.9 A

2.13.4.10 C

2.13.5.1

Star type	Energy source	Where found on HR diagram	Main features
Red giants	Helium fusion in the core. Shell burning of products of the inner shell in shell surrounding the core.	Middle to top right.	Large, high luminosity despite relative coolness.
Red supergiants	As above.	Top middle to right.	
Blue supergiants	As above.	Top middle to centre.	
White dwarfs	No energy source – heat is produced by gravitational collapse.	Bottom left to middle.	Small, dense, extremely hot.
Black dwarfs	No source (see main features).	None exist, so not found on HR diagram.	None detected to date – hypothesised to be the end stage of any evolutionary path. Universe not yet old enough for any to have formed.
Neutron stars	Huge pressure causes electron collapse into protons forming neutrons.	Not placed on HR diagram.	Radius no larger than 10 km, extremely dense.
Black hole	No energy source.	Not placed on HR diagram.	So dense with gravitational field so strong that no information can escape – hence the concept of 'black'.

2.13.5.2 (a) The Hertzsprung-Russell diagram is an *observational* diagram. It places stars on a grid according to their luminosity, absolute magnitude and colour/temperature. Since neither neutron stars nor black holes can be seen – they are only known to exist from non-visible measurements – they have no position on the diagram.

(b) All classification systems are purely arbitrary and will differ from viewpoint to viewpoint on the nature of the items being classified. In the case of the blue stars, they are all so big and so hot that many consider them all to be supergiants, rather than making an artificial distinction based on their size.

(c) There are two distinct ranges of sizes for red giants. Some are relatively less massive and smaller than others and these tend to fall into two distinctly different positions on the HR diagram. Hence the two divisions.

2.13.5.3 The fusion of lower atomic mass elements requires huge temperatures and pressure and releases enormous amount of energy, and hence this can occur in the cores and the surrounding layers of the supergiants. However, fusion of elements beyond iron is endothermic and requires an enormous input of energy. Stars, even the supergiants, do not have this amount of energy and so fusion beyond iron cannot occur.

2.13.5.4 Requiring temperatures well in excess of those produced in the supergiants elements beyond iron can form only during nova and supernova explosions of the larger supergiants at the end of their supergiant phase.

2.13.5.5 (a) A second generation star is one which has been formed from the core of a nova or supernova explosion where enough material remains from the explosion of a previous, supergiant star to form a much smaller, new main sequence star.

(b) Our Sun contains elements with atomic masses greater than iron which can only be formed as the result of a nova or supernova explosion of a supergiant star. The temperatures in the supergiants are not great enough to produce elements beyond iron on the periodic table.

2.13.5.6 (a) A black dwarf is the predicted end point in the evolutionary pathway of stars that result in white dwarf stars.

(b) The predicted life cycle of a star which will result in the formation of an eventual black dwarf is longer than the current age of the Universe. There has not been enough time for them to form (yet).

(c) It is a logical hypothesis that white dwarf stars will eventually cool down and eventually form a cold remnant – a black dwarf.

(d) They will be cold (no emitted electromagnetic radiation), dark and small and therefore reflecting very small amounts of received radiation.

2.13.5.7 Black holes are detected because they 'pull in' light from around them (within their Schwarzschild radius) and therefore form a space within the Universe which gives out no radiation. This dark area of space indicates the existence of a black hole at its centre.

2.13.6.1 (a) They are all main sequence stars.

(b) P

(c) T

(d) P

(e) T

(f) P

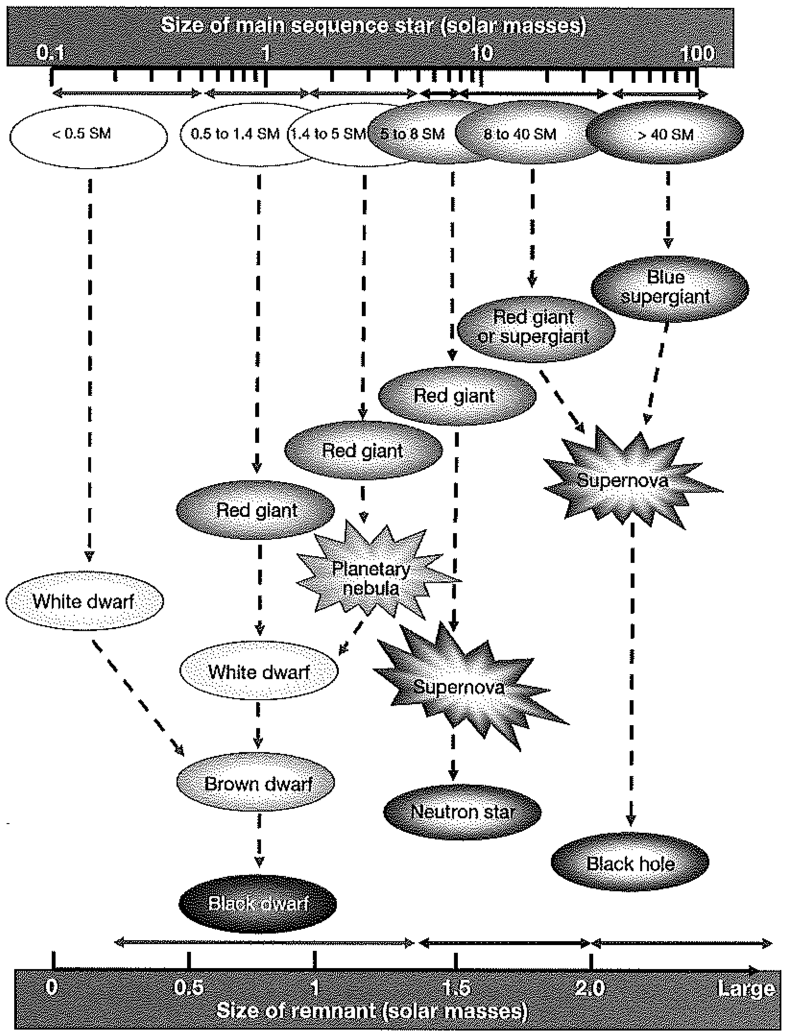
(g) R, S and T

(h) R, S and T

(i) T

2.13.6.2 A represents a gas cloud or nebula, or perhaps a protostar – the birth of a star from the nebula material of space or form that formed by a previous nova or supernova. When the star itself forms it begins as a main sequence star (B), fusing hydrogen to helium. When 10% of the core hydrogen has been fused, the energy produced is not enough to prevent gravitational collapse and the star shrinks, greatly increasing its core temperature to the point that helium fusion starts. The star expands outwards very quickly forming a red giant (C). At the end of its life cycle, the red giant expands to form a diffuse planetary nebula (D), the core of which cools to become a white dwarf which has no energy source apart from gravitational collapse (E).

2.13.6.3



2.13.6.4 These stars will cease hydrogen fusion when their central 10% has been fused. However, because of their larger mass, the heating effects associated with their contraction are enough to initiate helium fusion and so a red giant will form. As helium fuses to carbon the star expands and becomes hotter. When its central helium is depleted it contracts, becomes hotter and may, if the temperature of the core rises sufficiently, start to fuse carbon and begin a second red giant stage before gravitational collapse into a white dwarf when these reactions cease. Again, the white dwarf will eventually become a brown, then a black dwarf.

2.13.6.5 These stars are very luminous and lose mass very rapidly because their radiation pressure outwards is so high that it tends to strip off their own gaseous envelopes before they can expand to red supergiants. Instead, they maintain an extremely high temperature and their blue-white colour as blue supergiants. A massive core does not build up as convection carries and disperses higher mass elements throughout the star. If no iron core is available to contract and cause a supernova the star will simply contract inwards and because of its mass, form a black hole. If an iron core is present, gravitational forces will collapse it until its internal forces become too great and it will explode rapidly outwards resulting in what we call a supernova explosion.

2.13.6.6 D

2.13.6.7 C

2.13.6.8 A

2.13.6.9 B

3.1.1.1 (a) The charged particles of matter (quarks), although, if you say deficiency or excess of electrons compared to protons of matter, this is a good answer also.

(b) By losing electrons.

(c) By gaining electrons.

(d) Electrons are in orbitals around nuclei and are easily detached and made to flow. Protons are in the nuclei of atoms and require nuclear magnitude energies to release them. It is only electron movement that is responsible for electrical phenomena.

(e) The coulomb (C).

(f) One coulomb is equal to the charge on 6.25×10^{18} electrons.

(g) +1.0 mC

3.1.1.2 (a) One ampere of current is equal to the flow of one coulomb of electrons past a point in the circuit each second.

(b) The ampere (A).

(c) Electron flow is a physical fact. Conventional current is a hypothetical movement of positive charge in the opposite direction to electron flow.

(d) All the directions for electric field and current and electric potential were defined before the discovery of electrons on the assumption that current was a flow of positive charge. When this turned out to be in error, the 'incorrect' conventions were retained, and a new concept of 'conventional' current (that is – current as it has always been assumed to be) was introduced.

3.1.1.3 (a) 1.2×10^{20} electrons

(b) 15 s

3.1.1.4 (a) 1.92 A

(b) 1.5 mA

(c) 7.8×10^{17} electrons

(d) 2.19×10^{10} electrons

3.1.1.5 (a) Same

(b) Same

(c) Same

(d) Same

3.2.1.1 (a) Energy is required to separate positive and negative charges.

(b) When charges are separated work is done.

(c) Whenever work is done to separate objects a potential energy difference is set up between them.

(d) This is how an electrical potential energy difference is produced.

(e) Potential difference is defined as the work done per unit positive test charge in moving the charge from one point in an electric field to another point in the field.

(f) The direction of the potential difference at P is given by the direction of the force which would act on a positive test charge placed at P.

(g) If a charge is positive, then work needs to be done on it to move it against the field.

(h) If a charge is negative, then work will need be done on it to move it in the direction of the field.

(i) If a charge is positive, then work is done on it by the field to move it in the direction of the field.

(j) If a charge is negative, then work is done on it by the field to move it in the opposite direction of the field.

(k) In general, electrical potential difference is directed away from positive charge and towards negative charge.

(l) The direction of electrical potential difference is the direction in which a positive test charge would move if it was placed in the potential field.

(m) The potential energy difference set up when charges are separated from each other can be used as the driving force to produce an electric current in a circuit.

3.2.1.2 D

3.2.1.3 C

3.2.1.4 A

3.2.1.5 D

3.2.1.6 B

3.2.1.7 C

3.2.1.8 C

3.2.1.9 A

3.2.1.10 D

3.2.1.11 B

3.2.1.12 C

3.2.1.13 D

3.2.1.14 A

3.2.1.15 A

3.2.1.16 B

3.2.1.17 B

3.3.1.1
(a) The electrical potential difference between two points in a conductor carrying a current is equal to the product of the current flowing and the resistance of the conductor between the two points.
(b) $\Delta V = RI$
(c) Ohm ampere (Ω A)
(d) One ohm ampere = 1 volt

3.3.1.2
(a) 12 V
(b) 2 A
(c) 4.5 Ω

3.3.1.3
(a) 6 V
(b) 12 V
(c) 27 V
(d) 33 V
(e) 0 V (no resistance between P and Q)

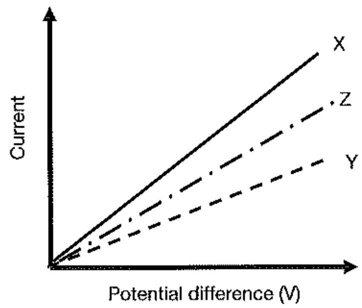
3.3.1.4
(a)

Run	V1	V2	V3	V4	A1	A2	A3	A4	A5
1	3.3	8.5	13.5	25.0	1.8	1.6	1.8	1.7	1.7
2	3.4	8.2	13.4	25.3	1.6	1.7	1.5	1.7	1.8
3	3.3	8.3	13.3	24.7	1.7	1.7	1.7	1.6	1.5
Average	3.33	8.33	13.4	25.0	1.7	1.67	1.67	1.67	1.67

(b) The current flowing through each resistor in a series only circuit is the circuit current, $I_{circuit} = I_1 = I_2 = \ldots$
(c) The sum of the potential difference of the resistors connected in series in a series only circuit is equal to the potential difference on the power source, $V_{source} = V_1 + V_2 + \ldots$
(d) The total resistance in a series only circuit equals the sum of all the resistors connected in series, $R_{total} = R_1 + R_2 + \ldots$
(e) The potential difference across resistors connected in series is directly proportional to the value of the resistor, $V \infty R$
(f) A1 = 0 V
A2 = 3.33 V
A3 = 11.67 V
A4 = 0 V
A5 = 25 V

3.3.2.1
(a) Electrical resistance is the loss of energy as electrons travel through a conductor due to collisions between them and lattice particles in the conductor.
(b) Resistors are used to change electricity into alternate, useful forms of energy and to reduce current flow through a circuit in order to protect circuit components or devices connected externally to the circuit.
(c) Electrical resistance is due mainly to inelastic collisions between current electrons and conductor lattice particles.
(d) Ohm, Ω
(e) One ohm is the resistance of a conductor if an applied voltage of 1 volt causes a current of 1 ampere to flow through it.
(f) Resistance = voltage across the resistor ÷ current flowing through it $\left(R = \frac{V}{I}\right)$.

3.3.2.2 (a)
(b)
(d)



(c) Y, being twice as long, will have twice the resistance of X and therefore only half as much current will flow for each applied voltage.

(d) Note that your graph for (d) could be as shown or even lower than Y. The important concept is that it lies below X.

(e) At a higher temperature the conductor will have a higher resistance so applied voltages will produce less current than for X.

(f) Need power source, ammeter, voltmeter, circuit wires and either various lengths of test wire or a long length and two circuit wires with alligator clips that can be connected to the wire at desired lengths. Apply different voltages to each length and record currents flowing. Repeat measurements to determine reliability. Controls include the type of wire used, its temperature, its diameter, the meters used.

3.3.2.3 (a) Increasing the temperature of a conductor gives its particles increased kinetic energy. When current electrons collide with lattice particles the collision will be more violent and more energy will be lost by the electrons – the resistance is increased.

(b) The longer the conductor, the more lattice particles current electrons have to negotiate and the more collisions they will have. Resistance is directly proportional to the length.

(c) Some conductor atoms release their valence electrons more readily (less energy) than others. This is reflected in their inherent resistance.

(d) The larger the cross-sectional area of a conductor, the more room there is for current electrons to move through the conductor. However, there are also more lattice particles and this works to increase resistance. Of greater effect is the increased circumference which means surface electrons have much more room to flow along the surface of the conductor.

3.3.2.4

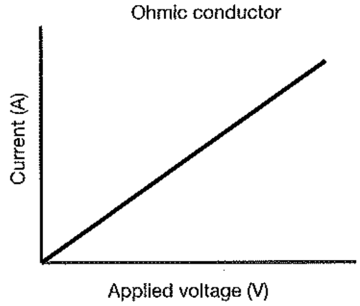
Electrical device	Input energy	Useful output energy
Toaster	Electrical	Heat energy
Fan	Electrical	Kinetic energy
Car hoist	Electrical	Gravitational potential energy
Car alarm	Electrical	Sound energy
Light globe	Electrical	Light energy

3.3.2.5 A 10 gauge wire is wider than 18 gauge wire and thus has less resistance. The lesser resistance of 10 gauge wire means that it can allow a larger current. Thus, 10 gauge wire is used in circuits which are protected by 15 ampere fuses and circuit breakers. On the other hand, the thinner 18 gauge wire can support less current owing to its larger resistance; it is used in circuits which are protected by 10 ampere fuses and circuit breakers.

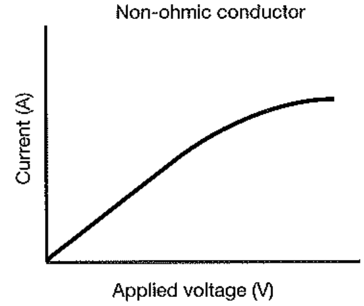
3.3.3.1 (a) An ohmic conductor obeys Ohm's law, i.e. the current flowing through it is directly proportional to the applied voltage. A non-ohmic conductor does not obey Ohm's law.

(b)

Ohmic conductor



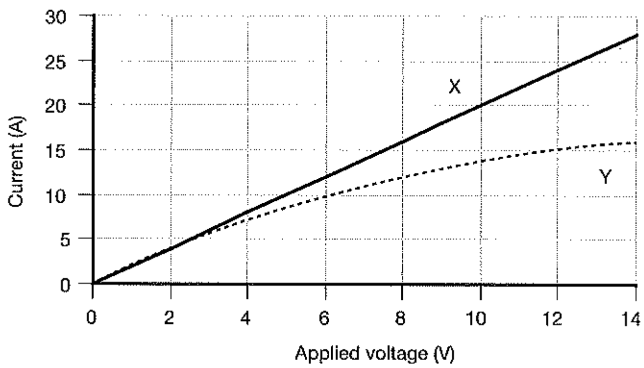
Non-ohmic conductor



(c) Resistance remains constant.

(d) Resistance increases due to the heating effects of the current.

3.3.3.2 (a)



(b) For conductor X, the current is directly proportional to the applied voltage.

For Y, as the applied voltage increases, the current increases but by lesser amounts.

(c) X is an ohmic conductor, Y is a non-ohmic conductor.

(d) 0.5 Ω

(e) By using the inverse gradient of the line of best fit because individual values have experimental error associated with them.

(f) 0.7 Ω

(g) Using the gradient of lines of best fits in graphs improves reliability because the outliers are more readily identified (the value for current for X at 8 V) and can be removed from consideration.

(h) By repeating all measurements at least three times and using average values for all currents produced.

3.4.1.1 (a) Power is a measure of the rate at which work is done or energy is transformed.

(b) The electrical potential difference between two points in a conductor carrying a current is equal to the power dissipated in a resistor per unit current.

(c) $\Delta V = \frac{V}{I}$ or on rearranging $P = \Delta VI$

(d) Watts per ampere (W A^{-1})

(e) One watt per ampere = 1 volt

(f) Watts, J s^{-1}, V A (volt ampere)

3.4.1.2 (a) The power of an appliance is equal to the product of the voltage across it and the current flowing through it, i.e. $P = VI$

(b) $P = VI$, $P = I^2R$, $P = \frac{V^2}{R}$

(c) 3.6×10^6 J

(d) The cost per joule would be so low that it would not be sensible to charge consumers in this way. The unit is too small to be useful for this purpose.

3.4.1.3 (a) 5.83×10^{-6} cents

(b) Electrical power is a measure of the rate at which electrical energy is used.

3.4.1.4 (a) 3.33 A

(b) 72 Ω

(c) 43.75 hours

3.4.1.5 (a) 54 W

(b) 810 J

(c) 38.88 kWh

3.4.1.6 (a) 3.33 A

(b) 120 000 J

(c) $2.92

3.4.1.7 (a) 0.167 A

(b) 1440 Ω

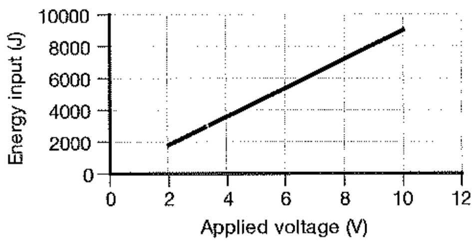
(c) 30 V

(d) Calculations show the globe is designed to carry 1.33 A for maximum performance. A 240 V circuit would cause 10.67 A to flow through it. This would probably melt the filament and cause the bulb to blow.

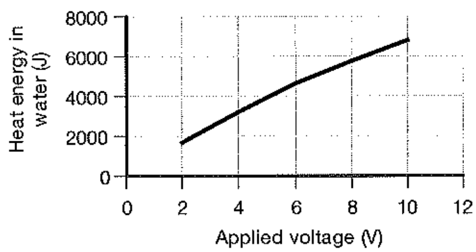
3.4.1.8
(a) 504 W
(b) It would be operating at only about 20% efficiency. The water in it would take 5 times as long to boil.
(c) 10 A
(d) 4.58 A

3.4.2.1
(a)

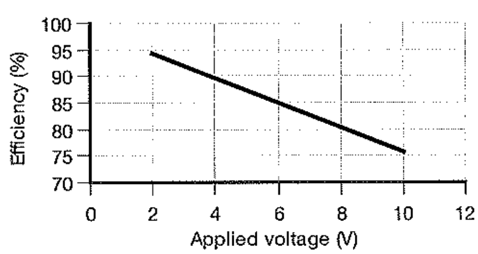
Mass water heated (g)	Applied voltage (V)	Current flowing (A)	Time current flowed (min)	Temperature rise (°C)	Input electrical power (W) $P_i = VI$	Input electrical energy (J) $E = Pt$	Heat energy absorbed by water $\Delta H = mc\Delta T$ (J)	Efficiency of coil (%)
100	2	1.5	10	4.09	3.0	1800	1709.6	94.98
100	4	1.5	10	7.66	6.0	3600	3201.9	88.9
100	6	1.5	10	11.01	9.0	5400	4602.2	85.2
100	8	1.5	10	13.86	12.0	7200	5793.5	80.5
100	10	1.5	10	16.32	15.0	9000	6821.8	75.8

(b)


(c) Energy input is directly proportional to the applied voltage.

(d)


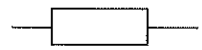
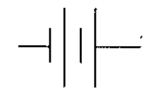
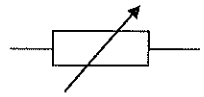
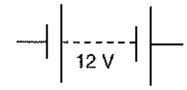
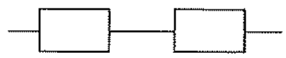
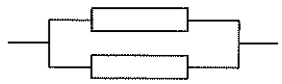
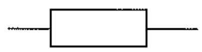
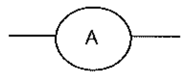
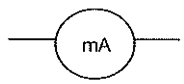
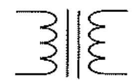
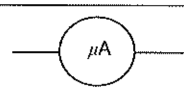
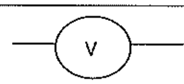
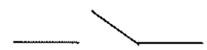
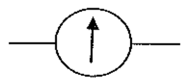
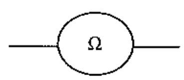
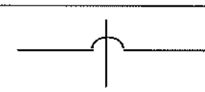
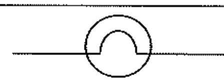
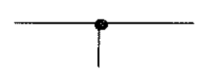
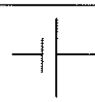
(e) Because the graph is not a straight line, it is not possible to state the relationship between the variables, the best we can do is: As the applied voltage increases, the energy produced by the coil and absorbed by the water increases almost proportionally.

(f)


(g) Efficiency of the coil is inversely proportional to the applied voltage.

(h) The applied potential difference would need to be kept constant and the variable resistor adjusted to vary the current flowing in the circuit. The same measurements and calculations would be done, but graphs would be against current rather than applied voltage.

3.4.3.1

Component	Diagram of component	Component	Diagram of component
Resistor		3.0 V battery	
Variable resistor		12 V battery	
Two resistor in series		Variable DC power supply	
Two resistors in parallel		DC terminals (power supply)	
Fuse		AC terminals (power supply)	
Ammeter		Coil	
Milliammeter		Transformer	
Microammeter		Closed switch	
Voltmeter		Open switch	
Galvanometer		Circuit wire	
Ohmmeter		Two wires crossing but not connected	
Globe		Two wires crossing and connected	
1.5 V DC cell		Earth connection	

3.4.4.1

(a) An electrical circuit is simply the device we use to change electrical energy into other forms of energy.

(b) So that we use the correct amount of electrical energy in a circuit, designers need to make measurements so that the circuit is designed accurately.

(c) In this way, appliances connected into the circuit will not be damaged by excess current flow.

(d) It also ensures that the appliances have the current required for them to operate efficiently.

(e) At home some low voltage appliances need only a small current.

(f) These are therefore connected via a transformer to the power point in your home.

(g) If they were connected directly to the power supply, the current flowing through them would be too large and they would burn out very quickly.

(h) The instrument used to measure current is the ammeter.

(i) Ammeters measure current flowing in amperes.

(j) Other current measuring instruments include milliammeters and microammeters.

(k) These are always connected into circuits in series.

(l) In this way they measure the current flowing through that part of the circuit by measuring the current flowing through themselves.

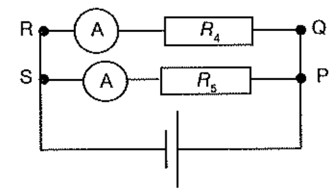
(m) So that they don't affect the resistance of the circuit, it is essential that ammeters have negligible resistance.

3.4.4.2 A Only one ammeter needed and it can be placed anywhere in the circuit since all resistors are in series.

B Two ammeters needed.

For R_4, connected anywhere along the wires PQRS.

For R_5, anywhere along the wire SP.

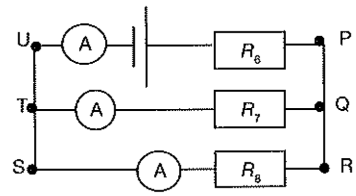


C Three ammeters needed for the easiest result.

For R_6, anywhere along wire TUPQ.

For R_7, anywhere along TU.

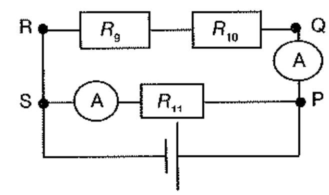
For R_8, anywhere along QRST.



D Two ammeters needed.

For R_9 and R_{10}, connected anywhere along the wires PQRS.

For R_{11}, anywhere along the wire SP.

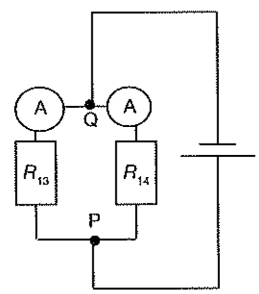


E Two ammeters needed.

For R_{13}, connected anywhere along the wires PQ and R_{13}.

For R_{14}, anywhere along the wire PQ and R_{14}.

F Only one ammeter needed and it can be placed anywhere in the circuit since all resistors are in series.



3.4.4.3

Amps (A)	Milliamps (mA)	Microamps (μA)
0.1	100	100 000 (1.0×10^5)
0.75	750	750 000 (7.5×10^5)
2.6	2600	2 600 000 (2.6×10^6)
0.2	200	200 000 (2.0×10^5)
1.5	1500	1 500 000 (1.5×10^6)
10.6	10 600	10 600 000 (1.06×10^7)
0.001	1.0	1000
0.04	40	4×10^4
2.0	2000	2×10^6

3.4.5.1 Answers will vary, but these would be good answers:

(a) A battery is a source of electrical energy.

(b) A battery is often described as an electron 'pump'.

(c) A battery is a source of electrons.

(d) Electrons flow through a circuit from a battery because of forces of electrostatic repulsion.

(e) The ability of a battery to 'pump' electrons through a circuit is called its voltage.

(f) Voltage is a measure of how much energy a battery can supply to a circuit.

(g) The instrument used to measure voltage is the voltmeter.

(h) Voltmeters measure electrical potential difference across two points in a circuit.

(i) To do this they must be connected across those two points.

(j) Voltage can also be expressed as joules per coulomb.

(k) Anything connected across two points in a circuit is said to be connected in parallel.

(l) The symbol for voltage is V.

(m) Voltage is measured in volts.

(n) The electrical potential difference between two points is equal to the work done per unit charge moving from one point to the other.

(o) In order to minimise current through them, voltmeters have very high internal resistance.

(p) This is required so that they do not affect the current flowing through the resistor they are connected across.

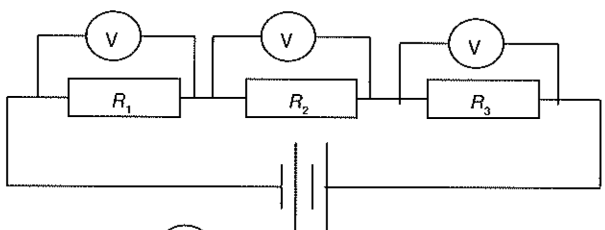
3.4.5.2 (a) Voltmeters are connected into circuits in parallel.

(b) They are connected in parallel so that they measure potential difference across or between two points.

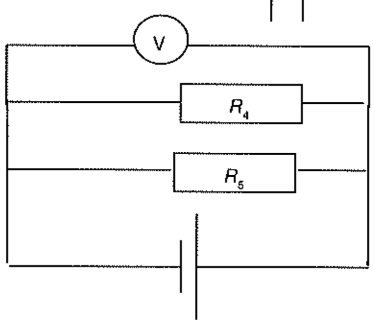
(c) If they were connected in series, they could not measure potential difference across two positions in the circuit, and because of their very high internal resistance, no current, (or negligible current) would flow in the circuit.

3.4.5.3

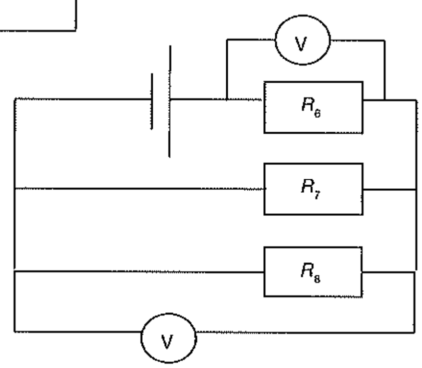
A



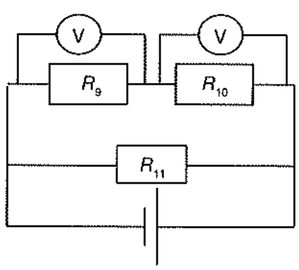
B



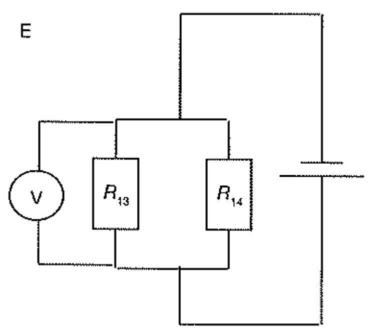
C



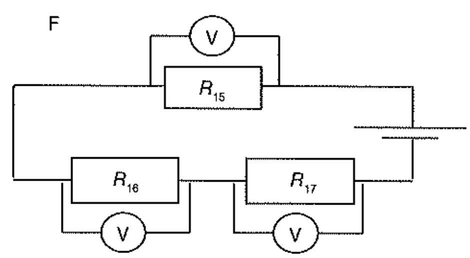
D



E



F



3.4.5.4

A	Circuit contains a 3 V battery and three resistors connected in series with each other.
B	Circuit contains a 1.5 V cell and two resistors connected in parallel with each other.
C	Circuit contains a 1.5 V cell and three resistors, two connected in parallel with each other and these connected in series with the third.
D	Circuit contains a 1.5 V cell and three resistors, two connected in series and these connected in parallel with the third.
E	Circuit contains a 1.5 V cell and three resistors connected in series with each other.
F	Circuit contains a 1.5 V cell and two resistors connected in parallel with each other.

3.5.1.1 A S
B S
C C
D P
E P
F S
G S
H P
I S
J C
K P
L C
M S
N S
O S
P S
Q S
R S
S P
T P

3.5.1.2

	Description of circuit
A	Circuit contains 1.5 V cell and two resistors connected in series with each other.
B	Circuit contains a 1.5 V cell and three resistors, two connected in series and these connected in parallel with the third.
C	Circuit contains a 1.5 V cell and three resistors, two connected in parallel with each other and these connected in series with the third.
D	Circuit contains 1.5 V cell and two resistors connected in series with each other.
E	Circuit contains 1.5 V cell and two resistors connected in parallel with each other.
F	Circuit contains a 1.5 V cell and two resistors connected in parallel with each other.
G	Circuit contains a 1.5 V cell and two resistors connected in parallel with each other.

3.5.2.1 (a), (b)

Point	Potential
A	10
B	16
C	6
D	0

Point	Potential
E	6
F	18
G	0
H	18

Point	Potential
I	9
J	7.5
K	0
L	4.5

Point	Potential
M	0
N	8
O	20
P	36
Q	0
R	36

Point	Potential
S	0
T	15
U	25
V	45
W	50
X	0
Y	0
Z	50
AA	50

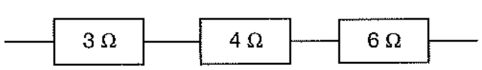
BC	6
FG	18
GH	18
JK	7.5

IL	4.5
NQ	12
PQ	16
PM	36

YZ	50
UX	25
VS	45
WT	35

SZ	50
UY	25

3.5.3.1 (a)



(b) 13 Ω

3.5.3.2 (a) 17 Ω

(b) $I_{5\,\Omega} : I_{10\,\Omega} : I_{2\,\Omega} = 1 : 1 : 1$

(c) $V_{5\,\Omega} : V_{10\,\Omega} : V_{2\,\Omega} = 5 : 10 : 2$

3.5.3.3 2.5 A

3.5.3.4 15 Ω

3.5.3.5 18 V

3.5.3.6

Resistor	0.5 Ω	2.0 Ω	2.5 Ω	3.0 Ω	4.0 Ω	5.0 Ω
Current through each resistor (A)	4	4	4	4	4	4
Voltage across each resistor (V)	2	8	10	12	16	20

3.5.3.7

	Circuit A			Circuit B		Circuit C		
Total circuit resistance (Ω)	12			18		13		
Circuit current (A)	3			2		2		
Current flowing through each resistor (A)	2 Ω	6 Ω	R Ω	6 Ω	12 Ω	3 Ω	4 Ω	6 Ω
	3	3	3	2	2	2	2	2
Value unknown resistor (Ω)			4					
Potential difference across each resistor (V)	6	18	12	12	24	6	8	12
Potential of power source (V)	36			36		26		

3.5.3.8

A Total resistance = 4 Ω
Circuit current = 3 A
Current through 4 Ω = 3 A
Potential across 4 Ω = 12 V

B Total resistance = 6 Ω
Circuit current = 3 A
Current through 2 Ω = 3 A
Current through 4 Ω = 3 A
Potential across 2 Ω = 6 V
Potential across 4 Ω = 12 V

C Total resistance = 5 Ω
Circuit current = 4 A
Current through 5 Ω = 4 A
Potential across 5 Ω = 20 V
Potential of source = 20 V

D Total resistance = 11 Ω
Circuit current = 3 A
Current through 2 Ω = 3 A
Current through 3 Ω = 3 A
Current through 6 Ω = 3 A
Potential across 2 Ω = 6 V
Potential across 3 Ω = 9 V
Potential across 6 Ω = 18 V
Potential of source = 33 V

E Total resistance = 6 Ω
Circuit current = 7 A
Current through 4 Ω = 7 A
Current through *R* Ω = 7 A
Potential across 4 Ω = 28 V
Potential across *R* Ω = 14 V
Value of resistor *R* = 2 Ω

F	Total resistance	= 11 Ω
	Circuit current	= 2 A
	Current through 4 Ω	= 2 A
	Current through 5 Ω	= 2 A
	Current through *R* Ω	= 2 A
	Potential across 4 Ω	= 8 V
	Potential across 5 Ω	= 10 V
	Potential across *R* Ω	= 4 V
	Potential of source	= 22 V
	Value of resistor *R*	= 2 Ω

3.5.4.1

Run	V1	V2	V3	V4	A1	A2	A3	A4	A5
1	3.4	8.8	12.3	25.0	1.7	1.8	1.9	1.8	1.8
2	3.4	8.6	12.2	24.0	1.8	1.8	1.7	1.7	1.8
3	3.6	8.8	12.4	24.4	1.7	2.7	1.7	1.8	1.8
4	3.5	8.7	12.5	24.5	1.7	1.8	1.6	1.6	1.6
Average	3.5	8.7	12.4	24.5	1.7	1.8	1.7	1.7	1.7

3.5.4.2 Totals divided by number of readings except for the reading on A2 where the 2.7 is an outlier and has not been included.

3.5.4.3 (a) The circuit current is the same as the current flowing through each series resistor.

(b) Potential difference of the circuit is equal to the sum of the potential differences across the series resistors.

(c) Total resistance is equal to the sum of the resistor in series.

(d) Potential difference across each resistor in series is directly proportional to the values of each individual resistor.

(e) Voltage at: A1 = 0

A2 = 3.4 V

A3 = 12.1 V

A4 = 0

A5 = 24.5 V

3.5.4.4 For resistors connected in series:

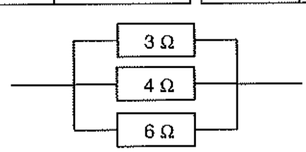
$$I_{circuit} = I_{R_1} = I_{R_2} = I_{R_3} = \ldots\ldots$$
$$V_{circuit} = V_{R_1} + V_{R_2} + V_{R_3} + \ldots\ldots$$
$$R_{circuit} = R_1 + R_2 + R_3 + \ldots\ldots$$

3.5.4.5 For each resistor, the ratio of the potential difference across the resistor to the value of the resistor is equal to the current flowing through it, so the statement is correct.

3.5.5.1

Point	Potential	Point	Potential	Point	Potential	Point	Potential	Point	Potential
A	36	I	15	Q	18	Y	12	FF	0
B	0	J	0	R	0	Z	0	GG	18
C	36	K	15	S	12	AA	12	HH	0
D	0	L	0	T	0	BB	0	II	18
E	0	M	18	U	12	CC	12	JJ	0
F	9	N	0	V	0	DD	0	KK	18
G	9	O	18	W	12	EE	18	LL	0
H	9	P	0	X	0				

3.5.6.1 (a)



(b) 1.33 Ω

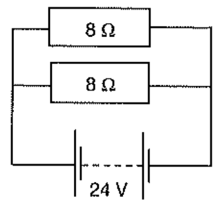
3.5.6.2 (a) 1 : 1 : 1

(b) 1 : 1 : 1

3.5.6.3 (a) 6 : 3 : 2

(b) 1 : 1 : 1

3.5.6.4 Several possible answers here. The diagram here is an example:



3.5.6.5

A	Total resistance	= 4 Ω
	Circuit current	= 10 A
	Current through 5 Ω	= 8 A
	Current through 20 Ω	= 2 A
	Potential across 5 Ω	= 40 V
	Potential across 20 Ω	= 40 V
B	Total resistance	= 4 Ω
	Circuit current	= 15 A
	Current through 6 Ω	= 10 A
	Current through 12 Ω	= 5 A
	Potential across 6 Ω	= 60 V
	Potential across 12 Ω	= 60 V
C	Total resistance	= 2 Ω
	Circuit current	= 12.5 A
	Current through 2.5 Ω	= 10 A
	Current through 10 Ω	= 2.5 A
	Potential across 2.5 Ω	= 25 V
	Potential across 10 Ω	= 25 V
D	Total resistance	= 2 Ω
	Circuit current	= 6 A
	Current through 6 Ω	= 2 A
	Current through *R* Ω	= 4 A
	Potential across 6 Ω	= 12 V
	Potential across *R* Ω	= 12 V
	Value of resistor *R*	= 3 Ω
E	Total resistance	= 2.67 Ω
	Circuit current	= 6 A
	Current through *R* Ω	= 4 A
	Current through 8 Ω	= 2 A
	Potential across *R* Ω	= 16 V
	Potential across 8 Ω	= 16 V
	Value of resistor *R*	= 4 Ω
F	Total resistance	= 5 Ω
	Circuit current	= 18 A
	Current through 6 Ω	= 15 A
	Current through 30 Ω	= 3 A
	Potential across 6 Ω	= 90 V
	Potential across 30 Ω	= 90 V
G	Total resistance	= 1.63 Ω
	Circuit current	= 11 A
	Current through 9 Ω	= 2 A
	Current through 6 Ω	= 3 A
	Current through 3 Ω	= 6 A
	Potential across 9 Ω	= 18 V
	Potential across 6 Ω	= 18 V
	Potential across 3 Ω	= 18 V
	Potential of source	= 18 V

H Total resistance = 1.25 Ω
Circuit current = 24 A
Current through R Ω = 15 A
Current through 5 Ω = 6 A
Current through 10 Ω = 3 A
Potential across R Ω = 30 V
Potential across 5 Ω = 30 V
Potential across 10 Ω = 30 V
Value of resistor R = 2 Ω

I Total resistance = 1.85 Ω
Circuit current = 13 A
Current through 4 Ω = 6 A
Current through 8 Ω = 3 A
Current through R Ω = 4 A
Potential across 4 Ω = 24 V
Potential across 8 Ω = 24 V
Potential across R Ω = 24 V
Value of resistor R = 6 Ω

3.5.6.6 (a)

Circuit	A	B	C	D	E	F	G	H	I	J
Total resistance (Ω)	1.5	2.0	8.0	75	4.0	3.0	1.5	1.5	1.25	3.0

(b) For two (only) identical resistors in parallel, the total resistance of the parallel combination is half the value of one resistor.

(c) For three identical resistors it will be one third the value of one. (*Note:* The pattern continues: for four identical resistors it will be one quarter, for five it will be one fifth the value of one etc.)

(d)

	Resistor	A	B	C	D	E	F	G	H	I	J
Current in resistor (A)	2 Ω									10	
	3 Ω	5							8		
	4 Ω		5				9		6		
	4.5 Ω							4			
	5 Ω					8				4	
	9 Ω										2
	10 Ω									2	
	12 Ω						3		2		
	16 Ω			1							
	20 Ω					2					
	150 Ω				2						

3.5.7.1

Average	12.0	12.0	12.0	3.0	1.0	2.0

3.5.7.2 (a) The circuit current is equal to the sum of the currents flowing through the parallel resistors.

(b) The potential difference across resistors connected in parallel is equal to the potential of the power source.

(c) The total resistance of a parallel circuit is equal to the inverse of the inverse sum of the values of the individual resistors.

(d) The potential across resistors connected in parallel is independent of the value of the resistor.

(e) Potential at A1 = 0
Potential at A2 = 0
Potential at A3 = 12 V

3.5.7.3 For resistors connected in parallel:

$$I_{circuit} = I_{R_1} = I_{R_2} = I_{R_3} = \ldots\ldots$$

$$V_{source} = V_{R_1} + V_{R_2} + V_{R_3} + \ldots\ldots$$

3.5.7.4 $\frac{1}{R_T} = \frac{1}{R_1} + \frac{1}{R_2}$ or $R_T = \frac{R_1 \times R_2}{R_1 + R_2}$

$\frac{1}{R_T} = \frac{1}{6} + \frac{1}{12}$ or $R_T = \frac{6 \times 12}{6 + 12}$

So, $R_T = 4\ \Omega$ $= 4\ \Omega$

Which is correct for this circuit. $\left(R = \frac{V}{I} = \frac{12}{3}\right)$.

3.5.7.5 Reliability is checked by taking multiple readings and increased if the average (excluding outliers) is used in all calculations.

3.5.8.1

A
Resistance = 4 Ω (6 and 3 in parallel, and this combination in series with the 2)
Circuit current = 12 A
Current through 6 Ω = 4 A
Current through 2 Ω = 12 A
Current through 3 Ω = 8 A
Potential across 6 Ω = 24 V
Potential across 2 Ω = 24 V
Potential across 3 Ω = 24 V
Potential of source = 48 V

B
Total resistance = 5 Ω (4_1 and 4_2 in parallel, and this combination in series with the 3)
Circuit current = 8 A
Current through 3 Ω = 8 A
Current through 4 Ω_1 = 4 A
Current through 4 Ω_2 = 4 A
Potential across 3 Ω = 24 V
Potential across 4 Ω_1 = 16 V
Potential across 4 Ω_2 = 16 V

C
Total resistance = 10 Ω (12 and 6_2 in parallel, and this in series with the 6_1)
Circuit current = 9 A
Current through 6 Ω_1 = 9 A
Current through 12 Ω = 3 A
Current through 6 Ω_2 = 6 A
Potential across 6 Ω_1 = 54 V
Potential across 12 Ω = 36 V
Potential across 6 Ω_2 = 36 V
Potential of source = 90 V

D
Total resistance = 8 Ω (6_1 and 6_2 in parallel, and this in series with the 5)
Circuit current = 6 A
Current through 6 Ω_1 = 3 A
Current through 6 Ω_2 = 3 A
Current through 5 Ω = 6 A
Potential across 6 Ω_1 = 18 V
Potential across 6 Ω_2 = 18 V
Potential across 5 Ω = 30 V
Potential of source = 48 V

E
Total resistance = 5 Ω (4 and 12 in parallel, and this in series with the 2)
Circuit current = 12 A
Current through 2 Ω = 12 A
Current through 4 Ω = 9 A
Current through 12Ω = 3 A
Potential across 2 Ω = 24 V
Potential across 4 Ω = 36 V
Potential across 12 Ω = 36 V
Potential of source = 60 V

F Total resistance = 14 Ω (4_1 and 4_2 in parallel, and this in series with the 12)
Circuit current = 6 A
Current through 12 Ω = 6 A
Current through 4 Ω_1 = 3 A
Current through 4 Ω_2 = 3 A
Potential across 12 Ω = 72 V
Potential across 4 Ω_1 = 12 V
Potential across 4 Ω_2 = 12 V
Potential of source = 84 V

G Total resistance = 6 Ω (3 and 6 in parallel, and this combination in series with the 4)
Circuit current = 2 A
Current through 3 Ω = 1.33 A
Current through 6 Ω = 0.67 A
Current through 4 Ω = 2 A
Potential across 3 Ω = 4 V
Potential across 6 Ω = 4 V
Potential across 4 Ω = 8 V
Potential of source = 12 V

H Total resistance = 4.67 Ω (4 and 8 in parallel, and this in series with the *R*)
Circuit current = 6 A
Current through 4 Ω = 4 A
Current through 8 Ω = 2 A
Current through *R* Ω = 6 A
Potential across 4 Ω = 16 V
Potential across 8 Ω = 16 V
Potential across *R* Ω = 12 V
Potential of source = 28 V
Value of resistor *R* = 2 Ω

I Total resistance = 6 Ω (6_1 and 6_2 in parallel, and this in series with the 3)
Circuit current = 3 A
Current through 6 Ω_1 = 1.5 A
Current through 6 Ω_2 = 1.5 A
Current through 3 Ω = 3 A
Potential across 6 Ω_1 = 9 V
Potential across 6 Ω_2 = 9 V
Potential across 3 Ω = 9 V
Potential of source = 18 V
Reading on meter A = 1.5 A

J Total resistance = 4 Ω (8 and 4 in series, and this in parallel with the 6)
Circuit current = 9 A
Current through 8 Ω = 3 A
Current through 4 Ω = 3 A
Current through 6 Ω = 6 A
Potential across 8 Ω = 24 V
Potential across 4 Ω = 12 V
Potential across 6 Ω = 36 V
Potential of source = 36 V
Reading on meter A = 3 A

3.6.1.1 (a) Transformers are used to increase or decrease the voltage supplied to homes so that the correct voltage is used with particular electrical devices.

(b) Step-up transformers increase the supply voltage to a higher value.

Step-down transformers decrease the supply voltage to a lower voltage.

(c) Step-up transformers have more coils in their output coil than in their input coil.

Step-down transformers have fewer coils in their output coil than in their input coil.

(d) The voltage is directly proportional to the number of turns in the coils.

3.6.1.2 Changing current in the primary coil induces a changing magnetic field in the iron core which in turn induces a current in the secondary coil. Induction in the output coil is proportional to the ratio of the coils in the primary and secondary coils.

3.6.1.3
(a) Step-down
(b) 500

3.6.1.4
(a) Step-up
(b) 500 V
(c) 3 A

3.6.1.5
(a) 180
(b) 100 mA

3.6.1.6
(a) 1 : 125
(b) 125 : 1
(c) The 30 000 V coil

3.6.1.7
(a) Input is 240 volts AC voltage with frequency 50 cycles per second, producing 10 amperes of current. Output, because of other electrical components built into transformer circuit, is 12 volts direct current voltage producing 0.5 amperes of current. (Note that the output from the transformer component of the device would be 12 V, 300 A.)
(b) Step-down
(c) Changing AC input to DC output.
(d) 100
(e) 200 A
(f) Output circuit must have a resistor in circuit to reduce current to required value (1 ampere).

3.6.1.8
(a) Step-down
(b) Number of coils in output circuit is fewer than the number in the input coil.
(c) 360 is input, 18 is output.
(d) 11V

3.6.1.9
(a) Same
(b) Same
(c) Current
(d) Each side of this transformer cancels the effect of the other – input and output voltages and currents are identical. There is no point in having a device which simply cancels itself out in a circuit.
(e) Transformer at a substation needs to be a step-down transformer so that input power is a high voltage to minimise energy losses through heating effects and output voltage is at an appropriate lower voltage for transmission to district substations or homes.

3.6.1.10 As seen in previous problems, transmission at low voltages and high currents over almost any distance from power stations is impossible – so transformers needed for this. Power supply to isolated areas would not be possible. Many would be relying on their own generators or solar cells.

3.6.2.1 Answers will vary, for example:

Device	Useful form of energy produced
Toaster	Heat
Blender	Kinetic
Light globe	Light
Radio	Sound

3.6.2.2

Device	Heat energy useful or not useful?	Most useful form(s) of energy produced
Dishwasher	Useful	Heat and kinetic
TV	Not useful	Light and sound
Refrigerator motor	Not useful	Kinetic
Frypan	Useful	Heat
Microwave oven	Useful	Microwaves (electromagnetic)
Motor in clothes dryer	Not useful	Kinetic

3.6.2.3 Radio:

A sound energy
B microphone
C electrical energy
D transmitter
E electromagnetic (radio) energy
F antenna
G electrical energy
H speaker
I kinetic energy
J sound energy

Television:

A sound energy
B microphone
C electrical energy
D light energy
E video camera
F electrical energy
G transmitter
H electromagnetic energy
I antenna
J electrical energy
K television
L kinetic energy of electrons
M TV screen
N light energy
O speaker
P kinetic energy
Q sound energy

3.6.3.1 (a) AC electricity cannot be stored. AC has to be produced as it is needed. AC power stations are huge, highly polluting and therefore usually a fair distance from populated areas.

(b) DC cannot be transformed easily to high voltage and low current for transmission like AC. Therefore too much energy would be lost during transmission.

(c) Heating effects due to the resistance of the transmission lines.

3.6.3.2 (a)

Transmission voltage (V)	100 000	250 000	350 000	500 000
Power generated (MW)	40	40	40	40
Current in lines $I = \frac{P}{V}$	400	160	114.3	80
Total resistance of lines (Ω)	240	240	240	240
Power loss in lines $P = I^2R$ (MW)	38.4	6.14	3.14	1.54
Power left after losses (MW)	1.6	33.86	36.86	38.46
Power lost (%)	96	15.4	7.85	3.85
Power getting to consumers (%)	4	74.6	91.15	96.15

(b) If electricity is transmitted at low line voltages more will be lost in heating effects and so the net cost to the consumers will be higher.

(c) Transmitting electricity above 500 000 V would cause the lines and towers to spark. A lot of energy would be lost through this sparking, apart from the added danger it would pose to humans and wild life.

3.6.3.3 (a) X will be a step-up transformer so that the transmission voltage is high and the transmission current relatively lower so that energy losses are reduced.

(b) Y will be a step-down transformer to reduce voltage in the lines to values safe for consumer use.

3.6.3.4

Structure	How this protects against lightning strikes
Shield conductors	Non-current carrying wires (usually two) at the very top of transmission towers. They connect into the earth cable and carry lightning strikes to ground or along their length in preference to transmission lines being hit.
Insulation chains	Saucer-shaped stacks of ceramic material used to insulate transmission lines from the tower and each other at lengths depending on the voltage to prevent electrical discharges between wires and the tower.
Earth cable	Runs from the top of the pole down into the earth to carry lightning strikes to earth thus protecting the tower and the transmission lines.
Metal tower	Being metal, and set deeply into the ground, the tower itself is an earth protection against lightning strikes.
Distance between towers	The distance between towers is at least 150 m, which is enough to protect each tower from adjacent towers in case one is hit by lightning.

3.6.4.1 C

3.6.4.2 A

3.6.4.3 D

3.6.4.4 C

3.6.4.5 B

3.6.4.6 D

3.6.4.7 A

3.6.4.8 C

3.6.4.9 (a)

Appliance	Voltage of appliance (V)	Power appliance develops (W)	Current drawn by appliance (A)
Coffee machine	240	1200	5.0
Frypan	240	1800	7.5
Kettle	240	1500	6.25
laptop	18	75.6	4.2
Microwave	240	600	2.5
Rice cooker	240	500	2.083
Toaster	240	800	3.33

(b) Four devices only. Maximum power that can be drawn from the circuit = VI = 2400 W. So, the laptop, rice cooker, microwave, and toaster add to 1975.6 W, although any of these could be replaced by the coffee machine without exceeding the 2400 W.

(c) Minimum three appliances, for example, the frypan and either the microwave, rice cooker or toaster, but this still allows power for the laptop as well.

3.6.4.10 (a) Parallel circuits isolate appliances and lights to their own section of the circuit so that if they malfunction, or some other appliance blows, the only appliance affected is the one that blows.

(b) Series circuits are seldom used in homes except in some appliances like strings of fairy lights because if one globe or device blows, all in the series circuit will stop working.

3.6.4.11 (a) Lighting, power and dedicated circuits.

(b) The different circuits carry different amounts of current so that the devices connected to them function efficiently. Having high current circuits for all circuits simply because a few appliances require the higher currents would decrease safety.

(c) Dedicated then power then lighting least.

(d) Larger current wires (the dedicated circuits) use thicker wires (less resistance per unit length) to minimise heating effects.

3.6.4.12 (a) P = dedicated

Q = power

R = lighting

(b) All in parallel except in the dedicated circuit in which the appliance is in series with the rest of the circuit – but given there is only one appliance per circuit, this is the only way it can be connected.

(c) How water system, air conditioning unit, stove.

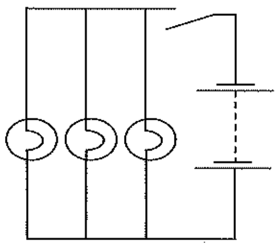
3.6.4.13 (a) Room Q.

(b) Each switch is connected in series with the globe it operates. Each switch and globe combination is connected in parallel with the other switch and globe combination.

(c) The other will stay on with the same brightness because the potential difference across it will not change.

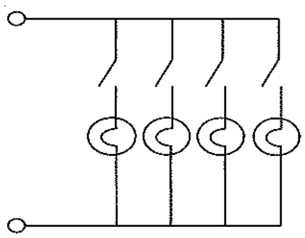
(d) Parallel.

(e)



(f) Four.

(g)



3.6.4.14 (a) 2

(b) 3

(c) 3

(d) Stove, hot water system, air conditioner.

(e) The labels indicate the maximum current the circuit can carry (C10 = 10 amperes etc).

(f) The stove draws up to 32 A.

(g) 80 amperes. The label on the 'normal supply main switch' indicates that it is designed to control up to 80 amperes.

(h) The home has a solar hot water system.

(i) If the total current in the power and/or lighting circuits exceeds 40 A then this switch will throw. This does not mean that any of the individual circuits is overloaded, simply that in this home, the maximum current allowed in the lighting and power circuits overall is 40 A.

3.7.1.1 The fluorescent, compact fluorescent and LED globes are more efficient than the other two types in the table. You may classify the LED as the most efficient based on its longevity, despite its higher cost. The LED is also the most efficient in terms of light output per unit power used. Least efficient is the incandescent globe in all properties apart from cost closely followed by the halogen globes. The fluorescent and compact fluorescent would be second choice after the LED, or perhaps for many people, first choice because of their lower cost despite their shorter longevity.

3.7.1.2 (a) The diagram shows the relative efficiencies of incandescent and fluorescent lamps with equal light energy output and their sources of inefficiencies.

(b) The fluorescent lamp produces the same light energy output for 25% of the energy input, and is therefore the more efficient of the two.

(c) It is interesting that the energy losses in the same areas are the same percentages, which would suggest the same inefficiencies, however, the deciding factor is the energy input and the fact that the same light energy output comes from a 25% input energy for the fluorescent lamp. The statement is therefore incorrect.

(d) Part of the driving force for this is global warming and the need to reduce greenhouse gases. The more efficient any electrical device is, the less energy it uses and the less pollution will be produced as electrical energy production is minimised, especially in an ever increasing global population and the spread of electrical energy into third world communities which raises demand significantly.

3.7.2.1 Answers will vary, for example:

	Device running on electricity or use of electricity	Impacts this has had on your life
(a)	Cooking	Without electricity, cooking would be a more complicated undertaking and take more time which would impact on our time for other activities.
(b)	Home cleaning	Without electricity, cleaning would be more labour intensive and take more time. Again, leaving less time for other activities.
(c)	Washing	Be it clothes, dishes or ourselves, again, without electricity these activities would take longer and involve more personal effort, taking time away from other, more enjoyable activities.
(d)	Lighting	Imagine your home lit only by candles, or kerosene lanterns or even gas lamps. Night activities would be significantly darker and less enjoyable
(e)	Transport	Imagine life without electric trains, cars, planes – all relying on electricity to operate internal systems to keep them running efficiently and safely. Travelling times would be longer and less comfortable.
(f)	Health	Without electricity we would have no X-rays, MRI machines, ultrasound etc. Diagnosis of medical problems would be much more difficult and health problems more significant.
(g)	Research	Imagine your study research without the use of computers which obviously rely on electricity. Your ability to collect information would be seriously limited and take much longer.
(h)	Learning	Doing well in any task involves knowing lots about the task. Electricity drives the computers that allow you to collect and store the information you need to do well at school.
(i)	Leisure	No Mind Craft. No Angry Birds. No iTunes. Imagine all the leisure things in your life that would disappear if electricity was not available. Maybe you would get more exercise and be fitter!
(j)	Communication	Imagine your life without your phone, Facebook, the internet in general. Your ability to keep in touch with friends would be limited.

3.7.2.2 Answers will vary, for example:

When answering 'impact' questions, give at least two examples and then how they affect you and then what other things you can do because of them ... many possible answers, e.g. Electricity allows the use of computerised games and learning. This provides entertainment at home and opportunities to reinforce school work. This can lead to better results at school and better employment opportunities down the track. These in turn will allow a better adult lifestyle.

The domestic use of electricity has enabled me to use many handyman tools and learn many carpentry skills. This has provided a hobby for leisure time as well as allowing me to build useful items which I might otherwise either not have or have to pay for. This leaves my pocket money free for other pursuits. In the future these skills may save money in adult life allowing for available money to be used on other things – clothes, education for children, leisure activities.

3.7.2.3 Answers will vary, for example:

Atmospheric pollution has increased enormously as more and more demand for electrical power has been met over the years. This will have contributed to asthma problems for many people and reduced the quality of their lifestyle, directed available money into medical expenses and reduced spending on other areas such as leisure activities or consumables.

Pollution from the burning of fossil fuels to work steam turbines produces heat pollution as well as air pollution. Heat pollution from coolant water into natural streams has affected water life adversely. This reduces the ability of all organisms in the waters to survive and affects the quality of the water. This in turn reduces the enjoyment many people get from water activities (fishing, swimming) and may have long-term effects on larger food chains.

Ugly power lines crossing the countryside, ugly power poles and lines in streets also offering impact sites for accidents so increasing death toll from transport accidents. Ugly transformers in substations and pole-mounted in streets. Possible harmful effects from high frequency electromagnetic radiations associated with high AC voltage transmission.

The development of hydroelectricity schemes impacted severely on specific areas of the environment, requiring huge dams, flooding of forests, valleys and even towns.

3.8.1.1 (a) *Critical evaluation* of these motives involves your analysing them in terms of their possible importance to Edison, and the likelihood that these would have influenced his actions. You may decide they are legitimate motives, or that a man of Edison's standing would not stoop 'so low'. Discuss your answer with your friends to get different viewpoints.

(b) In an *assessment* you are required to make a judgement as to whether the tactics are fair or not. You should support your assessment with statements of fact, or opinions of how you think an eminent scientist should behave in these circumstances. Again, discuss answer with your friends to get differing ideas.

(c) Opinions here will also differ. Discuss your answer with friends to see what they have to say.

3.8.1.2 (a) The Chicago World Fair and the Niagara Falls projects.

(b) They were very publicly visible projects – people could see their success – they very clearly demonstrated the efficiencies of AC transmission of power. The World Fair in particular, would have been a very impressive demonstration of a new technology to people.

(c) The final outcome would not have been any different – perhaps delayed a little. The deciding factor in the debate was the ability of AC to be transformed and so transmitted over long distances without significant loss of power.

(d) Distribution generators would need to be very numerous throughout the city. Household appliances requiring different voltages (TVs, computers, toys) to the normal 240 V supply would need separate generators and separate supply lines.

3.8.1.3 (a) DC cannot be transformed (easily).

(b) Any distribution system using DC would be significantly more expensive than its AC counterpart, requiring many more generators to cope with distances and duplicate generators for different supply voltages. Transmission line costs would be higher as more inputs to each house would be needed. Cost factors may limit the availability of differing supplies to some households and so deprive those homes of the variety of appliances we take for granted.

3.8.2.1 (a) A cell is specifically a single 1.2 to 2.0 volt power supply device while a battery is formed when two or more cells are connected in series with each other, or a device is specifically built which has several cells.

(b) Both supply electrical energy to a circuit. If the electrolyte in the cell is a paste, the cell is referred to as a dry cell. If the electrolyte is a solution, the cell is called a wet cell.

3.8.2.2

Type of battery
Alkaline cell
Carbon-zinc
Lead-acid
Lithium ion
Nickel cadmium
Nickel metal hydride

Type of battery	
Lead-acid	Least efficient
Carbon-zinc	
Nickel cadmium	
Nickel metal hydride	
Alkaline	
Lithium ion	Most efficient

3.8.2.3 (a) Memory effect: refers to the property that if the battery is not fully discharged, it 'remembers' the level of the discharge and only discharges to that level when recharged in the future. This reduces its efficiency.

(b) Discharge rate: refers to the rate at which a rechargeable battery uses its energy load before needing to be recharged. Batteries with a low discharge rate will be more efficient than those with a high discharge rate.

(c) Energy density: refers to the energy supplied per unit mass, Batteries with low energy density need to be larger to provide the same energy as batteries with high energy density. This makes low energy density batteries less convenient, which could be regarded as decreasing their efficiency.

3.8.2.4 (a) Number or recharges, memory effect, perhaps energy density.

(b) Y then X then Z.

(c) Technically, with a voltage of 3.6, Z would be considered a battery, while X and Y are cells.

(d) Least efficient would be X. Its energy density and number of recharges is comparable with X, but it will self-discharge faster, has a memory effect and is polluting when discarded. However, it is only about half the cost to buy and use. In addition, if run down and recharged correctly, the memory effect is not a disadvantage.

(e) Z. Despite its higher cost, it has the appropriate voltage (note that three of the others could be used in series to gain the same voltage), many more recharges and a high energy density.

Unit 2 Linear Motion and Waves

1.1.1.1

Quantity	Definition	Scalar or vector
Distance travelled	How far an object has moved along a particular path from its staring position.	Scalar
Displacement	How far an object is from its starting position measured in a straight line with direction.	Vector
Speed	A measure of the rate of travel of an object.	Scalar
Velocity	A measure of the rate of change of displacement of an object.	Vector
Acceleration	A measure of the rate of change in velocity of an object.	Vector

1.1.1.2 'Uniform' used in this context means 'constant'. If the motion is uniform then the velocity of the object is constant. If the acceleration is uniform, then the object is accelerating at a constant rate and in a constant direction.

1.1.1.3 Because velocity has both magnitude and direction, then an object travelling at uniform speed can still accelerate (change its velocity) if its direction of travel changes, for example, a car turning a corner.

1.1.1.4 (a) Tapes A and C.

(b) Tape B shows that the object first accelerated, sped up, then decelerated (slowed down).

Tape D shows that the object was decelerating (moving with ever decreasing speed).

1.1.1.5

Constant velocity	The velocity an object which travels the same displacement in every period of time.
Average velocity	The constant velocity an object would need to travel at so as to travel the same displacement in the same time.
Instantaneous velocity	The velocity of an object in the instant of time we consider it. This will vary from instant to instant depending on road and traffic conditions etc.
Initial velocity	The velocity of an object when we first consider it. The object's velocity at the start of its journey.
Final velocity	The velocity of an object at the end of its journey or when we finish our consideration of its motion.

1.1.1.6

Vector	Displacement (m)	Direction as compass direction	Direction as bearing
A	95	N 53° W	307°
B	26	N 20° W	340°
C	60	N 15° E	15°
D	23	E 44° S	134°
E	50	S 30° W	210°
F	91	W 21° S	249°

1.1.1.7 Car may slow down (reaches a corner, up a hill, speed bump, pedestrian crossing etc) or speed up (down hill, moving away from stop lights, as it starts etc).

1.1.1.8 Average speed is a calculated value equal to the total distance travelled divided by the total time taken while instantaneous speed is the actual speed of a vehicle at a particular instant of time.

1.1.1.9 Scalars have magnitude only while vectors have both magnitude and direction.

1.1.1.10 Examples include:

Scalar quantities	Symbol	SI unit	Vector quantities	Symbol	SI unit
Mass	m	kg	Displacement	s	m (+ direction)
Time	t	s	Velocity	v	m s^{-1} (+ direction)
Speed	speed	m s^{-1}	Acceleration	a	m s^{-2} (+ direction)
Distance	d	metre (m)	Force	F	N (+ direction)
Length	l	metre (m)	Momentum	p	kg m s^{-1} (+ direction)

1.1.1.11 Distance travelled is a measure of how far an object has moved from its starting point. Displacement indicates how far, in a straight line, an object is from its starting point, and the direction of its finish position from its starting position.

1.1.1.12 A bearing indicates the direction an object is travelling (or has travelled) as an angle measured clockwise from north.

1.1.1.13

Object travelling by	Distance travelled (km)	Displacement (km)
Road 1	75	50 east
Road 2	50	50 east
Road 3	150	50 east

1.1.1.14

Wombat	Distance travelled (m)	Displacement (m) (directions as compass readings)	Displacement (m) (directions as bearing)
1	55	55 N 64° W	55 bearing 296°
2	58	58 N 65° E	58 bearing 065°
3	30	30 S 46° E	30 bearing 134°
4	32	32 S 22° W	32 bearing 202°

1.1.2.1 SI units are units of measurement which form the Standard System of Units. These are units for the measurement of quantities which have been agreed on internationally and used so that communications of quantities between nations is easier. It is the modern form of the metric system.

1.1.2.2

- No full stops are used after units.
- All units are lower case unless they are named in honour of a person (e.g. amperes = A). The only exception is L for litre to avoid confusion with some typeface number 1's or 'l'.
- If a combination of units is used, e.g. metre per second, then there are three acceptable formats:
  - (i) m/s (use a slash between the m and the s^{-1})
  - (ii) $m.s^{-1}$ (a full stop between the m and the s^{-1})
  - (iii) $m\ s^{-1}$ (a space between the m and the s^{-1})

1.1.2.3 For large measurements, it is more sensible to use units which better suit that measurement. For example, we would not measure the distance to the next galaxy in metres. Light years, or parsecs are much more sensible units. While they are not SI, there is international agreement on their use.

1.1.2.4

Quantity	SI unit (name and symbol)	Quantity	SI unit (name and symbol)
Mass	kilogram (kg)	Electric potential difference	volt (V)
Length	metre (m)	Electrical resistance	ohm (Ω)
Time	second (s)	Electric current	ampere (A)
Displacement	metre (m)	Speed	metres per second ($m\ s^{-1}$)
Force	newton (N)	Velocity	metres per second ($m\ s^{-1}$)
Energy	joule (J)	Acceleration	metres per second per second ($m\ s^{-2}$)
Power	watt (W)	Temperature	kelvin (K)
Momentum	kilogram metre per second ($kg\ m\ s^{-1}$ or newton second (N s)	Volume	litre (L)

1.1.2.5

Unit prefix	Symbol	Meaning	Unit prefix	Symbol	Meaning
deca	da	10^1	deci	d	10^{-1}
hecta	h	10^2	centi	c	10^{-2}
kilo	k	10^3	milli	m	10^{-3}
mega	M	10^6	micro	μ	10^{-6}
giga	G	10^9	nano	n	10^{-9}
tera	T	10^{12}	pico	p	10^{-12}
peta	P	10^{15}	femto	f	10^{-15}
exa	E	10^{18}	atto	a	10^{-18}
zeta	Z	10^{21}	zepto	z	10^{-21}
yotta	Y	10^{24}	yocto	y	10^{-24}

1.2.1.1 (a) Average speed = $\dfrac{\text{total distance travelled}}{\text{total time taken}}$

(b) Average speed = $\dfrac{\text{total speed + final speed}}{2}$

(c) Average velocity = $\dfrac{\text{displacement}}{\text{total time taken}}$

(d) Average velocity = $\dfrac{\text{initial velocity + final velocity}}{2}$

(e) Acceleration = $\dfrac{\text{change in velocity}}{\text{time taken for change to occur}}$

(f) Acceleration = $\dfrac{v - u}{t}$

(g) Final velocity = $v = u + at$

1.2.1.2 1.69 m s^{-1}

1.2.1.3 223.2 m s^{-1}

1.2.1.4 Average speed = 48.5 km h^{-1}, average velocity = 48.5 km h^{-1} east

1.2.1.5 Average speed = 0.57 m s^{-1}, average velocity = 0.57 m s^{-1} from X to Y

1.2.1.6 (a) 39 km

(b) 8 km bearing 135°

(c) 26 km h^{-1},

(d) 5.33 km h^{-1} bearing 135°

1.2.1.7 (a) They will be the same, 60 km h^{-1}.

(b) At P instantaneous velocity is 60 km h^{-1} bearing 160° while at Q it is 60 km h^{-1} bearing 055°

(c) 60 km h^{-1}

(d) 40 km h^{-1} bearing 086°

1.2.1.8 (a) 90 km (given)

(b) 2 hours (given)

(c) Distance travelled by A each hour = $\dfrac{90}{2}$ = 45 km h^{-1}

(d) 45 km h^{-1} (from (c))

(e) Average speed of B = $\dfrac{\text{total distance travelled}}{\text{time taken}} = \dfrac{200}{3} = 66.7$

(f) Average speed of C = $\dfrac{\text{total distance travelled}}{\text{time taken}} = \dfrac{150}{5}$ = 30 km h^{-1}

(g) Instantaneous speed can vary because of road conditions, hills, corners etc.

(h) A = 45 km h^{-1} away from X towards Y or bearing 135°

B = 66.7 km h^{-1} away from X towards Y, bearing 135°

C = 30 km h^{-1} away from X towards Y, bearing 135°

1.2.1.9 (a) Zero

(b) 180 km h^{-1} (50 m s^{-1})

(c) 90 km h^{-1} (25 m s^{-1})

(d) 90 km h^{-1} (25 m s^{-1}) If we assume the acceleration is constant, then its speed at the halftime point will be half its final speed.

(e) 90 km h^{-1} (25 m s^{-1})

1.2.1.10 (a) 37.5 m s^{-1}, 37.5 m s^{-1} N

(b) 80 m s^{-1}, 80 m s^{-1} S

(c) 32 m s^{-1}, 32 m s^{-1} E

(d) 5.6 m s^{-1}, 5.6 m s^{-1} SE

(e) 24 m s^{-1}, 24 m s^{-1} bearing 147°

(f) 12 s, 50 m s^{-1} bearing 112°

(g) 5.33 s, 15 m s^{-1} bearing 347°

(h) 5 hours, 5 km h^{-1} bearing 220°

(i) 32.5 s, 80 m s^{-1} NE

(j) 6 minutes, 55 km h^{-1} S

(k) 105 km E, 60 km h^{-1}

(l) 135 m NE, 7.5 m s^{-1}

(m) 2700 m SW, 9 m s^{-1}

(n) 1.05m W, 0.3 m s^{-1}

(o) 5.6 m bearing 045°, 28 m s^{-1}

1.2.1.11 The equations for uniformly accelerated motion apply whenever an object is moving with constant acceleration in a straight line, regardless of the number of forces acting on it. The only force considered is the resultant or net force.

1.2.1.12 (a) There are several schools of thought here. Positive acceleration is usually thought of as acceleration in the direction of motion of an object (i.e. it increases the velocity of the object). On this definition, negative acceleration would be against the motion (slowing the object down, or increasing its velocity in the opposite direction to the original velocity). Another school of thought is that positive acceleration is always north, or east, or vertically downwards, depending on the nature of the particular problem.

(b) If we regard the acceleration due to gravity as a positive acceleration, then a cannonball shot into the air would be considered to have a negative velocity and a positive acceleration.

(c) We would usually regard the velocity of a car as given to us in a problem as a positive velocity (unless defined otherwise), so if that car is braking, then it has a positive velocity and a negative acceleration.

1.2.1.13 9.8 m s^{-2}

1.2.1.14 (a) While we would probably use 9.8 as our value in a calculation of the motion of this falling object, we would technically be incorrect. 9.8 is the average value of g at the surface. To two decimal places, this value would be less at altitude (actually 9.74 at 30 000 m). So, the object would fall with an acceleration which commenced at 9.74 m s^{-2} and this would increase to 9.8 m s^{-2} at ground level. So the statement is technically incorrect.

(b) Air resistance, which, acting against the motion of the object will decrease its rate of acceleration. The falling object will eventually reach a constant (terminal) velocity when air resistance upwards equals gravitational force downwards.

1.2.1.15 (a) 18 m s^{-1} N

(b) 36 m s^{-1} N

(c) 54 m s^{-1} N

1.2.1.16 383.3 m s^{-2} against motion

1.2.1.17 39 s

1.2.1.18 (a) 28 m s^{-1}

(b) 64 m s^{-1}

1.2.1.19 6.5 m s^{-1} N

1.2.1.20 15 m s^{-1} S

1.2.1.21 (a) 3.91 s

(b) 38.3 m s^{-1}

(c) 19.17 m s^{-1}

1.2.1.22 (a) 8.33 m s^{-1}

(b) 3.54 m

1.2.1.23 (a) 147 m s^{-1}

(b) 1102.5 m

1.2.2.1 Vector quantities require both a measure and a direction. That is, they have both magnitude and direction. Scalar quantities only have magnitude. It is incorrect to give a direction for a scalar quantity.

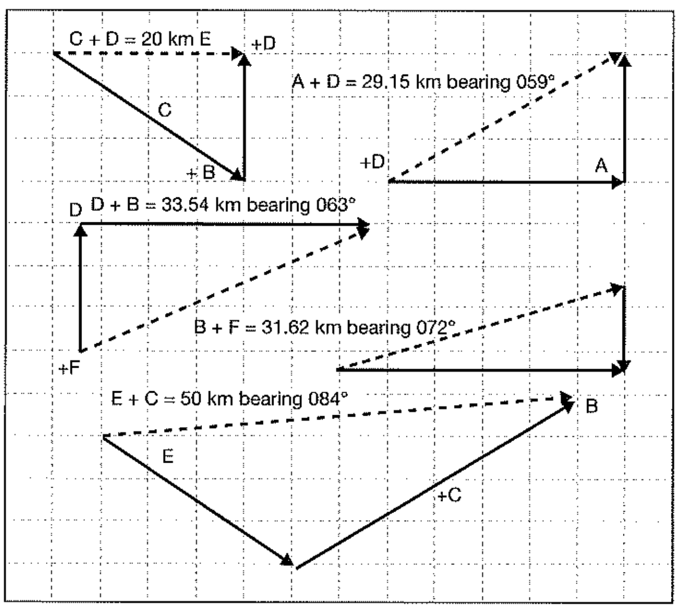
1.2.2.2 For example: (Your table may include others – check with text or teacher if unsure.)

Scalar quantities			Vector quantities		
Quantity	**Symbol**	**Unit**	**Quantity**	**Symbol**	**Unit**
Mass	m	kilogram (kg)	Weight	W_F	newton (N)
Time	t	second (s)	Electric field	E	volts per metre (V m^{-1})
Length	L	metre (m)	Magnetic field	B	tesla (T)
Electric charge	Q	coulomb (C)	Torque	τ	newton metre (N m)
Temperature	T	kelvin (K)	Displacement	s	metre (m)
Distance travelled	d	metre (m)	Velocity	v	metres per second (m s^{-1})
Speed	Speed	metres per second (m s^{-1})	Force	F	newton (N)
Energy	E	joule (J)	Acceleration	a	metres per second per second (m s^{-2})
Work	W	joule (J)	Impulse	I	newton seconds (N s or kg m s^{-1})
Power	P	watt (W)	Momentum	p	kilogram metres per second (kg m s^{-1})

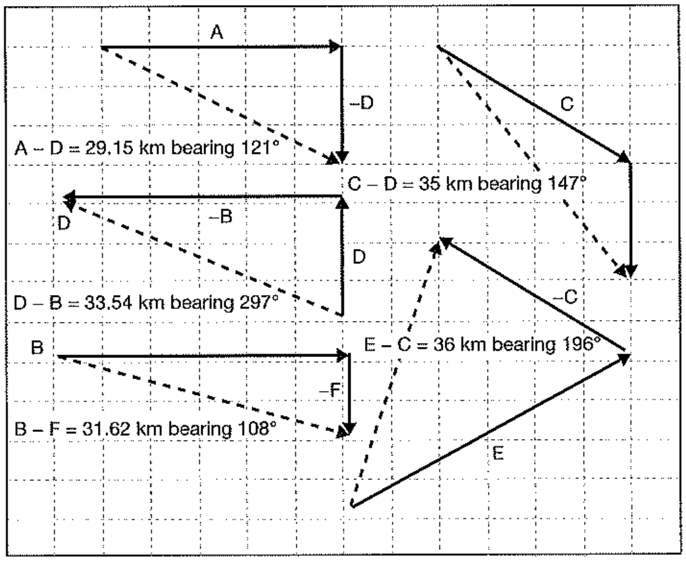
1.2.2.3 (a) 13 left
(b) 37 right
(c) 21 right

1.2.2.4 Maximum value is 11 units (when both vectors act in the same direction). Minimum value is 1 unit (or –1 unit) when the vectors act directly in opposite directions.

1.2.2.5



1.2.2.6



1.2.2.7 A

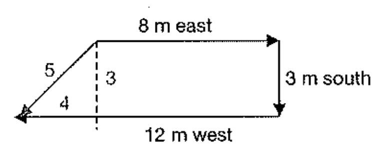
1.2.2.8
(a) 90 km h^{-1}
(b) 64.3 km h^{-1} bearing 143°
(c) 5 hours

1.2.2.9 B

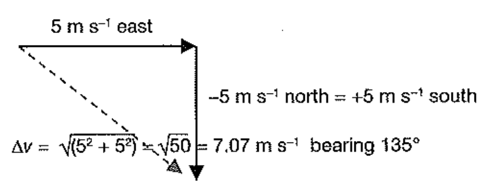
1.2.2.10
(a) 82.2 N
(b) 25.5 N
(c) 200 N at 37° to 160 N force
(d) 50.4 N at 52.5° up from 100 N force
(e) 93.3 N
(f) 193.2 N at 15° to each force
(g) 600 N
(h) 112.5 N at 53.2° upwards from the 50 N force

Note that the vector diagrams shown in the answers to these questions are not to scale.

1.2.3.1 From vector diagram, displacement = 5 m bearing 233°



1.2.3.2
(a) Change in speed = 0
(b) Change in velocity = final velocity – initial velocity
= 5 m s^{-1} east – 5 m s^{-1} north
= from diagram = 7.07 m s^{-1} bearing 135°
(c) Acceleration = $\frac{\Delta v}{t} = \frac{7.07}{0.05}$
= 141.1 m s^{-2} bearing 135°



1.2.3.3 From vector diagram (which is the equivalent of a 3-4-5 right-angled triangle):

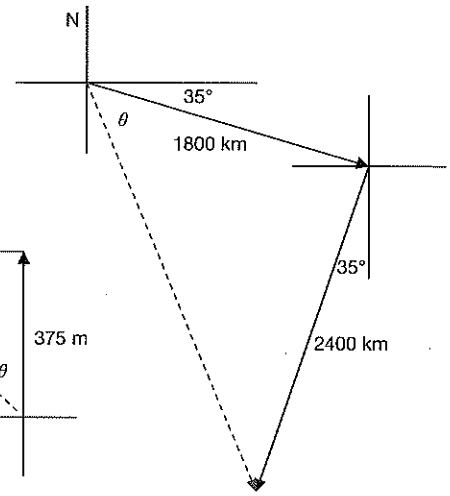
Resultant displacement = $\sqrt{1800^2 + 2400^2}$ = 3000 km

Tan $\theta = \frac{2400}{1800} = 1.33$

From which $\theta = 53°$

So bearing of displacement = 125 + 53 = 178

Which makes the displacement = 3000 km bearing 178°



1.2.3.4 (a) 30 km h^{-1} = $\frac{30000}{3600}$ = 8.33 m s^{-1}

Distance travelled north = 8.33 × 45 = 375 m

20 km h^{-1} = $\frac{20000}{3600}$ = 5.55 m s^{-1}

Distance travelled west = 5.55 × 60 = 333.3 m

Total distance = 708.3 m

(b) Displacement is given by the vector diagram

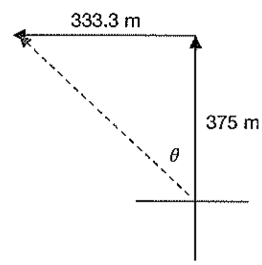
= $\sqrt{375^2 + 333.3^2}$

= 501.7 m

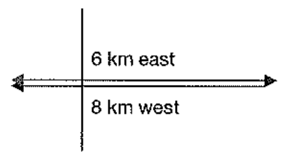
Tan $\theta = \frac{333.3}{375}$

From which $\theta = 41.6°$

Therefore displacement = 501.7 m bearing 322.6°



1.2.3.5 From vector diagram, displacement = 2 km west



1.2.3.6 (a) Distance travelled north = 50 × 4 = 200 km

Distance travelled west = 75 × 4 = 300 km

Therefore, average speed = $\frac{\text{total distance travelled}}{\text{time taken}}$

= $\frac{500}{8}$ = 62.5 km h^{-1}

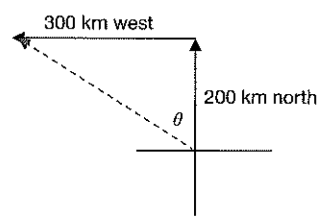
(b) From vector diagram, displacement = $\sqrt{200^2 + 300^2}$ = 360.56

Tan $\theta = \frac{300}{200} = 1.5$

From which $\theta = 56.3°$

So, displacement = 360.56 km bearing 303.7°

(c) Average velocity = $\frac{\text{displacement}}{\text{time}} = \frac{360.56}{8}$ = 45.07 km h^{-1} bearing 303.7°



1.2.3.7 (a) From the vector diagram:

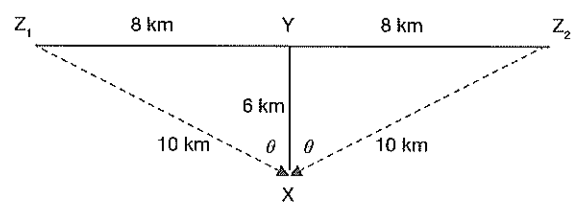
Z can be 8 km west of Y or 8 km east of Y

(b) From the vector diagram:

$\theta = 53°$

X can be 10 km bearing 127° (90 + 53) or

10 km bearing 233° (180 + 53) from Z



1.2.3.8 (a) For every second she swims, the current will carry her 0.6 m downstream.

(b) It will increase it = vector sum of 0.6 m s^{-1} downstream and 1.5 m s^{-1} across

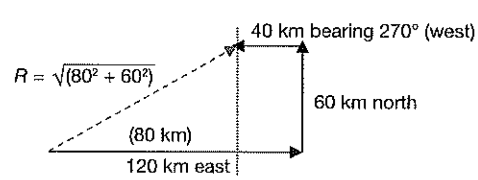
= 2.61 m s^{-1} at 21.8° downstream from straight across.

(c) She needs to head 21.8° upstream from straight across.

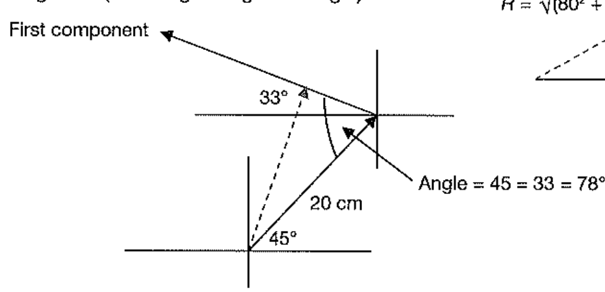
(d) Time = $\frac{\text{distance}}{\text{speed across}} = \frac{100}{1.5}$ = 66.67 s

1.2.3.9 From the vector diagram:

Displacement = 100 km bearing 053° (3-4-5 right-angled triangle)



1.2.3.10 From the vector diagram:



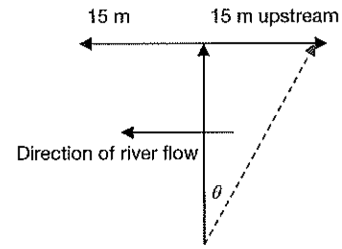
1.2.3.11 (a) Swimming at 1.6 m s^{-1} for 25 s makes the river $= 25 \times 1.6 = 40$ m wide.

(b) Being carried downstream 15 m in 25 s means the river flows at $\frac{15}{25} = 0.6$ m s^{-1}

(c) From the vector diagram:

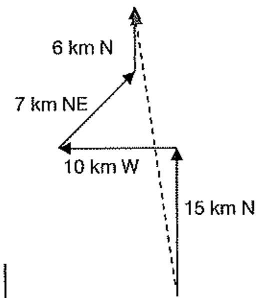
$\theta = \tan^{-1}\left(\frac{15}{40}\right) = 20.6°$

So swimmer must head 20.6° upstream from straight across.



1.2.3.12 From the vector diagram:

When drawn to scale, the resultant scales to be 26.44 N bearing 348.8°



1.2.3.13 (a) Average speed $= \frac{\text{total distance}}{\text{total time}}$

$= \frac{240 + 180}{10}$

$= 42$ km h^{-1}

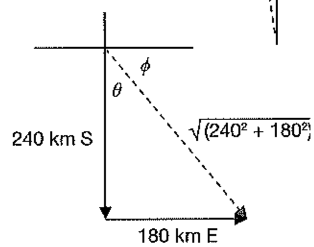
(b) Average velocity $= \frac{\text{diplacement}}{\text{time}}$

From vector diagram, displacement = 300 km

$\theta = \tan^{-1}\left(\frac{180}{240}\right) = 36.9°$

Therefore $\phi = 53.1°$

So displacement = 300 km bearing 143.1°



1.2.3.14 (a) Maximum values = 4 + 9 = 13 units when acting in the same direction.

(b) Minimum values = 4 – 9 = –5 units When acting in opposite directions.

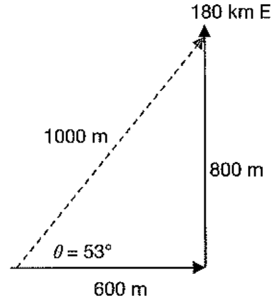
1.2.3.15 (a) Distance east = 12 × 50 = 600 m

Distance north = 16 × 50 = 800 m

Total distance = 1400 m

(b) From the vector diagram:

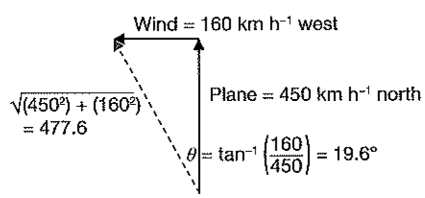
Displacement = 1000 m bearing 037°



(c) Average speed $= \frac{\text{distance}}{\text{time}} = \frac{1400}{100} = 14$ m s^{-1}

(d) Average velocity $= \frac{\text{displacement}}{\text{time}}$

$= \frac{\text{1000 m bearing 0.37°}}{100} = 10$ m s^{-1} bearing 037°



1.2.3.16 (a) From the vector diagram:

Direction = N 19.6° W or

Bearing 340.4°

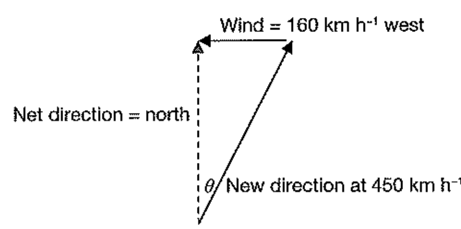
(b) Speed = 477.6 km h^{-1}

(c) (Note that bearing 019.6° is not correct.)

From new vector diagram:

$\theta = \sin^{-1}\left(\frac{160}{450}\right) = 20.8°$

So the plane must head on a bearing of 020.8°.



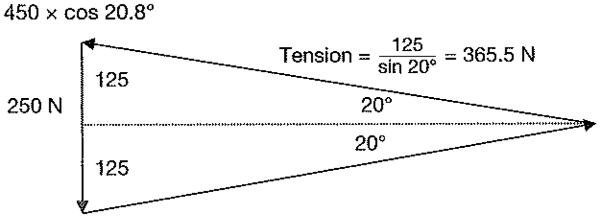
(d) From new vector diagram

$\text{Cos } \theta = \frac{\text{new speed}}{450}$

Therefore net speed = 450 × cos 20.8°

= 420.7 km h^{-1}

1.2.3.17 From vector diagram:



1.2.4.1 You need to draw vectors to add F_1 and F_2 and show that the third vector is equal to F_3. If all angles are drawn correctly, F_3 will be vertical. If you get lost, ask your teacher for help. Note diagrams below not drawn to scale.

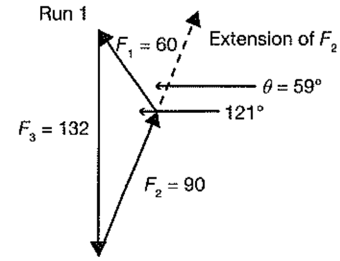
Run 1:

From the cosine rule, $F_3 = \sqrt{90^2 + 60^2 - 2 \times 90 \times 60 \times \cos 121°}$

Therefore, $F_3 = \sqrt{8100 + 3600 - (-5562.4)}$

$= 131.4$ g

That is, the results are consistent with a practical value for $F_3 = 132$ g with an error of $\left(\frac{0.6}{132}\right) \times 100\% = 0.04\%$



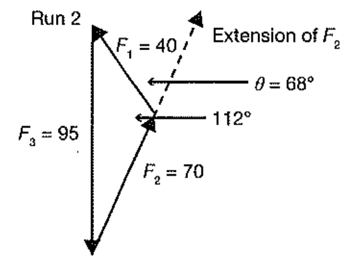
Run 2:

From the cosine rule, $F_3 = \sqrt{40^2 + 70^2 - 2 \times 40 \times 70 \times \cos 112°}$

Therefore, $F_3 = \sqrt{1600 + 4900 - (-2098)}$

$= 92.7$ g

That is, the results are consistent with a practical value for $F_3 = 95$ g with an error of $\left(\frac{2.3}{95}\right) \times 100 = 0.24\%$



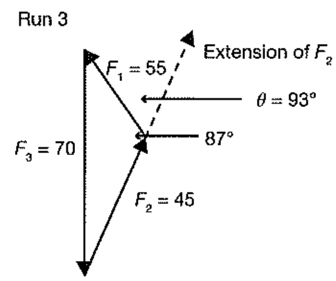
Run 3:

From the cosine rule, $F_3 = \sqrt{45^2 + 55^2 - 2 \times 45 \times 55 \times \cos 87°}$

Therefore, $F_3 = \sqrt{(2025 + 3025 - (259)}$

$= 69.2$ g

That is, the results are consistent with a practical value for $F_3 = 70$ g with an error of $\left(\frac{0.8}{70}\right) \times 100 = 0.11\%$



1.2.5.1 B

1.2.5.2 A

1.2.5.3 B

1.2.5.4 B

1.2.5.5 B

1.2.5.6 C

1.2.5.7 A

1.2.5.8 D

1.2.5.9 B

1.2.5.10 D

1.2.5.11

(a) Object starts at the zero displacement position then travels uniformly at 1.5 m s^{-1} east for 14 s then stops. After 14 s it is at a displacement of 21 m east.

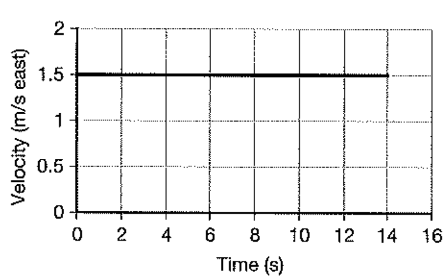
(b) Object starts at the zero displacement position then travels uniformly at 2.5 m s^{-1} for 8 s until it is at a displacement of 20 m N. It then stops and stays at this displacement for 6 s.

(c) Object starts at a displacement of 12 m W then travels uniformly at 1.5 m s^{-1} east for 18 s at which time it is at a displacement of 15 m east.

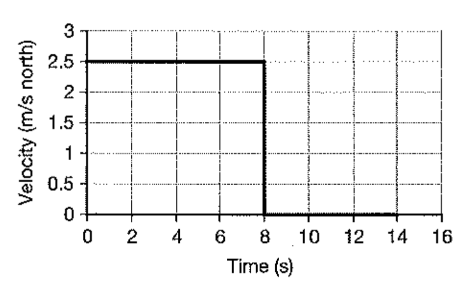
(d) Object starts at a displacement of 20 m north then travels uniformly at 2.5 m s^{-1} south for 14 s at which time it is at a displacement of 15 m south.

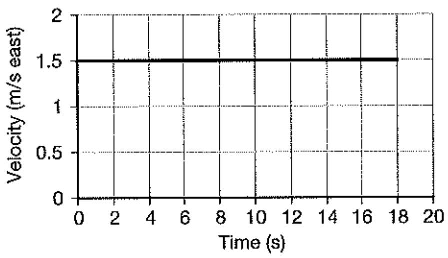
1.2.5.12

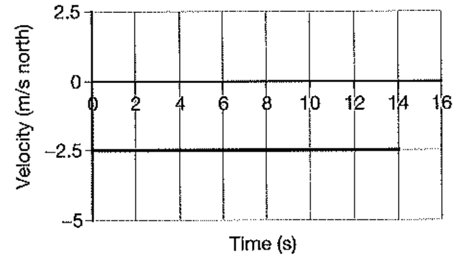
(a)



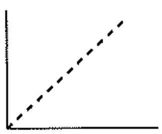
(b)

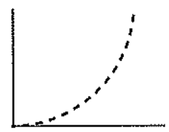


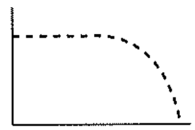
(c)


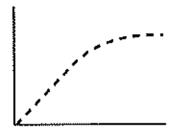
(d)


1.2.5.13 *Note*: In answers below, time on *x*-axis, displacement on *y*-axis.

(a)


(b)


(c)


(d)


1.2.5.14

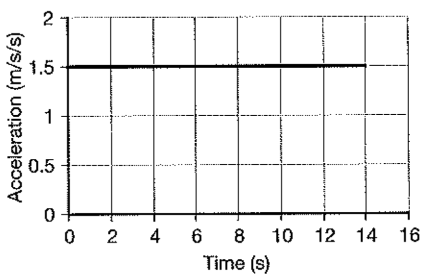
(a) Object starts from rest and accelerates uniformly at 1.5 m s^{-2} north for 14 s. After this time it is moving at 21 m s^{-1} north and has a displacement of 147 m north of its starting position. During this time its displacement changes by 147 m N.

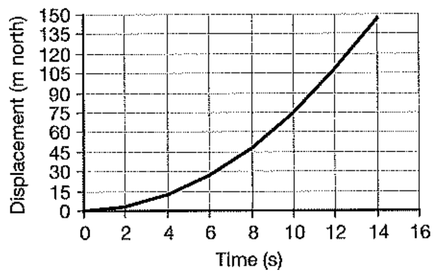
(b) Object starts at a velocity of zero then accelerates uniformly at 2.5 m s^{-2} south for 8 s. At the end of this time it is moving at 20 m s^{-1} south. It then moves with this velocity for a further 6 s. During the 14 s it travels a total distance of 200 m ending up at a displacement of 200 m south of its starting position.

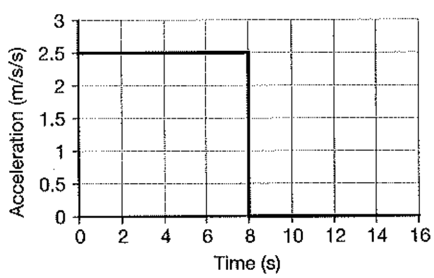
(c) Object starts at a velocity of 12 m s^{-1} west then accelerates uniformly at 1.5 m s^{-2} east for 18 s. During this time it travels a total distance of 123 m ending up at a displacement of 27 m east of its starting position.

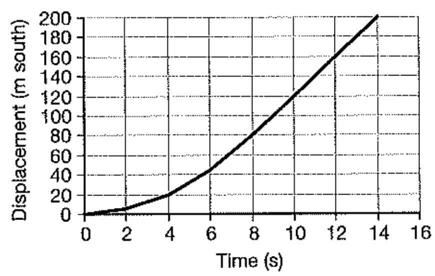
(d) Object starts at a velocity of 20 m s^{-1} E then accelerates uniformly 2.5 m s^{-2} W until it is moving at 15 m s^{-1} W. This takes 14 s. During the 14 s it travels a total distance of 125 m ending up at a displacement of 35 m E of its starting position.

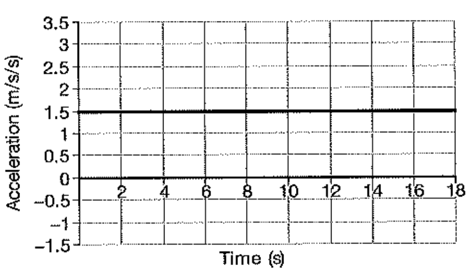
1.2.5.15

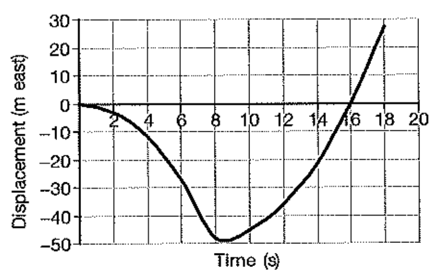
(a)


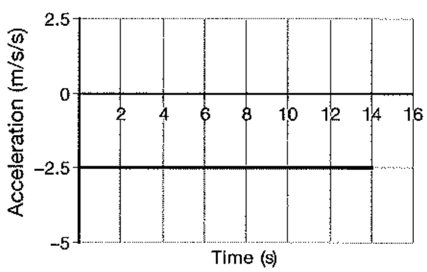


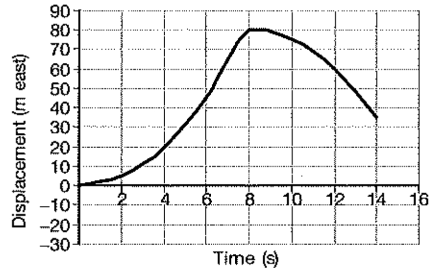
(b)




(c)




(d) 



1.2.5.16 (a) 8 m
(b) 15 m north
(c) 0.75 m s^{-1} north
(d) 0.75 m s^{-1} north
(e) 0.75 m s^{-1} north

1.2.5.17 (a) 7 m
(b) 20 m south
(c) 1.0 m s^{-1} south
(d) 1.33 m s^{-1} south

1.2.5.18 (a) 15 m
(b) 1.5 m s^{-1}
(c) 1.5 m s^{-1} south
(d) 10 m south
(e) 1.5 m s^{-1} south

1.2.5.19 (a) 34 m
(b) 56 m north
(c) 2.8 m s^{-1} north
(d) About 3.5 m s^{-1} north
(e) Car starts at displacement zero and travels for 20 s with increasing velocity (accelerating) until it reaches displacement 56 m north.

1.2.5.20 (a) 30 m
(b) 70 m south
(c) 3.5 m s^{-1}
(d) 3.5 m s^{-1} south
(e) About 2.25 m s^{-1} south

1.2.5.21 (a) 40 m
(b) 110 m north
(c) 5.5 m s^{-1}
(d) 5.5 m s^{-1} north
(e) About 8.9 m s^{-1} north

1.2.5.22 (a)

Time (s)	2	4	6	8	10	12	14
Displacement	4 m E	4 m E	8 m E	12 m E	16 m E	8 m E	0

(b) From $t = 0$ to 4 s, 4 to 10 s, and 10 to14 s (different values)

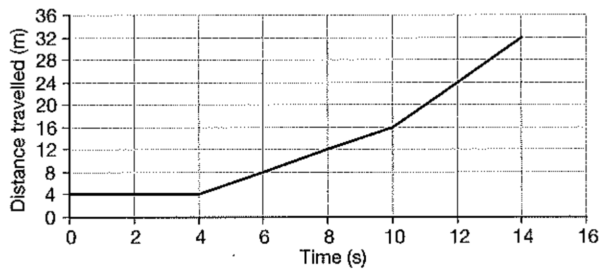
(c)

Time Interval	**Speed during time interval**	**Velocity during time interval**
$t = 0$ and $t = 2$	Zero	Zero
$t = 4$ and $t = 10$	2 m s^{-1}	2 m s^{-1} east
$t = 10$ and $t = 14$	4 m s^{-1}	4 m s^{-1} west

(d) 2.0 m s^{-1}

(e) 0.29 m s^{-1} west

(f)



1.2.5.23 (a)

Time	Instantaneous velocity
3	0
5	3 m s^{-1} S
8	3 m s^{-1} S
12.5	5.5 m s^{-1} N
16	5.5 m s^{-1} N
18	5.5 m s^{-1} N

(b)

Total distance travelled by object	51 m
Total displacement of object	15 m north
Average speed of object	2.83 m s^{-1}
Average velocity of object	0.83 m s^{-1} north

(c)

Section	A	B	C	D
How far did object travel in each section (m)	0	18 m	0	33 m
Average speed in each section (m s^{-1})	0	3 m s^{-1}	0	5.5 m s^{-1}
Average velocity in each section (m s^{-1})	0	3 m s^{-1} S	0	5.5 m s^{-1} N

1.2.5.24 (a)

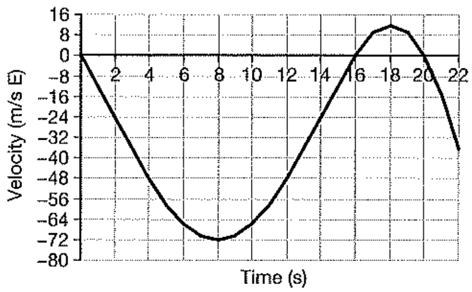
Times	Instantaneous acceleration (m s^{-2})	Instantaneous velocity (m s^{-1})
3	12 W	36 W
6	6 W	66 W
8	0	72 W
13	12 E	36 W
17	6 E	9 E
20	12 W	0
21	18 W	15 W

(b) 36 m s^{-1} W

(c) 0

(d) 60 m s^{-1} E

(e)



(f)

Total distance travelled by object	Approximately 678 m (count squares, each square = 16 m)
Total displacement of object	Approximately 538 m W
Average speed of object	Approximately 30.8 m s^{-1}
Average velocity of object	Approximately 24.5 m s^{-1} W

1.2.5.25

(a)	Initial velocity of car	0
(b)	Final velocity of car	20 m s^{-1} south
(c)	Find the velocity of the car at t = 7 s	14 m s^{-1} south
(d)	When was the car moving at 15 m s^{-1}?	After 7.5 s
(e)	What was the acceleration of the car?	2 m s^{-2} south
(f)	Change in displacement of the car	100 m south
(g)	Total distance travelled by the car	100 m
(h)	Average speed of the car	10 m s^{-1}
(i)	Average velocity of the car	10 m s^{-1} south

(j) We did not know where the car was at time zero. Just because its velocity was zero does not mean that it was at the zero displacement position.

1.2.5.26

(a)	What was the initial velocity of car?	20 m s^{-1} N	(f)	How far did it travel in the first 5 s?	50 m
(b)	What was the final velocity of car?	20 m s^{-1} S	(g)	What total distance did it travel?	100 m
(c)	What was its velocity at t = 7.0 s?	8 m s^{-1} S	(h)	What was its change in displacement?	0
(d)	At what time was its speed 16 m s^{-1}?	At t = 1 and t = 9	(i)	Calculate its average speed	10 m s^{-1}
(e)	What was its acceleration?	4 m s^{-2} S	(j)	Calculate its average velocity	0

1.2.5.27

(a)	What was the initial velocity of car?	0	(f)	How far did it travel in the first 25 s?	350 m
(b)	What was the final velocity of car?	60 m s^{-1} N	(g)	What total distance did it travel?	800 m
(c)	What was its velocity at t = 30.0 s?	34 m s^{-1} N	(h)	What was its change in displacement?	800 m N
(d)	At what time was its speed 45 m s^{-1}?	35 s	(i)	Calculate its average velocity	20 m s^{-1} N
(e)	What was its acceleration at t = 20 s?	1.2 m s^{-2} N	(j)	What was its acceleration at t = 35 s	2.6 m s^{-2} N

1.3.1.1

(a) To examine the relationship between the time it takes an object to fall from various heights and the height from which it is dropped.

(b) Object being dropped, stopwatch used, place where object dropped.

(c) Height object dropped from.

(d) Dependent variable = time to fall.

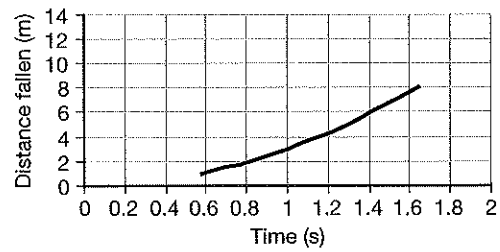
Independent variable = height from which object is dropped.

(e) Stopwatch may be running too fast or too slow. Systematic errors can be minimised by graphing results and using line of best fit analysis to calculate values required rather than using mathematical formulas and applying them individually to each reading and then taking an average.

(f) A typical random error would be an incorrect time due to lack of concentration of the person timing. This can be minimised by taking several readings for each measure and averaging them (and eliminating any outliers if they occur).

(g)

Distance object falls (m)	Time to fall (s)	(Time to fall)^2 (s^2)
1.0	0.58	0.336
2.0	0.82	0.672
3.0	1.00	1.00
4.0	1.15	1.32
5.0	1.29	1.66
6.0	1.41	1.99
7.0	1.53	2.34
8.0	1.64	2.70



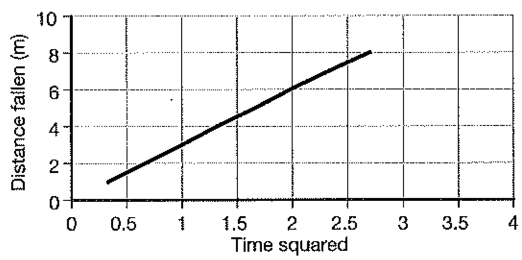
(h) Graph should curve gently upwards – will not be straight.

(i) From extrapolation of the back towards the origin, about 0.4 s.

(j) 1.05 s

(k) From extrapolation of the graph beyond the last reading, about 12 m.

(l) No. Conclusion can only be drawn from a straight line graph because we don't know the mathematical equation for the curve.



(m) See graph.

(n) Time to fall squared is directly proportional to height dropped from.

(o) 5.95 m s^{-2}

(p) No, gravity on Earth is 9.8 m s^{-2} or very close to this, depending on altitude.

(q) Yes. The graph is a straight line, so we know that the plotted variables are directly proportional to each other.

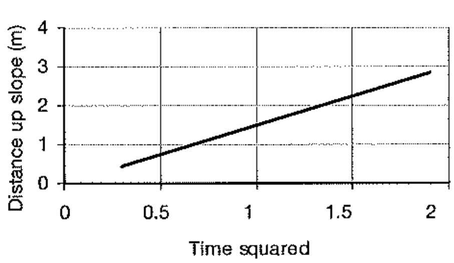
(r) Time to fall squared is directly proportional to height dropped from.

(s) Use average times to fall, repeat with objects of different mass, measure distances several times and use averages. All these strategies would improve the reliability of the results of the experiment.

1.3.1.2 (a)

Starting position	Distance up slope (m)	Average time to roll down (s)	(Average time)^2 (s^2)	Acceleration of ball down slope (m s^{-2})
P	0.5	0.55	0.30	3.33
Q	1.0	0.82	0.67	2.99
R	1.5	1.00	1.00	3.00
S	2.0	1.21	1.46	2.74
T	2.5	1.30	1.69	2.96
U	3.0	1.38	1.90	3.16
			Average value of acceleration of ball	3.03

(b)



(c) From the graph, acceleration = 2 × gradient = 3.13 m s^{-2}

(d) Calculating from a line of best fit is always more reliable than making mathematical averages because the graphical analysis minimises systematic errors, and ignores outliers.

(e) From $a = g \sin \phi$, 3.13 = 9.8 sin ϕ which gives ϕ = 18.6°

(f) The first one (Analysing an experiment 1) would be the more valid because it is measuring acceleration due to gravity as affected by air resistance which can be considered as negligible. The second experiment (Analysing an experiment 2) involves friction between the ball and the incline as well as air resistance, so it is measuring a less accurate value.

1.3.2.1 (a) From $s = ut + \frac{1}{2}at^2$

$200 = 0 + 0.5 \times 9.8 \times t^2$

From which t = 6.39 s

(b) From $v = u + at$

$v = 0 + 9.8 \times 6.39 = 62.6$ m s^{-1}

1.3.2.2 (a) From $v = u + at = 0 + (3 \times 4) = 12$ m s^{-1} N

(b) From $v = u + at = 0 + (3 \times 8) = 24$ m s^{-1} N

(c) From $v = u + at = 0 + (3 \times 12) = 36$ m s^{-1} N

1.3.2.3 From $v = u + at$, $29.4 = 0 + 3a$, therefore a = 9.8 m s^{-2} down

1.3.2.4 (a) From $s = ut + \frac{1}{2}at^2$, $50 = 0 + \frac{1}{2} \times 9.8 \times t^2$

Therefore t = 3.19 s

(b) From $v = u + at$, $v = 0 + 9.8 \times 3.19 = 31.3$ m s^{-1}

(c) Average speed = $\frac{u+v}{2}$ = 15.65 m s^{-1}

1.3.2.5 (a) From $s = ut + \frac{1}{2}at^2$, $147 = 0 + \frac{1}{2} \times 9.8 \times t^2$

Therefore t = 5.48 s

(b) From $v = u + at = 0 + 9.8 \times 5.48 = 53.7$ m s^{-1}

(c) Answers will be the same. Rate of fall is independent of mass.

(d) From $s = ut + \frac{1}{2}at^2$, $r = 0 + \frac{1}{2} \times 9.8 \times 2.5^2$

Therefore distance in 2.5 s = 30.625 m

1.3.2.6 (a) m s^{-1} = 1000 × km h^{-1} ÷ 3600 = 16.67 m s^{-1}

(b) From $v = u + at$, $16.67 = 0 + 9.8t$

Therefore time to fall = 1.7 s

From $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 9.8 \times 1.7^2 = 14.16$ m

(c) Average speed = $\frac{\text{distance}}{\text{time}} = \frac{14.16}{1.7}$ = 8.33 m s^{-1}

1.3.2.7 (a) From $v = u + at = 0 + 9.8 \times 20 = 196$ m s^{-1}

(b) From $s = ut + \frac{1}{2}at^2 = 0 + \frac{1}{2} \times 9.8 \times 20^2 = 1960$ m

(c) Average speed = $\frac{u+v}{2} = \frac{0+196}{2}$ = 98 m s^{-1}

1.3.2.8 (a) From $v = u + at$; $0 = 68.6 - 9.8t$; $t = \frac{68.6}{9.8}$ = 7 s

(b) Zero

(c) From $s = ut + \frac{1}{2}at^2 = 68.6 \times 7 - \frac{1}{2} \times 9.8 \times 7^2 = 240.1$ m

(d) Because time to rise = time to fall, time to fall = 7 s

1.3.2.9 (a) From $v = u + at$, at the top of the flight, $v = 0 = 49 + 9.8t$, so time to rise = 5 s

Therefore time of flight = 10 s

(b) From $s = ut + \frac{1}{2}at^2 = 49 \times 5 - \frac{1}{2} \times 9.8 \times 5^2 = 122.5$ m

(Remember, $a = -9.8$ when an object is going upwards)

(c) From $s = ut + \frac{1}{2}at^2 = 49 \times 3 - \frac{1}{2} \times 9.8 \times 3^2 = 102.9$ m

(d) From $v^2 = u^2 + 2as$

Velocity when object is 50 m up, $v = \sqrt{2 \times 9.8 \times 50}$ = 31.3 m s^{-1}

From $v = u + at$, $31.3 = 49 - 9.8t$, therefore t = 1.81 s

1.3.2.10 (a) From $v^2 = u^2 + 2as$, $v = \sqrt{-2.35^2 + 2 \times 9.8 \times 24.5} = 22.04$ m s^{-1}

(b) From $v = u + at$, 22.04 (down) = –2.35 (up) + 9.8t

Therefore t = 2.49 s

(c) Balloon rises at 2.35 m s^{-1} for 2.49 s = 5.85 m higher

Therefore altitude = 24.5 + 5.85 = 30.35 m

1.3.2.11 (a) Speed at highest point = 0

(b) From $v = u + at$, 0 = 14.7 – 9.8t

From which, t = 1.5 s

(c) From $s = ut + \frac{1}{2}at^2 = 14.7 \times 1.5 - \frac{1}{2} \times 9.8 \times 1.5^2 = 11.0$ m

Therefore, the stone fails to reach the height of the tree by 8.6 m

(d) Halfway back to the ground, stone has fallen 5.5 m, therefore

From $v^2 = u^2 + 2as$, $v = \sqrt{0^2 + 2 \times 9.8 \times 5.5} = 10.38$ m s^{-1}

1.3.2.12 (a) From $s = ut + \frac{1}{2}at^2$, $150 = 0 + 0.5 \times 9.8 \times t^2$

From which, t = 5.53 s

(b) From $v = u + at = 0 + 9.8 \times 5.53 = 54.2$ m s^{-1}

1.3.2.13 (a) From $v = u + at$, 0 = 49 – 9.8t

So time to rise = 5 s

So time of flight = 10 s

(b) From $v^2 = u^2 + 2as$,

Speed of arrow at 100 m is found from

$v = \sqrt{49^2 - 2 \times 9.8 \times 100} = 21.0$ m s^{-1}

From $v = u + at$, time to reduce velocity from 49 to 21 is

21 = 49 – 9.8t

Therefore t = 2.86 s

Therefore time above 100 m = 10 – (2 × 2.86) = 4.28 s

(c) From $s = ut + \frac{1}{2}at^2$,

Maximum height = 5 × 49 – 0.5 × 9.8 × 5^2 = 122.5 m

(d) 9.8 m s^{-2} downwards

1.3.2.14 9.8 m s^{-2}

1.3.2.15 Acceleration due to gravity on the surface of the Earth depends on several factors:

- The position on the Earth. If it is on top of a high mountain then the value of g will be slightly less because it will be further from the centre of mass of the Earth (more simply, from the centre of the Earth). If it is deep down a mine shaft or the bottom of an ocean trench, g will be greater as the position is closer to the centre of the Earth.
- The latitude of the position on the surface. Because the Earth is not a perfect sphere (flatter at the poles), the value of g will increase as we approach the poles as there is less distance between the surface and the centre.
- How close the object is to the equator. Because of the Earth's rotational spin, g will be less at the equator as the rotational speed at the equator is larger than anywhere else, and the reaction to the centripetal force decreases the gravitational force more here than as we approach the poles. (Note that this works against the previous factor.)
- g at the same distance from the centre of the Earth and at the same latitude will be slightly less on the surface of a deep ocean than on land due to the lesser mass of matter between it and the centre of Earth.

1.3.2.16 From $s = ut + \frac{1}{2}at^2$, the time for object A to hit the ground is:

$394 = 0 + 0.5 \times 9.8 \times t^2$

Which gives t = 8.97 s

From $s = ut + \frac{1}{2}at^2$, the time for object B to hit the ground is:

$394 = 8 \times t + 0.5 \times 9.8 \times t^2$ which rearranges to:

$4.9t^2 + 8t - 394 = 0$

Solving the quadratic gives t = 8.19 s

So B needs to be thrown 8.97 – 8.19 = 0.78 s after A is dropped.

1.3.2.17 (a) From $v = u + at$

$0 = 19.6 - 9.8t$, therefore $t = 2$ s

(b) From $s = ut + \frac{1}{2}at^2$

$s = 19.6 \times 2 - 0.5 \times 9.8 \times 2^2 = 19.6$ m

Therefore A rises 19.6 m

(c) Total distance for A to fall = 19.6 + 58.8 = 78.4 m

(d) From $s = ut + \frac{1}{2}at^2$

$78.4 = 0 + 0.5 \times 9.8 \times t^2$

So time for A to fall to the ground = 4 s

(e) Total time of flight for A = 2 + 4 = 6 s

(f) From $v = u + at = 0 + 9.8 \times 4 = 39.2$ m s^{-1}

(g) From $s = ut + \frac{1}{2}at^2$

$58.8 = 0 + 0.5 \times 9.8 \times t^2$

So B hits the ground after 3.46 s

(h) A hits the ground 6 – 3.46 = 2.54 s after B

1.3.2.18 While we would probably use 9.8 as our value in a calculation of the motion of this falling object, we would technically be incorrect. 9.8 is the average value of g at the surface. To two decimal places, this value would be less at altitude (actually 9.74 at 30 000 m). So, the object would fall with an acceleration which commenced at 9.74 m s^{-2} and this would increase to 9.8 m s^{-2} at ground level. So the statement is technically incorrect.

1.3.1.19 Air resistance, which, acting against the motion of the object will decrease its rate of acceleration.

1.3.2.20 (a) From $v = u + at$, $127.4 = 0 + 9.8t$, therefore $t = 13$ s

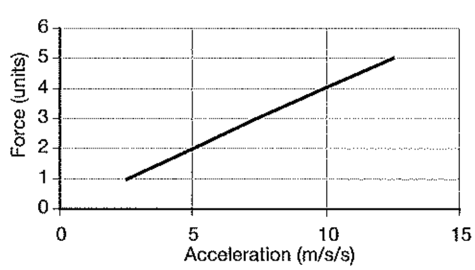
(b) Average speed of the object = $\frac{0+127.4}{2} = 63.7$ m s^{-1}

It falls for 13 s, so distance fallen = 63.7 × 13 = 828.1 m

1.4.1.1 (a)

Run	Accelerating force (N)	Time to travel 1.0 m (s)	Initial speed of trolley (m s^{-1})	Average speed of trolley (m s^{-1})	Final speed of trolley (m s^{-1})	Acceleration of trolley (m s^{-2})
1	*F*	0.89	0	1.12	2.25	2.53
2	2*F*	0.63	0	1.59	3.17	5.03
3	3*F*	0.52	0	1.92	3.85	7.40
4	4*F*	0.45	0	2.22	4.44	9.87
5	5*F*	0.40	0	2.50	5.00	12.5

(b)



(c) Dependent variable = time to travel 1 m across the bench

Independent variable = accelerating force

(d) Take multiple readings for each accelerating force and see if the times for each run are close to the same

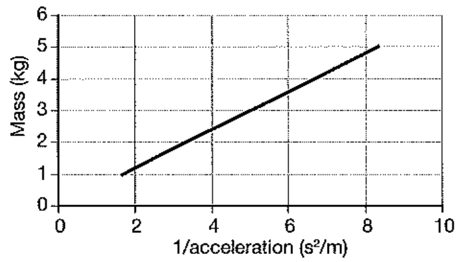
(e) To increase reliability, uncontrolled variables have to be minimised. You could use a highly polished surface and pulley surface to reduce friction (which has not been considered in the original experiment).

Using a light gate instead of a stopwatch would improve accuracy of the time measurement and therefore the reliability of the experiment.

(f) The acceleration produced by forces applied to a constant mass is directly proportional to the force applied.

1.4.1.2 (a)

Run	Trolley mass (kg)	Time to travel 1.0 m (s)	Initial speed of trolley (m s^{-1})	Average speed of trolley (m s^{-1})	Final speed of trolley (m s^{-1})	Acceleration of trolley (m s^{-2})	$(Acceleration)^{-1}$
1	1.0	1.83	0	0.55	1.09	0.60	1.67
2	2.0	2.58	0	0.39	0.78	0.30	3.33
3	3.0	3.16	0	0.32	0.63	0.20	5.00
4	4.0	3.65	0	0.27	0.55	0.15	6.67
5	5.0	4.08	0	0.25	0.49	0.12	8.33

(b) 

(c) The acceleration produced by a constant force acting on various masses is indirectly proportional to the mass.

(d) $F = ma$

(e) Experiment 1, force = 3.75 N (use gradient of graph)

Experiment 2, force = 0.6 N

1.4.2.1 An object at rest or in a state of uniform motion will stay at rest or in uniform motion until an unbalanced force acts on it.

1.4.2.2 Answers will vary, for example:

A space probe moving through space at a constant velocity (no engines firing) will continue to travel with the same velocity until the engines fire and provide a force to change that motion or until the probe enters a gravitational field which would also change its motion.

1.4.2.3 In the event of a sudden stop, the inertia of the person will cause him/her to continue forwards (Newton's first law) until they hit (the net force) the windscreen and go through it, or hit the steering wheel and dash and receive injuries.

1.4.2.4 If the car brakes suddenly the inertia of the object can cause it to continue to move (Newton's first law of motion). It will therefore 'fly' to the front of the car and can injure anyone it hits on the way.

1.4.2.5 Inertia is defined as the tendency of an object to resist any attempts to change its state of rest or uniform motion – in the cartoon as the car goes over the hump the dog moves upwards. As the car goes down the other side of the hump, the dog has a tendency to keep moving upwards until gravity pulls it down.

1.4.2.6 The racing harness stops the driver's body being 'thrown' sideways during a collision, especially if the car rolls. A lap-sash belt is not as good because the driver could be 'thrown' out of the belt in a violent sideways collision.

1.4.2.7 One reason is that the impulse of the airbag on the child, whose bones are softer can cause more injury than a similar impulse on an adult.

1.4.2.8 (a) Inertia of passenger causes their upper torso to continue to move forwards. The passenger feels as if he is experiencing a force in the opposite direction to the turn – the force on the passenger is actually applied in the direction of turn wherever the passenger is in contact with the structure of the bus – feet, bottom on seat, hand on handrail.

(b) Passenger feels pushed pack into his/her seat as the plane accelerates forwards due to his/her inertia – the seat puts a forwards accelerating force on the passenger.

(c) Inertia of passenger causes body to bend at the knees a little as the lift accelerates upwards because the inertia of the body keeps it in the same position until the upwards force transfers through his/her legs.

(d) Lift 'drops out' from under the passenger as the passenger's inertia keeps them stationary until gravity pulls them down to the lift's floor.

1.4.3.1 (a) There are several ways to state this, for example: The acceleration produced when a force acts on an object of constant mass will be directly proportional to the applied force. (Note that the statement 'net force is equal to the product of mass and acceleration, or $F = ma$' is not a statement of Newton's second law – it is the mathematical expression for it only.)

(b) There are several ways to state this, for example: The net force applied to an object is equal to the rate of change of momentum it produces.

(c) $F = ma$

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1.4.3.2 Answers will vary, for example:

If one engine of a space probe fires, it will produce an acceleration. If a second, identical engine also fires (same direction), then the acceleration will be double that from the one engine alone.

1.4.3.3

A $4\ \text{m s}^{-2}$

B $4\ \text{m s}^{-2}$

C $20\ \text{m s}^{-2}$

D 9 kg

E 9 kg

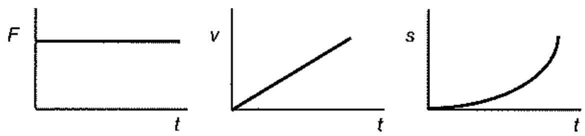
F 4 kg

G 21 N

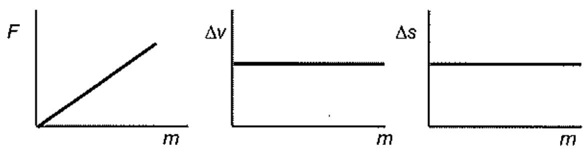
H 7.5 N

I 7 N

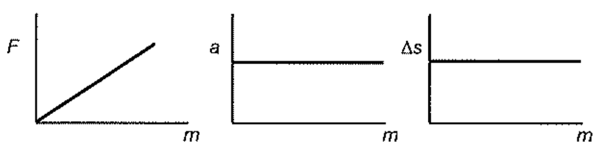
1.4.3.4



1.4.3.5



1.4.3.6



1.4.3.7 An unbalanced force.

1.4.3.8 Accelerating.

1.4.3.9 Speed it up, slow it down or change its direction of travel.

1.4.3.10

(a) A positive force acts in the direction of motion of an object causing it to go faster while a negative force acts against the direction of the motion of the object causing it to go more slowly.

(b) Accelerating from rest, accelerating away from a corner, coasting downhill.

(c) Braking as it approaches a corner or stop sign, coasting uphill.

1.4.3.11 33.335 N west

1.4.3.12 40 kg

1.4.3.13

(a) $2\ \text{m s}^{-2}$ north

(b) $10\ \text{m s}^{-1}$ north

(c) 25 m north

1.4.3.14 15 N (This and next 11 questions use $F = ma$)

1.4.3.15 5 N

1.4.3.16 30 N

1.4.3.17 $15\ \text{m s}^{-2}$

1.4.3.18 $3.09\ \text{m s}^{-2}$

1.4.3.19 $12\ \text{m s}^{-2}$

1.4.3.20 250 kg

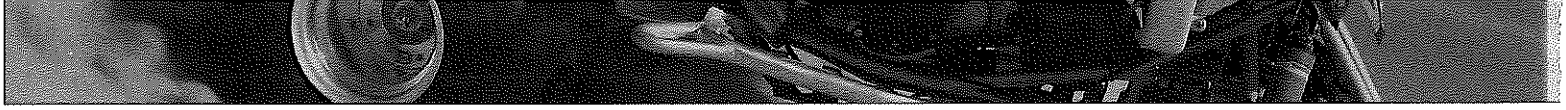
1.4.3.21 128 kg

1.4.3.22 14 N

1.4.3.23 $3\ \text{m s}^{-2}$

1.4.3.24 $5\ \text{m s}^{-2}$

1.4.3.25 210 kg



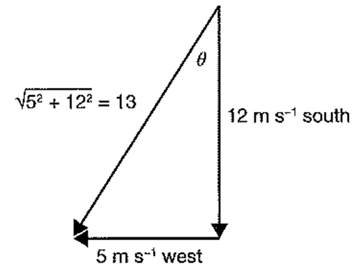
1.4.3.26 (a) $\Delta v = v - u = 12S - 5S = 12S + 5W$

$\phi = \tan^{-1} \frac{5}{12} = 22.6°$

$\Delta v = 13\text{ m s}^{-1}$ bearing 202.6°

(b) Acceleration $= \frac{\Delta v}{t} = \frac{13}{4} = 3.25\text{ m s}^{-2}$ bearing 202.6°

(c) From $F = ma = 2.5 \times 3.25 = 8.125$ N bearing 202.6°



1.4.3.27 (a) $\Delta v = v - u = 80\text{ m s}^{-1}$ north $- 60\text{ m s}^{-1}$ west

$= 80\text{ m s}^{-1}$ north $+ 60\text{ m s}^{-1}$ east

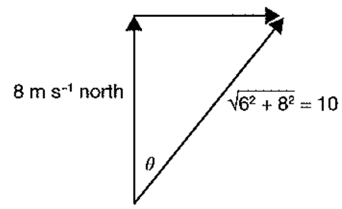
$\phi = \tan^{-1} \frac{6}{8} = 36.9°$

$\Delta v = 10\text{ m s}^{-1}$ bearing 36.9°

(b) Acceleration $= \frac{\Delta v}{t} = \frac{10}{1.25} = 8\text{ m s}^{-2}$ bearing 36.9°

(c) Bearing 36.9° or N 36.9° E

(d) From $m = \frac{F}{a} = \frac{12}{8} = 1.5$ kg



1.4.3.28 (a) $a = \frac{\Delta v}{t} = \frac{32}{0.02} = 1600\text{ m s}^{-2}$ east

(b) $F = ma = 750 \times 1600 = 1.2 \times 10^6$ N east

1.4.3.29 (a) From $a = \frac{\Delta v}{t}$

$t = \frac{\Delta v}{a} = \frac{1600}{40} = 40$ s

(b) From $F = ma = 20\,000 \times 40 = 8 \times 10^5$ N in the direction of the acceleration

(c) As the rocket burns fuel its mass decreases so less force is needed to maintain a constant acceleration, so the force decreases.

1.4.3.30 (a) Acceleration positive means acceleration in the direction of the motion, i.e. east.

From $v = u + at$, therefore $50 = u + (2.5 \times 4)$ from which $u = 40\text{ m s}^{-1}$

(b) Acceleration negative means acceleration in the opposite direction to the motion, i.e. west.

From $v = u + at$, therefore $50 = u + (-2.5 \times 4)$ from which $u = 60\text{ m s}^{-1}$ east

(c) From $F = ma = 500 \times 2.5 = 1250$ N

(d) 1250 N (acceleration and mass the same)

1.4.4.1 For every action there is an equal and opposite reaction.

1.4.4.2 Answers will vary, for example:

If a car crashes into a wall, then the car will put a force on the wall in the direction of its motion and the wall will apply an equal force on the car to oppose its motion.

1.4.4.3 (a) Action – downwards force of the hammer on the nail. Reaction – upwards force of the nail on the hammer.

(b) Action – forwards force of the racquet on the ball. Reaction – backwards force of the ball on the racquet.

(c) Action – downwards force of the boy on the trampoline. Reaction – upwards force of the trampoline on the boy.

(d) Action – forwards force of the gases produced by the explosive charge on the cannonball. Reaction – backwards force of the cannon ball on the gases in the explosion.

(e) Action – backwards force the person's foot puts on the road. Reaction – forwards force the road puts on the person's foot.

(f) Action – forwards force of the boxer's fist on the bag. Reaction – backwards force of the bag on the boxer's fist.

1.4.4.4 (a) From $F = ma = 20 \times 150 = 3000$ N

(b) From weight force $= mg$, $3000 = 9.8m$, therefore $m = 306.1$ kg

(c) Very unlikely. Not many mothers (or fathers) can lift 300 kg!

(d) Children should not be in traveling cars without being in seatbelts. Parents holding children will not be strong enough to protect them in the event of a collision.

(e) Newton's first law (an object will stay at rest or continue to move until an unbalanced force changes its motion). Newton's third law (to every action there is an equal and opposite reaction).

1.4.4.5 The dog shakes the bottle of drink causing much of the dissolved gas to come out of solution and build up pressure inside the bottle. When it takes the cork out of the bottle, the drink rushes out the opening putting an equal and opposite force backwards on the bottle and the baby holding it.

1.4.5.1 (a) From $v = u + at$

$1200 = 0 + 40t$

From which $t = 30$ s

(b) From $F = ma$

$25000 = 40m$

Therefore $m = 625$ kg

(c) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 40 \times 30^2$

Therefore s = 18 000 m (18 km) in the direction of the force.

1.4.5.2 From $F = ma$, $5000 = 500a$

Therefore acceleration = 10 m s^{-2} in the direction of the force.

From $v = u + at$

$v = 0 + 10 \times 8.75 = 87.5$ m s^{-1} in the direction of the force.

1.4.5.3 From $F = ma$, $24 = 0.3a$

Therefore acceleration = 80 m s^{-2} in the direction of the force.

From $v = u + at$

$45 = 10 + 80t$

Therefore t = 0.44 s

1.4.5.4 (a) From $F = ma$, $35 = 5a$

Therefore acceleration = 7 m s^{-2} in the direction of the force.

From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 7 \times 5^2$

Therefore s = 87.5 m in the direction of the force.

(b) From $v = u + at$

$v = 0 + 7 \times 5 = 35$ m s^{-1} in the direction of the force.

1.4.5.5 From $F = ma = \dfrac{m(v-u)}{t}$ for each problem, we get:

(a) $F = \dfrac{9(27-12)}{4} = 33.75$ N west

(b) $F = \dfrac{9(16-(-8))}{8} = 27$ N south

(c) $F = \dfrac{9(35-(-10))}{2.5} = 162$ N north

(d) $F = \dfrac{9(12-(-72))}{4} = 189$ N north

1.4.5.6 (a) From $v = u + at$

$32 = 0 + 9a$

Therefore acceleration = 3.56 m s^{-2} in the direction of the force.

(b) From $F = ma$

$6.4 = 3.56m$

Therefore mass of body = 1.8 kg

1.4.5.7 (a) From $F = ma$

$18 = 0.6a$

Therefore a = 30 m s^{-2} in the direction of the force.

Then from $v = u + at$

$60 = 10 + 30t$

From which $t = 1.67$ s

(b) From $s = ut + \frac{1}{2}at^2$

$s = 10 \times 1.67 + 0.5 \times 30 \times 1.67^2$

Therefore s = 58.5 m in the direction of the force.

1.4.5.8 From $F = ma$

$15 = 2.5a$

Therefore $a = 6$ m s^{-2} in the direction of the force.

From $v = u + at$

$v = 0 + 6 \times 0.6 = 3.6$ m s^{-1} in the direction of the force.

1.4.5.9 (a) From $F = ma$

$6 = 1.5a$

Therefore $a = 4$ m s^{-2} in the direction of the force.

From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 4 \times 9^2$

Therefore $s = 162$ m in the direction of the force.

(b) From $v = u + at$

$v = 0 + 4 \times 9 = 36$ m s^{-1} in the direction of the force.

1.4.5.10 (a) From $F = ma$

$3.6 = 0.3a$

Therefore $a = 12$ m s^{-2} in the direction of the force.

Then from $v = u + at$

$5 = 0.2 + 12t$

From which $t = 0.4$ s

(b) From $s = ut + \frac{1}{2}at^2$

$s = 0.2 \times 0.4 + 0.5 \times 12 \times 0.4^2$

Therefore $s = 1.04$ m in the direction of the force.

1.4.5.11 (a) From $v = \frac{s}{t} = \frac{900}{45} = 20$ m s^{-1}

(b) From $v = u + at$

$20 = 0 + 9a$

So acceleration = 2.22 m s^{-2} in the direction of the motion.

(c) From $v = u + at$

$0 = 20 + 5a$

So deceleration = 4 m s^{-2} against the direction of the force.

(d) Distance travelled while accelerating, from $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 2.22 \times 9^2 = 89.91$ m

While decelerating $s = 20 \times 5 - 0.5 \times 4 \times 5^2 = 50$ m

Therefore total distance between bus stops = 89.91 + 900 + 50 = 1039.91 m

1.4.5.12 (a) $a = \frac{v - u}{t} = \frac{4}{1.2} = 3.33$ m s^{-2}

(b) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 3.33 \times 1.2^2 = 2.4$ m in the direction of the force.

1.4.5.13 (a) From $a = \frac{F}{m} = \frac{75}{15} = 5$ m s^{-2} in the direction of the force.

(b) From $v = u + at$

$v = 0 + 5 \times 4 = 20$ m s^{-1} in the direction of the force.

(c) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 5 \times 4^2 = 40$ m in the direction of the force.

1.4.5.14 (a) From $v = u + at$

$18 = 0 + 9a$

Therefore $a = 2$ m s^{-2} in the direction of the force.

(b) From $m = \frac{F}{a} = \frac{4.5}{2} = 2.25$ kg

(c) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 2 \times 9^2 = 81$ m in the direction of the force.

1.4.5.15 (a) From $v = u + at$

$1200 = 0 + 30t$

So $t = 40$ s

(b) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 30 \times 40^2 = 24\ 000$ m in the direction of the force.

(c) From $m = \frac{F}{a} = \frac{25000}{30} = 833.33$ kg

1.4.5.16 (a) From km h^{-1} = m s^{-1} × 3.6

30 m s^{-1} = 108 km h^{-1}

So car is 28 km h^{-1} over the speed limit.

(b) From $v = u + at$

$0 = 30 + 12a$

So deceleration of the car = 2.5 m s^{-2}

(c) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 2.5 \times 12^2 = 180$ m in the direction of the motion.

1.4.5.17 (a) From $v^2 = u^2 + 2as$

$0 = 16^2 + 2 \times a \times 120$

Therefore $a = 1.067$ m s^{-2} against the motion.

(b) From $v = u + at$

$0 = 16 - 1.07t$

Therefore $t = 14.95$ s

(c) Car will move $s = 16 \times 14.95 = 239.2$ m

Therefore car is 239.2 – 120 = 119.2 m ahead of the hubcap.

1.4.5.18 (a) From $a = \frac{F}{m} = \frac{45}{25} = 1.8$ m s^{-2} in the direction of the force.

(b) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 1.8 \times 5^2 = 22.5$ m in the direction of the force.

(c) From $v = u + at$

$v = 0 + 1.8 \times 9 = 16.2$ m s^{-1} in the direction of the force.

1.4.5.19 (a) From $v = u + at$

$36 = 0 + 120a$

Therefore $a = 0.3$ m s^{-2} in the direction of the motion.

(b) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 0.3 \times 120^2 = 2160$ m in the direction of the motion.

(c) From $v = u + at$

$36 = 0 + 16a$

Therefore $a = 2.25$ m s^{-2} against the motion.

(d) From $s = ut + \frac{1}{2}at^2$

$s = 36 \times 16 - 0.5 \times 2.25 \times 16^2 = 288$ m in the direction of the motion.

(e) From km h^{-1} = m s^{-1} × 3.6

36 m s^{-1} = 129.6 km h^{-1}

So car is 29.6 km h^{-1} over the speed limit.

1.4.5.20 (a) From $v = u + at$

$v = 9 + 3.5 \times 6 = 30$ m s^{-1} east, then

From $v = u + at$

$v = 30 + 2 \times 4 = 38$ m s^{-1} east

So final velocity of the car is 38 m s^{-1} east.

(b) From $s = ut + \frac{1}{2}at^2$

$s = 9 \times 6 + 0.5 \times 3.5 \times 6^2 = 117$ m east, then

$s = 30 \times 4 + 0.5 \times 2 \times 4^2 = 136$ m east.

So total distance travelled = 117 + 136 = 253 m east.

(c) From $v = u + at$

$38 = 9 + 10a$

Therefore $a = 2.9$ m s^{-2} east

1.4.5.21 (a) Distance travelled by car = $vt = 20t$

Distance travelled by police car = $15t + 0.5 \times 1.5 \times t^2$

Since these distances will be equal when the police catch the car

$20t = 15t + 0.75t^2$

Therefore $5 = 0.75t$

So time for police to catch the car = 6.67 s.

(b) Distance each travelled by car in this time = 20 × 6.67 = 133.3 m

(Check: distance police travel = $15 \times 6.67 + 0.5 \times 1.5 \times 6.67^2 = 133.4$ m) – rounding off errors

1.4.5.22 (a) From $v = u + at$

Maximum speed = 0 + 1.4 × 5 = 7 m s^{-1}

(b) From $v = u + at$

$0 = 7 + 7.5a$

Therefore deceleration = 0.93 m s^{-2}

(c) Distance travelled while accelerating = $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 1.4 \times 5^2 = 17.5$ m

Distance travelled while decelerating = $s = ut + \frac{1}{2}at^2$

$s = 7 \times 7.5 - 0.5 \times 0.93 \times 7.5^2 = 26.34$ m

Therefore, distance travelled at constant speed = 150 – 17.5 – 26.34 = 106.16 m

1.4.5.23 (a) From $v^2 = u^2 + 2as$

$7.5^2 = 0 + 2 \times a \times 125$

Therefore $a = 0.225$ m s^{-2} north

(b) From $v = u + at$

$7.5 = 0 + 0.225t$

Therefore $t = 33.33$ s

(c) From $v^2 = u^2 + 2as$

$7.5^2 = 0 + 2 \times a \times 60$

Therefore $a = 0.47$ m s^{-2} south

(d) From $v = u + at$

$0 = 7.5 - 0.47t$

Therefore $t = 15.9$ s (rounding off error)

1.4.5.24 (a) From $v = u + at$

$v = 0 + 4.9 \times 3 = 14.7$ m s^{-1}

(b) From $v = u + at$

$24.1 = 0 + 4.9t$

Therefore t = 4.92 s

(c) From $v^2 = u^2 + 2as$

$24.1^2 = 0 + 2 \times 4.9 \times s$

Therefore $s = 59.3$ m in the direction of the motion.

(d) Sin 30° = 0.5

$4.9 = 0.5g$

Therefore, $a = g \sin \theta$

1.4.5.25 (a) From $v = u + at$

$72 = 0 + 4a$

Therefore $a = 18$ m s^{-2} in the direction of the force.

(b) From $m = \frac{F}{a} = \frac{18}{4} = 4.5$ kg

(c) From $s = ut + \frac{1}{2}at^2$

$s = 0 + 0.5 \times 18 \times 4^2 = 144$ m in the direction of the force.

1.4.6.1 (a) 3 m s^{-2} right

(b) A = 18 N, B = 27 N (both to right)

(c) 27 N right

(d) 27 N left

1.4.6.2 (a) 3 m s^{-2} left

(b) A = 18 N, B = 27 N (both to left)

(c) 18 N right

(d) 18 N left

1.4.6.3 A (a) 5 m s^{-2} right

(b) Both 5 m s^{-2} right

(c) X = 30 N right, Y = 40 N right

(d) 40 N (both directions)

B (a) 6.5 m s^{-2} right

(b) Both 6.5 m s^{-2} right

(c) X = 26 N right, Y = 39 N left

(d) 39 N both ways

C (a) 4 m s^{-2} right

(b) Both 4 m s^{-2} right

(c) X = 12 N right, Y = 36 N right

(d) 36 N (both directions)

D (a) 5 m s^{-2} left

(b) Both 5 m s^{-2} left

(c) X = 25 N left, Y = 10 N left

E (a) 7.5 m s^{-2} left

(b) All 7.5 m s^{-2} left

(c) X = 22.5 N left, Y = 39.5 N left, Z = 15 N left

(d) XY = 22.5 N both ways, YZ = 60 N both ways

F (a) 4 m s^{-2} left

(b) 4 m s^{-2} left

(c) X = 32 N left, Y = 20 N left

(d) 32 N (both directions)

G (a) 3 m s^{-2} right

(b) All 3 m s^{-2} right

(c) X = 12 N right, Y = 18 N right, Z = 24 N right

(d) XY = 42 N both ways, YZ = 24 N (both)

H (a) 3 m s^{-2} right

(b) All 3 m s^{-2} right

(c) W = 15 N right, X = 21 N right, Y = 18 N right, Z = 9 N right

(d) WX = 48 N, XY = 27 N, YZ = 9 N (all both directions)

1.4.7.1 A (a) 4 m s^{-2} to the left

(b) Both 4 m s^{-2} to the left

(c) P = 480 N, Q = 320 N both to the right

(d) 320 N both ways

B (a) 3 m s^{-2} to the right

(b) Both 3 m s^{-2} to the right

(c) P = 450 N, Q = 750 N both to the right

(d) 450 N both ways

C (a) 2.0 m s^{-2} to the left

(b) All 2.0 m s^{-2} to the left

(c) P = 120 N, Q = 60 N, R = 180 N all to the left

(d) String 1 = 240 N both ways, string 2 = 180 N both ways

D (a) 8 m s^{-2} to the right

(b) All 8 m s^{-2} to the right

(c) P = 64 N, Q = 48 N, R = 56 N, all to the right

(d) String 1 = 112 N both ways, string 2 = 64 N both ways

E (a) 1.5 m s^{-2} to the right

(b) All 1.5 m s^{-2} to the right

(c) P = 6 N, Q = 4.5 N, R = 18 N, S = 9 N all to the right

(d) String 1 = 28.5 N, string 2 = 10.5 N, string 3 = 6 N all both ways

F (a) 0.5 m s^{-2} to the left

(b) All 0.5 m s^{-2} to the left

(c) P = 9 N, Q = 20 N, R = 6 N, S = 8 N all to the left

(d) String 1 = 34 N, string 2 = 14 N, string 3 = 8 N all both ways

1.4.7.2 A (a) Acceleration = $\frac{\text{net force}}{\text{total mass}} = \frac{10 \times 9.8}{18} = 5.44$ m s^{-2}

(b) On the 10 kg, force = ma = 10 × 5.44 = 54.4 N down

On the 8 kg = force = ma = 8 × 5.44 = 43.52 N right

(c) Considering X, tension = accelerating force = ma = 8 × 5.44 = 43.52 N (acting both ways in the string)

B (a) Acceleration = $\frac{\text{net force}}{\text{total mass}} = \frac{2 \times 9.8}{18} = 1.09$ m s^{-2}

(b) On the 2 kg, force = ma = 2 × 1.09 = 2.18 N down

On the 16 kg = force = ma = 16 × 1.08 = 17.28 N right

(c) Tension = force accelerating X = 17.28 N both ways

C (a) Acceleration = $\frac{\text{net force}}{\text{total mass}} = \frac{9 \times 9.8}{17} = 5.2$ m s^{-2}

(b) For W, net force = ma = 8 × 5.2 = 41.6 N left

For X, net force = ma = 4 × 5.2 = 20.8 N down

For Y, net force = ma = 3 × 5.2 = 15.6 N down

For Z, net force = ma = 2 × 5.2 = 10.4 N down

(c) For string 1, tension = force accelerating W = 41.6 N both ways

For string 2, tension = $mg - ma$ = (5 × 9.8) – (5 × 5.2) = 23 N both ways

For string 3, tension = $mg - ma$ = (2 × 9.8) – (2 × 5.2) = 9.2 N both ways

D (a) Acceleration = $\frac{\text{net force}}{\text{total mass}} = \frac{2 \times 9.8}{18} = 1.09$ m s^{-2}

(b) For X, net force = ma = 9 × 1.09 = 9.81 N right

For Y, net force = ma = 7 × 1.09 = 7.63 N right

For Z, net force = ma = 2 × 1.09 = 2.18 N down

(c) For string 1, tension = force accelerating x = 9.81 N both ways

For string 2, tension = force accelerating x and Y = (16 × 1.09) = 17.4 N both ways

E (a) Acceleration = $\frac{\text{net force}}{\text{total mass}} = \frac{6 \times 9.8}{14} = 4.2$ m s^{-2}

(b) For X, net force = ma = 6 × 4.2 = 25.2 N down

For Y, net force = ma = 8 × 4.2 = 33.6 N left

(c) For string, tension = force accelerating Y = 33.6 N both ways

F (a) Acceleration = $\frac{\text{net force}}{\text{total mass}} = \frac{8 \times 9.8}{13} = 6.03 \text{ m s}^{-2}$

(b) For X, net force = $ma = 5 \times 6.03 = 30.15$ N right

For Y, net force = $ma = 5 \times 6.03 = 30.15$ N down

For Z, net force = $ma = 3 \times 6.03 = 18.09$ N down

(c) For string 1, tension = force accelerating X = 30.15 N both ways

For string 2, tension = $mg - ma = (3 \times 9.8) - (3 \times 6.03) = 11.31$ N both ways

G (a) Acceleration = $\frac{\text{net force}}{\text{total mass}} = \frac{2 \times 9.8}{28} = 0.7 \text{ m s}^{-2}$

(b) For X, net force = $ma = 16 \times 0.7 = 11.2$ N right

For Y, net force = $ma = 10 \times 0.7 = 7$ N right

For Z, net force = $ma = 2 \times 0.7 = 1.4$ N down

(c) For string 1, tension = force accelerating X = 11.2 N both ways

For string 2, tension = force accelerating X and Y $26 \times 0.7 = 18.2$ N both ways

1.4.7.3 In each question, taking the top string as string 1, the middle string as string 2 and the bottom string as string 3:

A From $T = mg + ma$ for the total mass suspended below each string (inertial force adds to weight force) we get

String 1 = 192 N

String 2 = 115.2 N

String 3 = 64 N

B From $T = mg - ma$ for the total mass suspended below each string (inertial force opposes weight force) we get

String 1 = 101.4 N

String 2 = 85.8 N

String 3 = 23.4 N

C From $T = mg$ for the total mass suspended below each string (acceleration = 0) we get

String 1 = 117.6 N

String 2 = 58.5 N

String 3 = 19.6 N

D From $T = mg - ma$ for the total mass suspended below each string we get

String 1 = 53.2 N

String 2 = 41.8 N

String 3 = 15.2 N

E From $T = mg$ for the total mass suspended below each string (acceleration = 0) we get

String 1 = 98 N

String 2 = 68.6 N

String 3 = 49 N

F From $T = mg + ma$ for the total mass suspended below each string we get

String 1 = 207 N

String 2 = 138 N

String 3 = 82.8 N

1.4.7.4 (a) Show gravity down on each mass, and tension in each side of the string acting both ways as shown in the diagram.

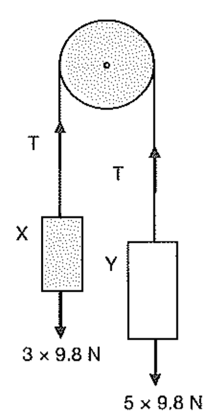
(b) From $a = \frac{\text{net force}}{\text{total mass}} = 2.45 \text{ m s}^{-2}$ Y moving down

(c) Force on X = $ma = 3 \times 2.45 = 7.35$ N up

Force on Y = $ma = 5 \times 2.45 = 12.25$ N down

(d) Considering X, $T = mg + ma = 29.4 + (3 \times 2.45) = 36.75$ acting both ways in string

(Check: considering Y, $T = mg - ma = 49 - 12.25 = 36.74$ N)



1.4.7.5 (a) Reading = $mg = 3 \times 9.8 = 29.4$ N

(b) Reading = $mg = 3 \times 9.8 = 29.4$ N

(c) Reading = $mg = 3 \times 9.8 = 29.4$ N

(d) Reading = $mg + ma = 29.4 + 6 = 35.4$ N

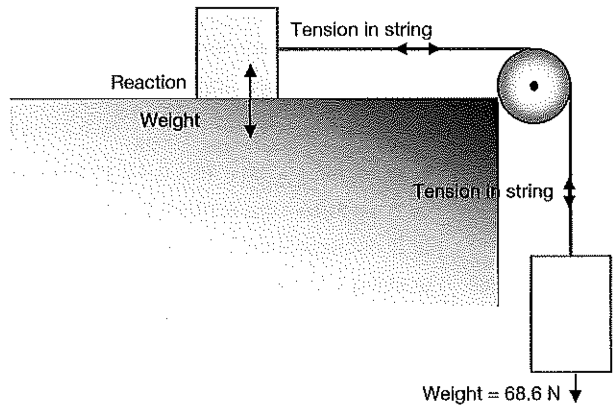
(e) Reading = $mg - ma = 29.4 - 6 = 23.4$ N

1.4.7.6 (a) Stopped or moving with constant velocity.

(b) Accelerating forwards.

(c) Accelerating backwards or braking.

1.4.7.7 (a) Show gravity down on each mass, reaction force up on mass x and tension in both bits of the string acting in both directions.

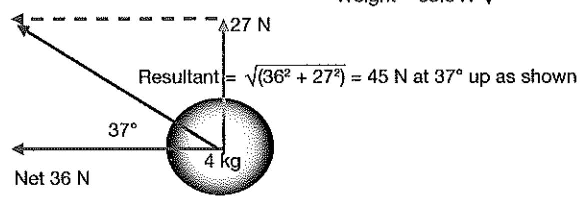


(b) $a = \dfrac{\text{accellerating force}}{\text{total mass}} = \dfrac{7 \times 9.8}{10}$
$= 6.86$ m s^{-2} with Y moving down.

(c) Force on X = ma
$= 3 \times 6.86 = 20.58$ N right
Force on Y = ma
$= 7 \times 6.86$ N down $= 48.02$ N down

(d) Tension = force pulling X = 20.58 N acting in both directions.

1.4.7.8 (a) 45 N at 37° up from original 45 N force.



(b) From $F = ma$, $a = \dfrac{F}{m} = $ 45 N bearing $\dfrac{307}{4}$
$= 11.25$ m s^{-2} in same direction as net force.

1.4.7.9 (a) $T = mg = 98$ N, 58.8 N

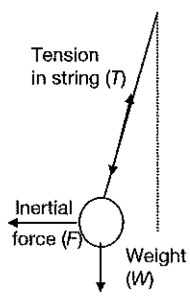
(b) $T = mg = 98$ N, 58.8 N

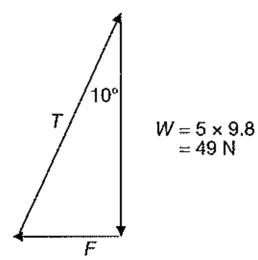
(c) $T = mg = 98$ N, 58.8 N

(d) From $T = mg + ma$, 128 N, 76.8 N

(e) From $T = mg - ma$, 68 N, 40.8 N

1.4.7.10 (a) Weight force vertically down, tension in string (acting both ways) and a third force (= ma) horizontally to the left (pulling the ball aside).





(b) From the vector diagram
$F = 49 \tan 10° = 8.64$ N

(c) $a = \dfrac{F}{m} = \dfrac{8.64}{5} = 1.73$ m s^{-2}

(d) From $\tan\theta = \dfrac{F}{49} = \dfrac{5 \times 2.5}{49}$
$\theta = 14.31°$ towards the front of the bus.

1.5.1.1 Momentum is the product of the mass and velocity of an object. It is a vector quantity and measure in kg m s^{-1} or (through impulse) N s.

1.5.1.2 In any isolated system (no external forces acting) the total momentum of the system always remains constant.

1.5.1.3 The impulse of a force is equal to the product of the force and the time it acts.

1.5.1.4 The impulse of a force is equal to the change in momentum it causes in the object it acts on.

1.5.1.5 Impulse = $Ft = mat$ = kg m s^{-2} s = kg m s^{-1} = mv = momentum

1.5.1.6 The momentum of an isolated system is always conserved.

1.5.1.7 From Newton's third law, during a collision, the force on object A is equal and opposite to the force on object B, i.e. $F_A = -F_B$.

Because the time involved is the same for each object, this means that the impulse on object A is equal and opposite to the impulse on object B.

That is, $(Ft)_A = -(Ft)_B$ or $I_A = -I_B$

Since impulse applied is equal to the change in momentum, we can see that the change of momentum of A is equal and opposite the change in momentum of B.

That is, $\Delta p_A = -\Delta p_B$. Therefore $\Delta p_A + \Delta p_B = 0$, therefore momentum is conserved.

1.5.1.8 (a) 1.12×10^4 N s east

(b) 6.89×10^{-23} N s west

(c) 22 000 N s south

(d) 30 N s north

(e) 712.5 N s south

1.5.1.9 Answers to (f) included alongside (a) to (e).

(a) 5 N s W (1.25 N W)

(b) 15.6 N s E (3.9 N E)

(c) 27 N s N (6.75 N N)

(d) 5 N s N (1.25 N N)

(e) 7.5 N s E (1.875 N E)

1.5.1.10 (a) 81 N s E

(b) 81 N s E

(c) 9 N s W

(d) 1.5 m s^{-1} W

1.5.1.11 (a) 144 N s

(b) 48 m s^{-1}

1.5.1.12 (a) 20 m s^{-2} east

(b) 16 N s east

(c) 40 m s^{-1} east

(d) 18 m s^{-1} east

1.5.1.13 (a) 12 N s backwards

(b) 12 N s backwards (Remember that impulse is equal to the change in momentum of the person – the initial momentum will be the same regardless of the time of the collision. The *average* force acting on her will change, but not the impulse.)

(c) 0.24 m s^{-1}

1.5.1.14 (a) 51 kg m s^{-1} away from the wall

(b) 51 kg m s^{-1} onto the wall

(c) 340 N away from the wall

1.5.1.15 (a) 800 N east

(b) 400 N east

(c) 4 N s east

(d) 80 m s^{-1} east

1.5.1.16 (a) 25 N s east

(b) Approximately 56 N s east

(c) 10 m s^{-1} east

(d) 22.4 m s^{-1} east

(e) (i) Approximately 46 to 47 m s^{-1} east

(ii) Approximately 58 to 59 m s^{-1} east

(iii) Approximately 28 to 29 m s^{-1} east

1.5.2.1 (a) 24 N s in direction of force

(b) 24 N s in direction of force

(c) 240 N

1.5.2.2 (a) 0.03 N s onto the lounge

(b) 0.03 N s onto the car

(c) 0.1 N onto the lounge

(d) 0.1 N onto the car

(e) Law of conservation of momentum

1.5.2.3 (a) 30 m s^{-1} away from car

(b) 15 N s away from car

(c) 300 N

(d) 1500 N

1.5.2.4 0.21 m s^{-1}

1.5.2.5 When the truck hits the wall it puts an impulse on the wall. This, while not collapsing the wall, will cause it to move (vibrate, shake) a little. Hence the momentum of the truck transfers to the wall and then into molecular motion of air particles in contact with the wall. The wall puts an equal but opposite impulse on the truck. This stops the truck.

1.5.2.6 2.0 m s^{-1} north

1.5.2.7 6500 kg

1.5.2.8 Car continues forwards at 33.3 km h^{-1}

1.5.2.9 The bike rebounds at 4.0 m s^{-1}

1.5.2.10 2.67 m s^{-1}

1.5.2.11 9.45 cm s^{-1}

1.5.2.12 0.26 m s^{-1}

1.5.3.1 The collision of the car with the plastic bumper (which will crumple on impact) takes longer than an identical collision involving the metal bumper. This longer impact time means that the force of collision is less and so the inertial effects of the collision on the passenger will be reduced. Less injury will therefore be a probable result.

1.5.3.2 (a) If you are travelling more slowly, then your momentum is less, so in the event of a collision, the impulse on you due to the collision will be less, both because you don't have as much momentum to lose and also because the time of the collision will probably be shorter.

(b) During a collision, as your inertia causes you to continue to move forwards, the airbag ensures that you lose your momentum slowly, by making the time of collision with the bag much longer than it would be if you hit the windscreen. Therefore, while the impulse will be the same, the force involved will be less.

(c) Crumple zones in vehicles are designed to increase the time of the collision and so while the momentum lost (the impulse imparted to the car and you) is the same, the forces involved will be smaller.

1.5.3.3 The racing harness stops the driver's body being 'thrown' sideways during a collision, especially if the car rolls. A lap-sash belt is not as good because the driver could be 'thrown' out of the belt in a violent sideways collision.

1.5.3.4 One reason is that the impulse of the airbag on the child, whose bones are softer, can cause more injury than a similar impulse on an adult.

1.5.3.5 Crumple zones are regions of deliberate weakness built into the structural members of cars so that when they collide with something, the structural members absorb a proportion of the energy of the crash and bend or crumple. This results in the momentum of the car being lost over a longer period of time and so reduces the maximum force on the car and its occupants. Less injury will therefore be a possibility.

1.5.3.6 Crumple bars, airbags, plastic components instead of all steel, seatbelts, engines that move downwards under the driver instead of straight back into the driver. These are all beneficial as they are all designed to reduce the forces impacting on passengers in vehicles either by extending the time of the collision or, in the case of the engine movement, totally avoiding an impact.

1.5.3.7 A = chicanes
B = low speed school zones
C = speed bumps

All increase safety of drivers and pedestrians by reducing the speeds at which vehicles travel, which in turn provides more time for drivers and pedestrians to react in times of emergency as well as reducing the impulses involved in collisions because the momentum of the vehicle is less.

1.6.1.1 Answers will vary, for example:

Work is done whenever a net force moves an object in the direction of the force.

1.6.1.2 Work is done whenever the energy of an object is changed.

1.6.1.3 The person throwing the ball does work on it to overcome the downwards force of gravity. The ball falls back down as gravity does work on it.

1.6.1.4 The formula is not appropriate at high altitudes because the value of the acceleration due to gravity, '*g*', changes significantly when displacements are large. The formula would be too inaccurate.

1.6.1.5 (a) The force/person doing the stretching.

(b) The restoring force due to interatomic forces of attraction between atoms in the material of the spring.

1.6.1.7 For example: When an object falls, gravity does work on it pulling it towards the ground.

1.6.1.8 For example: When an electric heater is turned on, electrical energy is converted into heat energy by the resistance of the heating coils. Work is done by the material in the coils.

1.6.1.9 From $W = Fs = 15 \times 6 = 90$ J

1.6.1.10 From $W = Fs = 150 \times 6000 = 900\,000$ J

1.6.1.11 (a) From $v^2 = u^2 + 2as$

$0 = 2^2 + 24a$

From which $a = -0.167$ m s^{-2} (against the motion)

From $W = Fs = ma, s = 0.8 \times 0.167 \times 12 = 1.6$ J

(b) $F = ma = 0.8 \times 0.167 = 0.13$ N against the motion

1.6.1.12 (a) From $F_c = \frac{mv^2}{s} = \frac{45 \times 3^2}{15}$ = 27 N towards the centre of the merry-go-round.

(b) Zero

(c) Zero

(d) There is no change in displacement in the direction in which the force is acting, therefore from $W = Fs$, with $s = 0$, no work is done by the force.

1.6.1.13 (a) 25 : 36 : 64 : 100. Note that no calculations need to be made here since cars are identical (i.e. equal masses), so the ratio of their kinetic energies is simply their velocities squared (regardless of units used).

(b) 25 : 36 : 64 : 100. Work done is equal to the change in energy of the objects involved. All cars lose all their *KE*.

(c) 5 : 6 : 8 : 10. Since their masses are equal, impulse = change in momentum (not energy), and $\Delta p = m(v - u)$ depends only on velocity change.

(d) Same ratio but forces all × 5.

(e) 5 : 6 : 8 : 10. Since $W = Fs = \Delta KE$, then the ratio of the work done must equal the ratio of Fs.

Therefore 25 : 36 : 64 : 100 = 5s : 6s : 8s : 10s. So ratio of stopping distances is also 5 : 6 : 8 : 10.

1.6.1.14 (a) Work done against friction = 6000 N

(b) Work is done against gravity by lifting the car through the equivalent of 2 m vertical displacement

$= mgh = 800 \times 9.8 \times 2 = 15\,680$ J, plus work done against friction (6000 N)

So total work done = 21 680 J

1.6.1.15 (a) Impulse = change in momentum = 20 N s north

(b) From $I = m(v - u)$

$20 = 0.6v$

Therefore, $v = 33.3$ m s^{-1} north

(c) From $I = F_{av}t$

$20 = F \times 2$

Therefore average force = 10 N north

(d) 16.67 m s^{-2} north

(e) 33.3 m

(f) 33.3 m north

(g) 333 J

1.6.1.16 Component of the force in the horizontal direction = 4500 cos θ

So work done by force = 4500 cos $\theta \times 28$ = 10 9119 J

1.6.2.1 (a) 6.0 J

(b) 2.94 J

(c) 300 N

(d) 1.5 J

1.6.2.2 (a) 1.25 J

(b) 510 g

1.6.2.3 Approximately 4.4×10^{10} J

1.6.2.4 (a) The unit for energy is the joule, which, from $GPE = mgh$ = kg m^2 s^{-2}.

The area under the graph = acceleration × distance = m^2 s^{-2}.

So, area under graph multiplied by mass = kg m^2 s^{-2} = joule = energy

(b) Approximately 346 J

(c) 6.7 m s^{-1}

1.6.2.5 (a) 60 000 J

(b) 48 000 J

(c) The energy of the cart does not change, so no work is done on it.

(d) Work done by the horses has been transferred to the wheels of the cart and the road in overcoming frictional forces.

1.6.2.6 (a) 66.67 N m^{-1}

(b) 40 J

1.6.2.7 (a) Work done = area under graph = $0.5 \times 3 \times 12 = 18$ N

(b) KE gained = work done = area = $(0.5 \times 3 \times 12) + (3 \times 12) + (0.5 \times 4 \times 12) = 78$ J

1.6.2.8 (a) from $F = ma$, $a = \frac{F}{m} = \frac{3}{1.5} = 2$ m s^{-2} east

(b) Work done = area under graph = $5 + 6 + (0.5 \times 3 \times 1) = 12.5$ J

(c) Work done = change in KE = 12.5 J

(d) From KE gained = $\frac{1}{2}mv^2 = 12.5$, $v = \sqrt{2 \times 12.5 \div 1.5} = 4.08$ m s^{-1} east

1.6.3.1 Power is a measure of the rate at which work is done or at which energy is changed into other forms. Power is measured in J s^{-1} or watts.

1.6.3.2 Yes, because by definition, power is a measure of the rate at which a device or force does work. If they do the same work at the same time then they develop the same power.

1.6.3.3 Y. X develops 3.33 W

1.6.3.4 20 W

1.6.3.5 Work done measures the total energy change in a system or done by a machine whereas power measures the energy change or work done per second.

1.6.3.6 20 000 W

1.6.3.7 1531.25 W

1.6.3.8 (a) 1080 W

(b) 3240 J

1.6.3.9 (a) 144 N s

(b) 48 m s^{-1}

(c) 216 W

1.6.3.10 (a) 25 N s

(b) 117.5 N s

(c) 47 m s^{-1}

(d) 345.16 W

1.6.4.1 Kinetic energy is energy possessed by an object (or a particle of matter) due to its mass and its velocity.

1.6.4.2 Gravitational potential energy is energy possessed by a mass due to its position in a gravitational field.

1.6.4.3 Answers will vary, for example: Energy cannot be created nor destroyed, but can be changed from one form to another.

1.6.4.4 The change in E_p of each block = $mg\Delta h$.

This is the same for each block.

From the law of conservation of energy, this E_p will change into E_K as the blocks slide down the inclines, so the E_K of each block at the bottom of the incline (or at any equal height down the incline) will be equal. Therefore, for each block: $mg\Delta h = \frac{1}{2}mv^2$

Rearranging $v^2 = 2g\Delta h$

Hence, the velocity of the block is independent of its mass and the length of the incline.

1.6.4.5 (a) 4.04×10^4 J

(b) 187.5 J

(c) 38.25 m s^{-1}

1.6.4.6 (a) 7.35×10^4 J

(b) 9.9 m s^{-1}

(c) 6.25×10^5 N m^{-1}

1.6.4.7 (a) 14.7 m

(b) 16.97 m s^{-1}

(c) 13.8 m s^{-1}

1.6.4.8 (a) A

(b) A

(c) C

(d) B

(e) D

(f) E

1.6.4.9 (a) 22.4 J

(b) 15 m s^{-1}

1.6.4.10 (a) 12.6 cm

(b) 81.67 N m^{-1}

(c) 0.64 J

(d) 9.77 cm

1.6.4.11 No. The law of conservation of energy is always followed. The problem here is that we cannot just consider the ball. We must also consider other objects in the *system* – that is, the court. In this situation the collision between the ball and the court is inelastic and some of the kinetic energy of the ball has been transferred to the court surface as heat energy, or dissipated into the air as sound associated with the collision. The ball may also not have regained its initial shape completely. Some energy may be absorbed (as molecular bonding energy) by the distortion of its shape.

1.6.4.12 (a) 3.6×10^{-2} J

(b) 6.9×10^{-2} m

(c) 4.9×10^{-1} m s^{-1}

1.6.4.13 Velocity of 100 g glider is 0.16 m s^{-1} rebound and velocity of 200 g glider is 0.333 m s^{-1} in original direction of the 100 g glider.

1.6.4.14 (a) 1125 J

(b) 4.92 m s^{-1}

(c) 1.24 m

(d) 99.98%

(e) Dissipated into the air as heat and sound energy, absorbed by the wooden block as it deforms and its structure changes as the bullet penetrates it.

1.6.4.15 The girl does work on the ball to throw it into the air giving it kinetic energy which changes to gravitational potential energy as it rises. As it rises gravity does work on it slowing and stopping its rise (although it still keeps moving horizontally), then accelerates it downwards towards the catcher. The girl catching the ball does work on it to stop it.

1.6.4.16 (a) 1.25 kg m s^{-1} towards the floor

(b) 1.20 kg m s^{-2} away from the floor

(c) No, momentum is conserved in all collisions. In this situation the momentum transferred from the ball to the floor has not been taken into account.

(d) 1.56 J

(e) 1.44 J

(f) No. Energy must be conserved in all situations. In this case, some of the kinetic energy has been converted into sound energy as the ball bounces, into heat energy as the ball deforms and into molecular motion of air particles as the ball collides and displaces them. What has not been conserved is the ball's kinetic energy.

1.6.4.17 Gravity does work on the bungee jumper converting gravitational potential energy into kinetic energy as he falls. When the bungee strap is taut the force of the jumper's weight (gravity) stretches it and the jumper's kinetic energy is converted into elastic potential energy in the strap. The restoring force in the stretched strap slows the jumper down then pulls him upwards again doing work against gravity. This gives the jumper kinetic energy (moving upwards) which changes to gravitational potential energy as he rises then this process repeats as he bobs up and down until at rest hanging on the end of the strap.

1.6.4.18 Kinetic energy of the car is partly absorbed by the tree as the car crashes into it and is partly absorbed by parts of the car as they bend and break and crumple, converting to heat and sound energy and kinetic energy of the particles in the matter of the car. The energy eventually dissipates into the surrounding air as increased molecular motion (the air in the vicinity of the crash warms slightly).

1.6.4.19 (a) 2402.5 J

(b) 2.58 m s^{-1}

(c) 20 J

(d) 99.16%

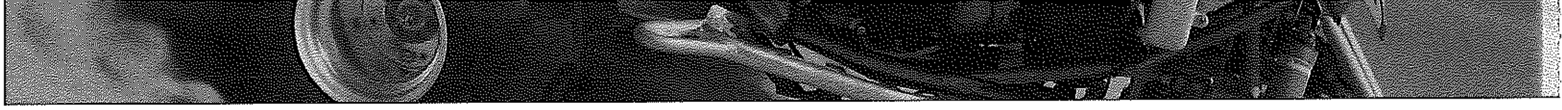
(e) Changed into sound energy during the collision, into deformation of the block as the bullet penetrates it and into gravitational potential energy as the block rises.

(f) 0.34 m

1.7.1.1 Energy can not be created nor destroyed but can be changed from one form to another.

1.7.1.2 The energy of an isolated system is always conserved.

1.7.1.3 The kinetic energy of an isolated system is conserved only if any collisions occurring within the system are elastic collisions.



1.7.1.4 An elastic collision is one in which both momentum and kinetic energy are conserved while in an inelastic collision only momentum is conserved.

1.7.1.5 (a) Car continues forwards at 33.3 km h^{-1} or 9.26 m s^{-1}.

(b) Collision is inelastic. Initial *KE* = 1.67×10^5 J, final = 1.01×10^5 J.

1.7.1.6 (a) 6500 kg

(b) Collision must be inelastic because objects couple (initial *KE* = 18 750 J, final = 9000 J).

1.7.1.7 (a) The bike rebounds at 4.0 m s^{-1}

(b) Collision is inelastic. Initial *KE* is 21 000 J, final is 8400 J.

1.7.1.8 (a) 1.33 m s^{-1}

(b) Collision is inelastic. Initial *KE* = 174.8 J, final = 150 J.

1.7.1.9 (a) 9.45 cm s^{-1}

(b) Collision is inelastic because carriages couple together. Initial *KE* = 2.77×10^{-1} J and final *KE* = 2.45×10^{-1} J.

1.7.1.10 (a) 0.26 m s^{-1}

(b) Inelastic. Initial *KE* = 0.048 J, final = 0.044 J.

1.7.2.1 (a) Momentum is conserved. If you have problems with this analysis, see your teacher for assistance. Note that in analysing this system for conservation of momentum you do not have to convert real measurements to scale measurements nor do you have to adjust the time to real time. Because both factors influence the before and after situations identically, they can be ignored in this case. However, for conservation of energy, real values need to be used.

(b) Kinetic energy is not conserved (Initial = 0.56 J, final = 0.06 J. Remember *KE* is not a vector quantity.) Collision is therefore inelastic.

1.7.2.2 (a) Mass Y = 300 g

(b) The collision is inelastic. No analysis needs to be done because any collision in which objects stick together after collision is inelastic. (Analysis will show initial kinetic energy = 0.67 J and final kinetic energy = 0.11 J.)

(c) Very long strings are used so that the motion approximates straight line motion. If they were short, the pendulum swing would be more the arc of a circle and different physics would be needed to work out velocities.

2.1.1.1

	Completed sentences
1.	A wave pulse is a single energy-carrying disturbance in a medium or cycle of an electromagnetic wave.
2.	A wave is a series of pulses which transfers energy from one place to another.
3.	Crests are the highest points on transverse waves. They are positions of maximum particle displacement.
4.	Troughs are the lowest points on transverse waves. They also represent positions of maximum particle displacement.
5.	The wavelength of a wave is the distance from a point on a wave pulse to the identical point on the next wave pulse.
6.	The frequency of a wave motion is equal to the number of waves that pass a fixed position each second.
7.	The frequency of a soundwave determines the pitch (highness or lowness) of the sound produced.
8.	The period of a wave is the time it takes for one complete pulse to pass a fixed point.
9.	Wave velocity is a measure of how fast the wave transfers energy from one place to another.
10.	The amplitude of a wave is the distance from the zero displacement position of the particles to the top of a crest or the bottom of a trough.
11.	The amplitude of a wave determines how much energy the wave is carrying. For a soundwave this determines the loudness or softness of the sound.
12.	A compression is a region of higher than normal density of the medium carrying a longitudinal wave.
13.	A rarefaction is a region of lower than normal density of the medium carrying a longitudinal wave.
14.	The displacement of a particle in a medium carrying a wave is a measure of how far a particle is from its rest or zero displacement position.
15.	The zero displacement position of a particle in a medium is its position when the medium is not transferring energy.

2.1.1.2

Position	Property
A	Crest
B	Wavelength
C	Zero displacement position
D	Crest
E	Amplitude
F	Zero displacement line
G	Zero displacement line

Position	Property
H	Trough
I	Wavelength
J	Amplitude
K	Wavelength
L	Rarefaction
M	Wavelength
N	Compression

2.1.1.3 A

2.1.1.4 D

2.1.1.5 A

2.1.1.6 A

2.1.1.7 A

2.1.1.8 B

2.1.1.9 B

2.1.1.10 C

2.1.1.11 D

2.1.1.12 B

2.1.1.13 B

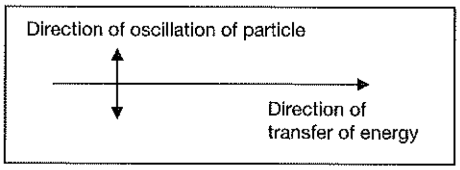
2.1.1.14 A

2.1.1.15 A

2.1.1.16 D

2.1.2.1 (a) Direction of oscillation of the particles in the medium is perpendicular to the direction of transfer of energy.

(b)



2.1.2.2 (a) 18 cm

(b) 12 cm

(c) 2

(d) 2

(e) 0.125 s

(f) 8 Hz

2.1.2.3

	Wave A	Wave B
Wavelength	1.875 cm	2.25 cm
Frequency	8 Hz	5 Hz
Period	0.125 s	0.20 s
Amplitude	1.0 cm	2.0 cm

2.1.2.4 (a) HK or EF

(b) FL OR JM

(c) E OR F

(d) M

(e) H or I or J or K or L

2.1.2.5 Y is better because it covers all possible examples of wavelength, for example, crest to crest, trough to trough etc. X is only one specific example, and by implication excludes other possible examples.

2.1.2.6 C

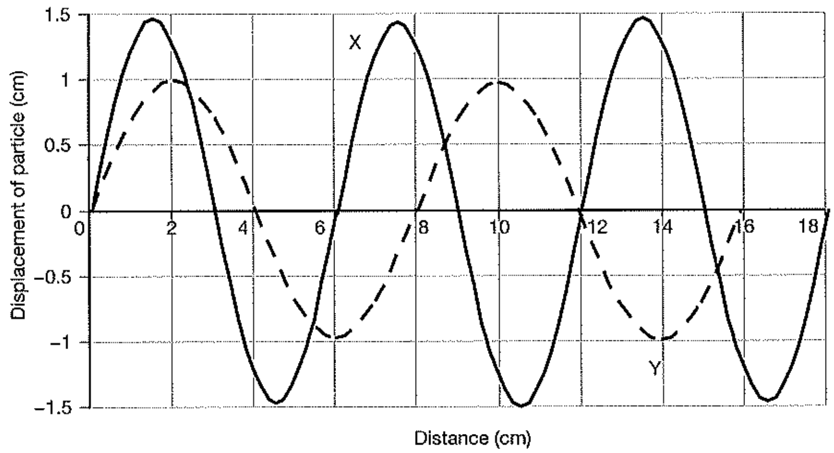
2.1.2.7 D

2.1.2.8 A

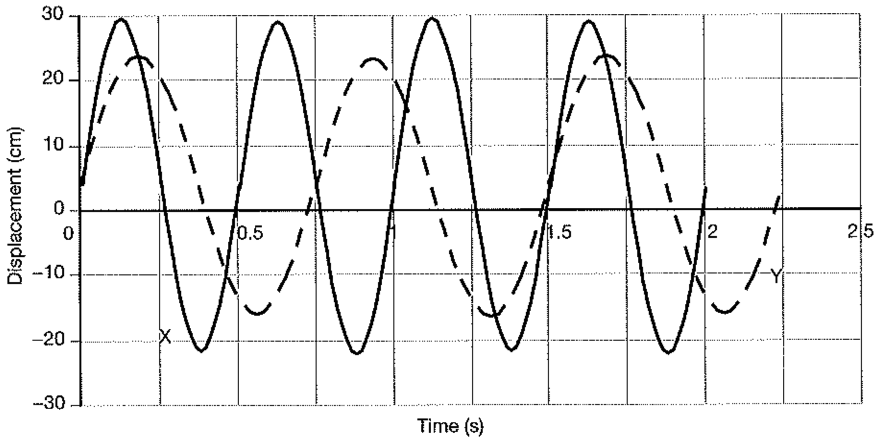
2.1.2.9 C

2.1.2.10 A

2.1.2.11



2.1.2.12



2.1.3.1
(a) T
(b) F
(c) T
(d) F
(e) T
(f) T
(g) T
(h) F
(i) T
(j) F
(k) T
(l) T

2.1.3.2
(b) They *do not pass* through a vacuum.
(d) In a longitudinal wave, particles oscillate back and forth *in the same direction* as energy transfer.
(h) The amplitude of a soundwave determines the loudness or softness of the sound, also known as its *intensity*.
(j) The frequency of a soundwave determines the highness or lowness of the sound, also known as its *pitch*.

2.1.3.3
(a) AH or BF or DJ = 7.2 cm
(b) From 3rd rest position line to point C or
From 7th rest position line to point E
(c) A or H or K
(d) D or E or I or J
(e) B or C or F or G
(f) All the lines which do not have particles on them.

2.1.3.4

(a) 8

(b) 8

(c) Half a wavelength.

(d) Wavelength.

(e) 0.2 s

(f) 5.0 Hz

(g) Back and forth in the same plane as the propogation of the energy (left/right).

(h) Amplitude.

(i) Stay the same unless the medium also changes. Speed is constant in a constant medium.

(j) Because we do not know the rest positions of the particles.

2.1.4.1

Completed sentences
All electromagnetic waves can travel through a vacuum.
Electromagnetic waves all travel at the speed of light (3×10^8 m s^{-1}) in a vacuum – they slow down a little in other media.
Electromagnetic waves are self-propagating alternating electric and magnetic fields.
Because the motion of the changing magnetic and electric fields are at right angles to the direction in which they carry energy, electromagnetic waves are also classified as transverse waves.
Because electromagnetic waves are really hard to draw, so we usually draw them as transverse matter waves.
The flaw in doing this is that the energy carried by a transverse wave is indicated by the amplitude of the wave.
In electromagnetic waves, the energy is directly proportional to the frequency of the photons which make up the radiation.
The wavelength of an electromagnetic wave is the distance between the peaks of successive magnetic or electric field pulses.
We usually refer to the intensity of an electromagnetic wave rather than to its amplitude.
The intensity of an electromagnetic wave depends on the number of photons in the beam.
Each photon will have energy dependent on its frequency.
The period of an electromagnetic wave is the time for one wavelength to pass a given point.
The frequency of an electromagnetic wave is the number of wavelengths that pass a point each second.
The frequency of electromagnetic waves is measured in hertz (Hz).

2.1.4.2

(a) Half a wavelength

(b) Wavelength

(c) Amplitude

(d) About 3.8×10^{-11} m

(e) 3

(f) 3

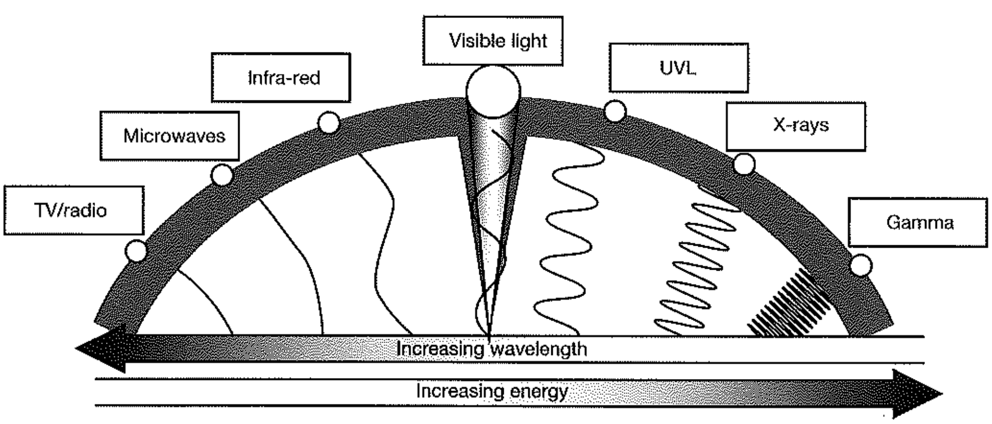
(g) 3×10^8 m s^{-1} because this is the speed of all electromagnetic waves in air.

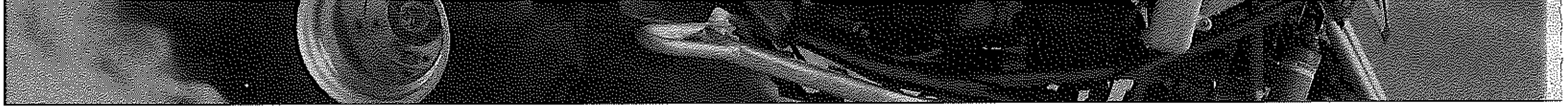
(h) About 7.9×10^{18} Hz

(i) Stay the same because the medium stays the same.

(j) Its wavelength would halve to maintain constant speed in the air.

2.1.4.3





2.1.4.4 A
2.1.4.5 B
2.1.4.6 C
2.1.4.7 D
2.1.4.8 B
2.1.4.9 D
2.1.4.10 C
2.1.4.11 A
2.1.4.12 C
2.1.4.13 D
2.1.4.14 B
2.1.4.15 A
2.1.5.1 Frequency = 2500
Velocity = 625 m s^{-1}
2.1.5.2 0.55 m
2.1.5.3 Wavelength = 0.83 m
Period = 2.6×10^{-3}
2.1.5.4 A
2.1.5.5 B
2.1.5.6 B
2.1.5.7 B
2.1.5.8 C
2.1.5.9 D
2.1.5.10 B
2.1.5.11 A
2.1.5.12 C
2.1.5.13 D
2.1.5.14 A
2.2.1.1

Wave	A	B	C	D
Period (s)	8	0.024	0.008	0.2
Frequency (Hz)	0.125	41.67	125	5
Wavelength (m)	1.25	1.25	1.25	1.25
Amplitude (m)	3.0	0.02	4	4
Velocity (m s^{-1})	0.156	52.1	156.25	6.25

2.2.1.2 Note wavelengths measure by measuring total length and dividing by the number of pulses per length.

Wave	Wavelength (cm)	Amplitude (cm)	Frequency (Hz)	Period (s)	Velocity (m s^{-1})
A	2.76	1.8	4	0.25	0.11
B	0.64	0.95	10	0.1	0.064
C	6.90	0.8	80	0.0125	5.52
D	2.78	-	250	0.004	6.95
E	1.50	-	23 256	4.3×10^{-5}	348.8
F	3.68	-	1000	1×10^{-3}	36.8

2.2.2.1 (a) Maximum period = 6.4 ms
(b) Frequency = 156.25 Hz
2.2.2.2 (a) Maximum period = 0.33 s
(b) Frequency = 3 Hz

2.2.2.3 (a) Wavelength = 8 cm
(b) Period = 8 s
(c) Frequency = 0.125 Hz
(d) Amplitude = 1.5 cm
(e) Speed = 0.01 m s^{-1}

2.2.2.4 (a) Wavelength = 0.16 m
(b) Period = 0.08 s
(c) Frequency = 12.5 Hz
(d) Amplitude = 1.5 cm
(e) Speed = 2.0 m s^{-1}

2.2.2.5 Maximum period = 6.4 s

2.2.2.6 (a) Wavelength = 0.12 m
(b) Period = 0.1067 s
(c) Frequency = 9.38 Hz
(d) Amplitude = 0.75 cm
(e) Speed = 0.875 m s^{-1}

2.2.3.1

	Completed sentences
(a)	A soundwave is a means of transporting energy without transporting matter.
(b)	A soundwave is a pressure wave. It can be considered as changes in pressure with respect to time.
(c)	If the frequency of a sound in a particular medium is increased then its wavelength will decrease.
(d)	In general, soundwaves travel fastest in solids and slowest in gases.
(e)	Soundwaves travel fastest in solids (compared to liquids and gases) because solids are more dense.
(f)	A highly elastic material has a strong tendency to return to its original shape if stressed, stretched, plucked or somehow disturbed.
(g)	A rigid material such as steel has a high elasticity and therefore sound tends to move through it at higher speeds than, say, through air.
(h)	The speed of a soundwave depends only on the properties of the medium through which it moves.
(i)	If all other factors are equal, a soundwave will travel fastest in the most dense materials.
(j)	Both low and high pitched sounds will travel through air at the same speed.
(k)	Doubling the frequency of the source of a soundwave will halve the wavelength but not alter the speed of the wave.
(l)	Both low and high pitched sounds will travel through air at the same speed.

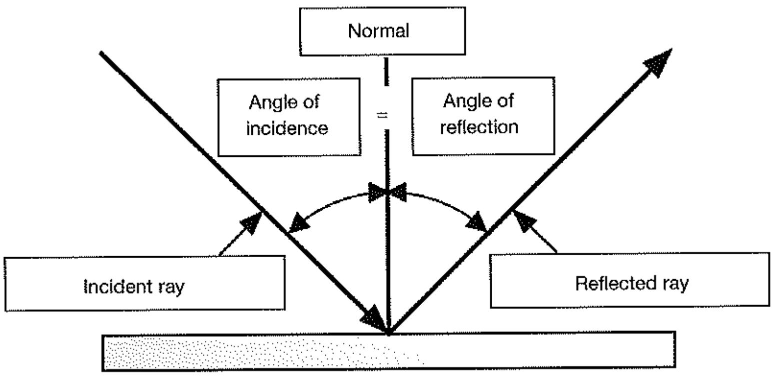
2.2.3.2 A
2.2.3.3 B
2.2.3.4 C
2.2.3.5 C
2.2.3.6 B
2.2.3.7 C
2.2.3.8 A
2.2.3.9 D
2.2.3.10 A
2.2.3.11 C
2.2.3.12 D

2.2.3.13 (a) 2×10^{-7} m
(b) 1.2×10^{-5} s
(c) 83 333 Hz
(d) 1.6×10^{-7} m
(e) 0.017 m s^{-1}

2.3.1.1

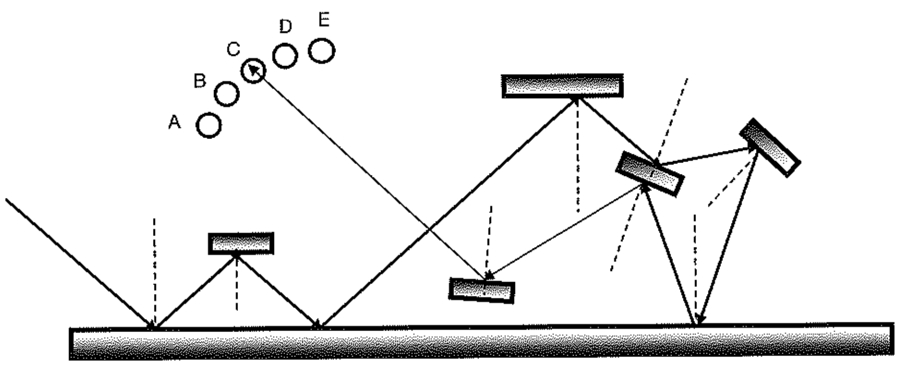
	Completed sentences
(a)	Reflection refers to the way waves 'bounce back' when they hit a surface.
(b)	We see things because reflected light from those things travels to our eyes.
(c)	The law of reflection states that the angle of incidence is equal to the angle of reflection.
(d)	The angle of incidence is the angle between the ray coming into the surface and the normal to the surface.
(e)	The ray coming into a surface is known as the incident ray.
(f)	The angle of reflection is the angle between the ray reflected away the surface and the normal to the surface.
(g)	The ray reflected away from a surface is known as the reflected ray.

2.3.1.2

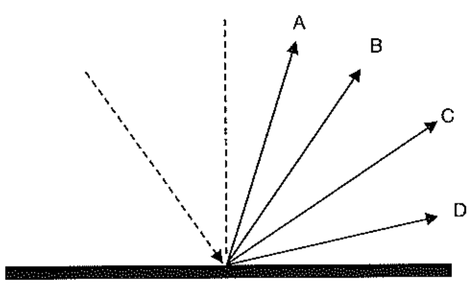


Plane mirror

2.3.1.3



2.3.1.4



Plane mirror

2.3.1.5 D

2.3.1.6 60°

2.3.1.7

(a) The reflective coating is stuck on the inside of the window. It has a mirrored finish (aluminium coating most commonly used) which reflects up to 90% of the Sun's energy back, preventing outside viewers from seeing the much lower intensity light passing through the tint from inside the home. With no reflecting surface inside the home, people inside can see straight through the film and see outside clearly.

(b) The privacy aspect of these tints do not work at night. Light from inside the home travels freely through the tint allowing outsiders to see in clearly.

2.3.1.8 D

2.3.1.9 D

2.3.1.10 A

2.3.1.11 B

2.3.1.12 C

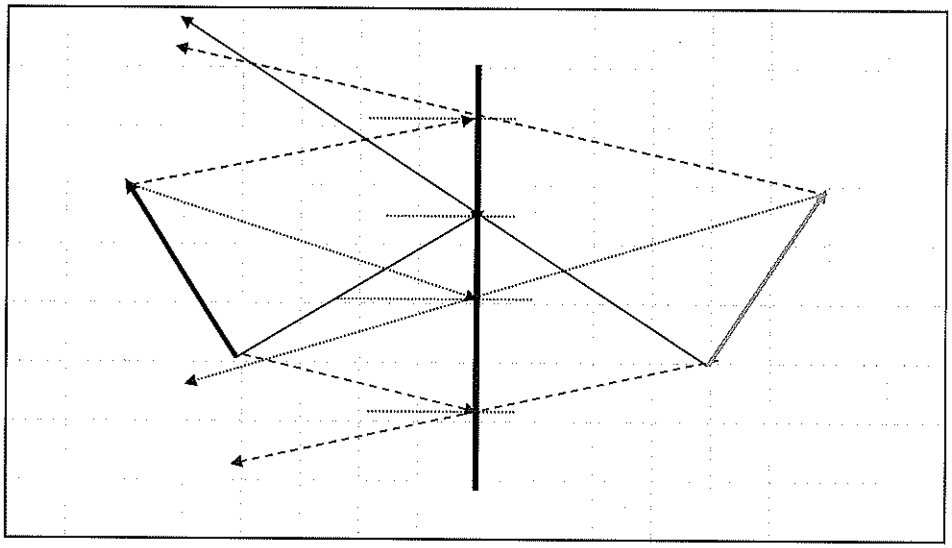
2.3.1.13 B

2.3.1.14 B

2.3.1.15 C

2.3.1.16 B

2.3.2.1



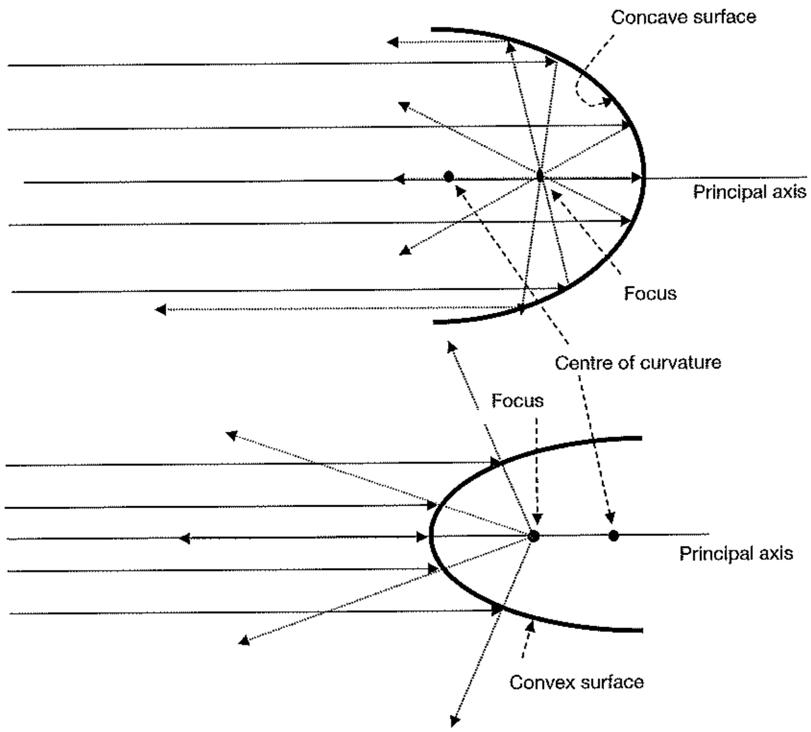
(a) An image formed in a plane mirror will be *as far behind the mirror* as the object is in front of the mirror.

(b) The size of the image formed by a plane mirror will be *the same* as that of the image it forms.

2.3.3.1

	Completed sentences
(a)	Surfaces which curve inwards (like a cave) are known as concave surfaces.
(b)	Surfaces which curve outwards are called convex surfaces.
(c)	Convex surfaces are converging surfaces.
(d)	Converging surfaces cause the rays to reflect and move closer together until they pass through the focus.
(e)	Convex surfaces reflect rays so that their backwards extension passes through the focus of the curve.
(f)	Convex surfaces are called diverging surfaces because after reflection the rays travel further apart.
(g)	When electromagnetic rays hit a concave surface they reflect so that they pass through the focus of the curve.
(h)	When electromagnetic rays hit a convex surface they reflect so that their extension backwards passes through the focus of the curve.
(i)	The ray travelling along the principal axis is reflected straight back (it hits the curved surface at 90°).
(j)	The principal axis is the (imaginary) line drawn through the centre of the lens.
(k)	The focus of a curved surface is the point on the principal axis through which reflected rays pass.
(l)	The focus of a curved surface is midway between the surface and the centre of curvature of the surface.

2.3.3.2



2.3.3.3 (a) Convex reflective surfaces, because they are diverging surfaces are used in situations where a wide angle view is beneficial, such as in car rear view mirrors, security mirrors in shops and mirrors on tight turns in roads so oncoming cars can see traffic around the corner before they get to it.

(b) Concave surfaces focus rays at their foci, and this can be utilised in solar barbecues and satellite collection dishes. They also send rays from the focus out, after reflection, parallel to the principal axis and are therefore useful in car headlights, torches and the like.

(c) Plane reflective surfaces do not have the distortion in images that we get with concave and convex surfaces, so they are useful when we want a true image. They are used in bathrooms, hairdressing salons, and anywhere mirrors are used.

2.3.3.4

	Concave or convex reflector	Reason for use in this application
A	Convex	Security mirror in store collects light rays from a wide area so workers can monitor what is going on in aisles that would otherwise not be in view.
B	Concave	Concave surface reflects light rays emitted backwards and sideways forwards and out of the shade to produce a greater intensity in the direction the light is needed.
C	Convex	Wide rear view mirror collects light from a much wider area than a plane mirror and reduces the 'blind spots' motorists usually have either side of their cars.
D	Convex	Light rays from the side streets are collected enabling people to see 'around the corners' before they move around them. This increases the safety of the roads.
E	Concave	The sun's rays are collected by the surface and reflected to the linear focus where a rotisserie and the food to be cooked is placed.
F	Concave	Rays collected from the transmitter are reflected to the focus of the dish. The focus will contain an electronic receiver to sent the signals to your TV or communications system.
G	Concave	Light emitted backwards and sideways from the globe at the focus is reflected parallel to the principal axis so that much more intense light is projected forwards.

2.3.4.1 (a) A ray from the object travelling parallel to the principal axis will reflect from the mirror to pass through the focus.

(b) A ray from the object that passes through the centre of curvature of a mirror is reflected back along its path.

(c) Any ray from the object passing through the focus will be reflected by the mirror to travel parallel to the principal axis.

2.3.4.2 Image is:

(a) Real if it occurs on the same side of the mirror as the object (R).

(b) Virtual if it occurs on the opposite side of the mirror to the object (V).

(c) Upright if it is the same way up as the object (U).

(d) Inverted if it is upside down (I).

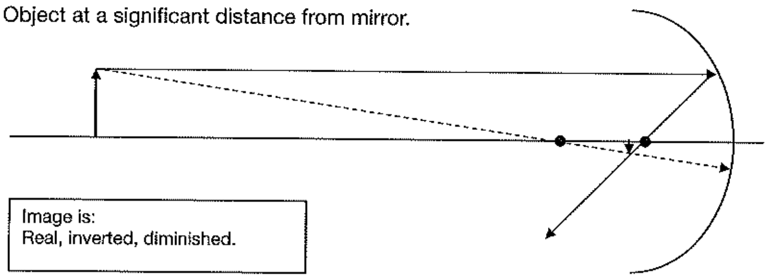
(e) Diminished if it is smaller than the object (D).

(f) Diminished if it is larger than the object (M).

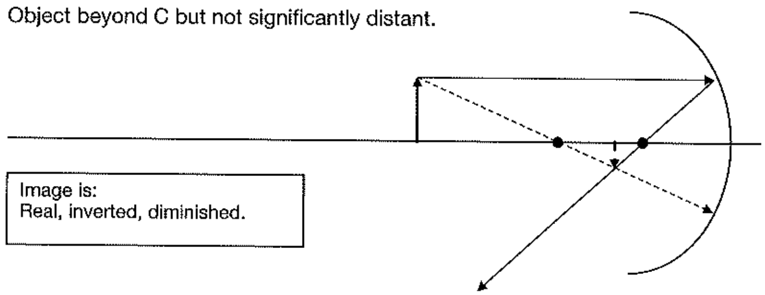
(g) Same size if it is the same size as the object (S).

2.3.4.3 Note that for simplicity of the diagrams, it is only necessary to determine the position of the top of the image because the bottom will be on the principal axis.

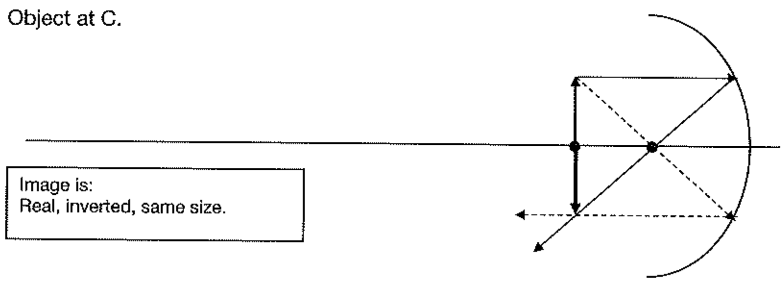
(a) Object at a significant distance from mirror.



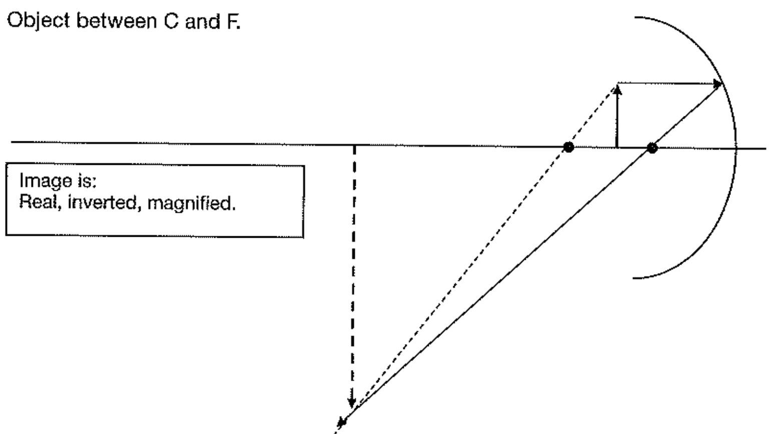
(b) Object beyond C but not significantly distant.



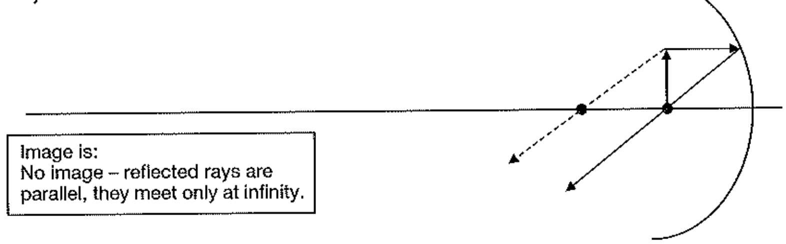
(c) Object at C.



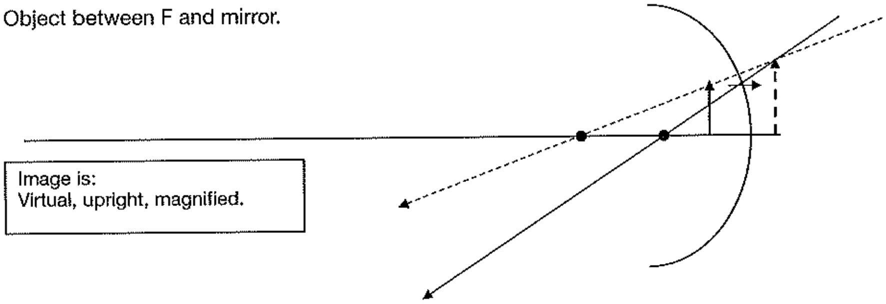
(d) Object between C and F.



(e) Object at F.



(f) Object between F and mirror.



2.3.5.1 (a) A ray from the object travelling parallel to the principal axis will reflect from the mirror so that its extension backwards passes through the focus.

(b) A ray from the object that passes through the centre of curvature of a mirror is reflected back along its path.

(c) Any ray from the object passing through the focus will be reflected by the mirror to travel parallel to the principal axis.

2.3.5.2 Image is:

(a) Real if it occurs on the same side of the mirror as the object (R).

(b) Virtual if it occurs on the opposite side of the mirror to the object (V).

(c) Upright if it is the same way up as the object (U).

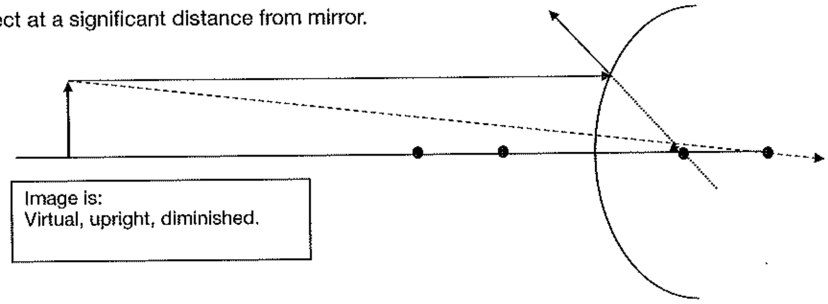
(d) Inverted if it is upside down (I).

(e) Diminished if it is smaller than the object (D).

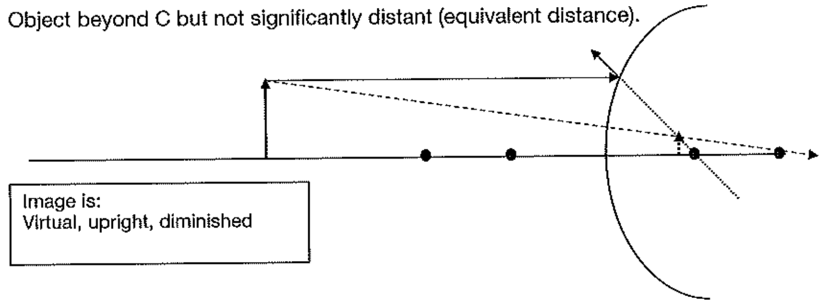
(f) Diminished if it is larger than the object (M).

(g) Same size if it is the same size as the object (S).

2.3.5.3 (a) Object at a significant distance from mirror.



(b) Object beyond C but not significantly distant (equivalent distance).



(c) Object at C (equivalent distance).

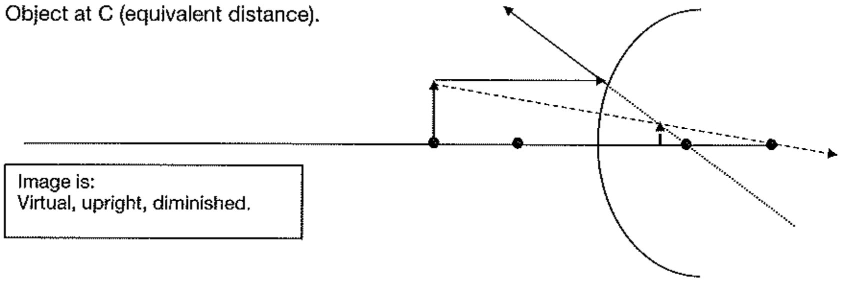


Image is:
Virtual, upright, diminished.

(d) Object between C and F (equivalent distance).

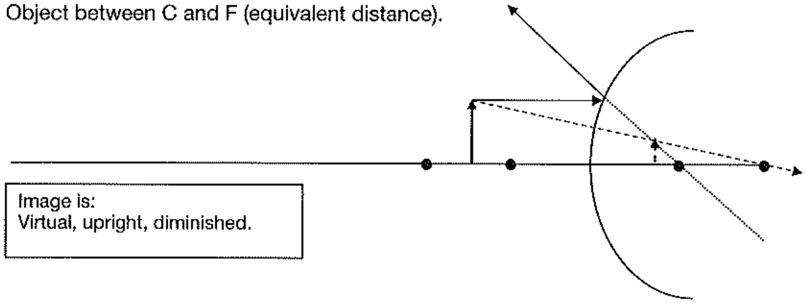


Image is:
Virtual, upright, diminished.

(e) Object at F (equivalent distance).

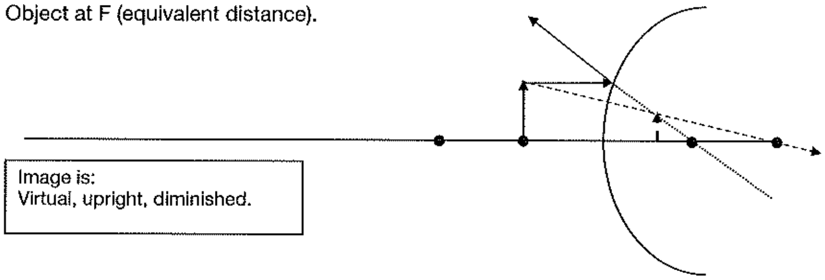


Image is:
Virtual, upright, diminished.

(f) Object between F and mirror (equivalent distance).

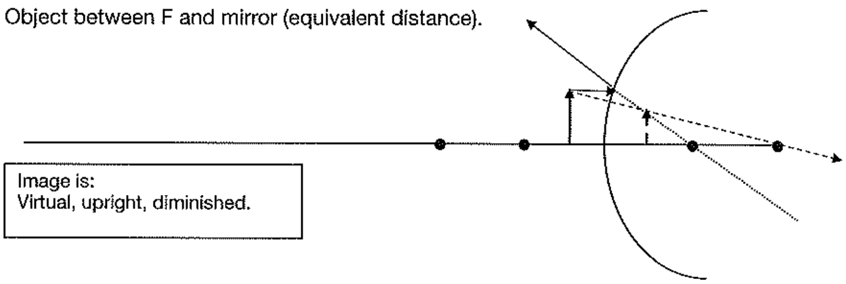


Image is:
Virtual, upright, diminished.

2.3.6.1 (a) 4.55 s

(b) 1443 m

2.3.6.2 6162.3 m

2.3.6.3 (a) 328.95 m s^{-1}

(b) 740 m

2.3.6.4 2380 m

2.3.6.5 (a) Soundwaves reflect within the tube of the stethoscope and travel to the doctor's ears.

(b) Soundwaves reflect along the length of the hearing aid trumpet to the person's ear.

(c) The soundwaves reflect from the animals and tell the bat and porpoise where, and how far away they are.

(d) Sound reflecting from the walls of the theatre provide a more complete sound experience for the audience.

(e) Acting like a concave mirror, the soundwaves moving backwards in the megaphone are reflected forwards adding to the intensity emitted.

(f) The soundwaves reflect off the shoal of fish and tell the fishermen in the boat where, and how far away they are.

(g) Sound from traffic reflects off the sound barrier and makes life for people behind the barrier quieter.

(h) Ultrasound waves reflect differently off different density tissue and enable the computer to form an image of the baby.

2.4.1.1 (a) Robert Hooke, 1665.

(b) Christiaan Huygens, 1690.

(c) He proposed that light waves simply slowed down when they entered a denser medium.

(d) Refraction.

(e) Thomas Young.

(f) The interference patterns observed from double slit diffraction could not be explained if light was a particle. The brightest spots would be directly opposite the holes if light was a particle, not midway between them.

(g) Sir Isaac Newton favoured a particle theory, and his huge reputation caused other scientists to hesitate before opposing his ideas.

2.4.1.2 If light was a particle, this would not happen, the particle would simply bounce back from the surface or, if small enough to fit in between the spaces of the matter particles refract through the material. As a particle it would not do both at the same time.

2.4.1.3 (a) Diffraction occurs when an edge or a small hole in a barrier acts as a point source of the wave and wavelets spread out in a circular path from the edge or hole.

(b) The wave theory argues that particles would simply travel straight ahead and could not form the diffraction patterns that are observed.

(c) Particle theory cannot explain diffraction.

2.4.1.4 (a) Dispersion is the breaking up of a band of electromagnetic radiation (e.g. visible light) into its component wavelengths when it undergoes refraction through a medium.

(b) Wave theory explains this by proposing that the different frequencies of the radiation undergo different amounts of refraction – higher frequencies are refracted (slowed down) more than lower frequencies – remember – 'blue bends best').

2.4.2.1 (a) Total internal reflection occurs when a wave is totally reflected back from a boundary into the medium. No energy from the wave refracts into the next medium.

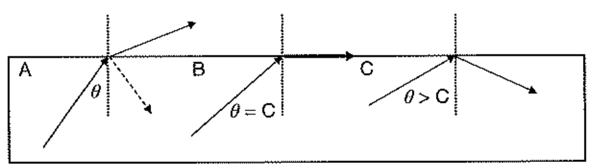
(b) The wave must be incident on the boundary from the more dense of the two media making up the boundary and the angle of incidence must be greater than the critical angle for the boundary.

(c) The critical angle is the minimum angle of incidence onto a boundary such that the angle of refraction for the wave will be 90°.

(d) A Reflection and refraction occurs.

B This is the critical angle, so light will travel along the surface of the boundary.

C Light will be totally internally reflected.



(e) Optical fibre or cable communications.

(f) From $\frac{\sin i}{\sin r} = n$, remembering that $r = 90°$

Sin $i = n$

And because $i = C$ in this example, then sin C = refractive index on the boundary.

2.4.3.1 (a) The term refraction refers to the change in speed of a wave when it passes from one medium into another.

(b) (i) During refraction the velocity of a wave changes.

(ii) During refraction the frequency of a wave remains constant.

(iii) During refraction the wavelength of a wave changes.

2.4.3.2 (a) Electromagnetic waves slow down as they pass from less dense to more dense media.

(b) Electromagnetic waves speed up as they pass from more dense to less dense media.

(c) The wavelength of electromagnetic waves is smaller in denser media.

(d) The wavelength of electromagnetic waves is larger in less dense media.

(e) The frequency of an electromagnetic wave passing from a less dense to a more dense medium stays the same.

(f) The frequency of an electromagnetic wave passing from a more dense to a less dense medium stays the same.

(g) The intensity of an electromagnetic radiation passing from a less dense to a more dense medium decreases due to some reflection from the boundary.

(h) The intensity of an electromagnetic radiation passing from a more dense to a less dense medium decreases due to some reflection from the boundary.

2.4.3.3 D

2.4.3.4 D

2.4.3.5 A

2.4.3.6 B

2.4.3.7 A

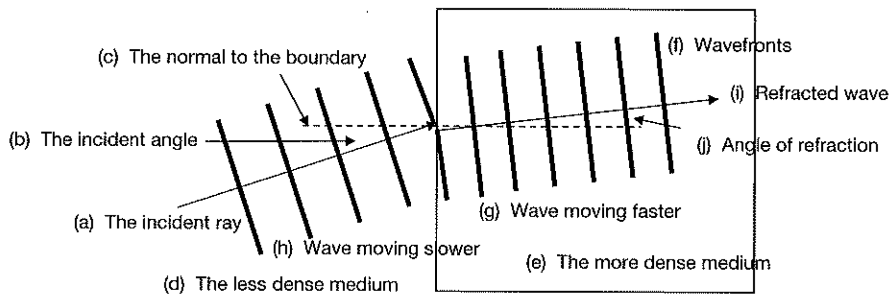
2.4.3.8 C

2.4.3.9 C

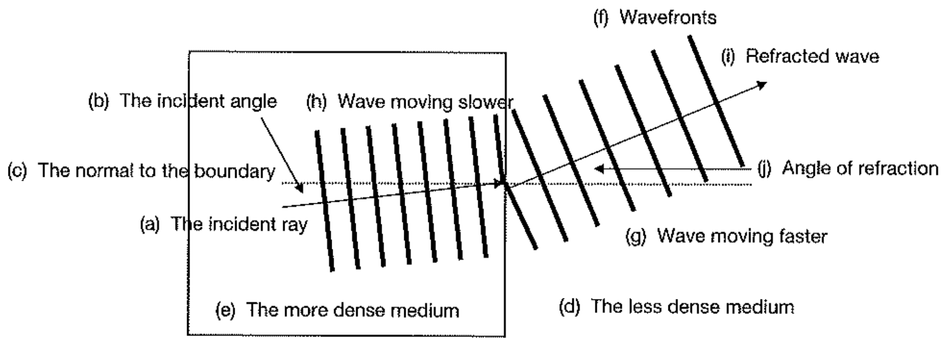
2.4.3.10 D

2.4.3.11 B

2.4.4.1 (A), (B)



(C)



2.4.4.2 In general, for electromagnetic radiations:

(a) When passing from less dense medium to more dense medium:

- The wavefront slows down.
- The wavelength decreases.
- Refraction is towards the normal.
- Angle of incidence is greater than angle of refraction.

(b) When passing from dense medium to less dense medium:

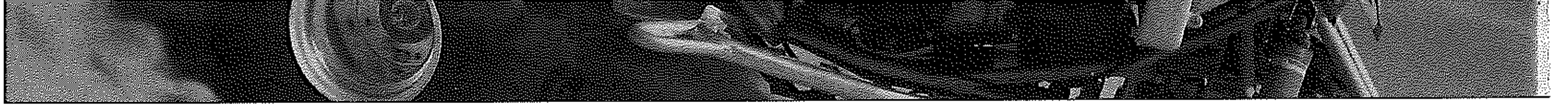
- Wavefront speed up.
- Wavelength increases.
- Refraction is away from normal.
- Angle of incidence is smaller than angle of refraction.

(c) When entering any medium at 90° (i.e. the incident angle is zero):

- Refraction will occur as usual .
- The direction of travel will *not* change.

2.4.4.3

Wave type	What causes refraction?	Identify the specific change
Sound	Changes in density of the media	Waves travel faster in denser media
Water	Changes in depth of the water	Waves slow down in shallow water
Earthquake compression	Changes in density of the rock	Waves travel faster in denser media



2.4.5.1 (a) The absolute refractive index of a medium is the ratio of the velocity of the wave in a vacuum to that of its velocity in the medium.

(b) Relative refractive index is the ratio of refractive indices at the boundary between two different media.

(c) Relative refractive index $= \dfrac{n_2}{n_1}$

(d) $\dfrac{\text{Relative refractive index of air}}{\text{glass boundary}} = \left(\dfrac{\text{relative refractive index of glass}}{\text{air boundary}}\right)^{-1}$

$_{air}n_{glass} = \dfrac{1}{_{glass}n_{air}}$

(e) Snell's law states that the relative refractive index of a boundary is equal to the sine of the angle of incidence divided by the sine of the angle of refraction.

Relative refractive index $= \dfrac{\sin i}{\sin r}$

(f) If medium 1 is less dense than medium 2 then the RRI of the boundary is greater than 1.

If medium 1 is more dense than medium 2 then the RRI of the boundary is less than 1.

(g) Relative refractive index $= \dfrac{\text{velocity of wave in medium 1}}{\text{velocity of wave in medium 2}} = \dfrac{v_2}{v_1} = \dfrac{\sin i}{\sin r} = \dfrac{\lambda_2}{\lambda_1} = \dfrac{n_2}{n_1}$

Where i = angle of incidence

r = angle of refraction

λ_1 = wavelength of wave in medium 1

λ_2 = wavelength of wave in medium 2

n_1 = absolute refractive index of medium 1

n_2 = absolute refractive index of medium 2

2.4.5.2

Medium	Absolute refractive index	Velocity of light in this medium (m s^{-1})	Medium	Absolute refractive index	Velocity of light in this medium (m s^{-1})
Vacuum	1.00	3×10^8	PET plastic	1.58	1.90×10^8
Air	1.00	3×10^8	Flint glass	1.65	1.82×10^8
Water ice	1.31	2.29×10^8	Bromine	1.66	1.81×10^8
Water	1.33	2.26×10^8	Sapphire	1.77	1.69×10^8
Amber	1.55	1.94×10^8	Diamond	2.42	1.24×10^8

For light travelling from:	The relative refractive index will be:
Vacuum to Earth's atmosphere	A = 1.00
Air to flint glass	B = 1.65
Air to sapphire	C = 1.77
Amber to air	D = 0.65
Diamond to sapphire	E = 0.73
Water to water ice	F = 0.98
Water ice to water	G = 1.02
Plastic to bromine	H = 1.05

2.4.5.3 (a) 1.94

(b) 1.94

(c) 1.55×10^8 m s^{-1}

2.4.5.4 (a) 0.52

(b) 19.53°

2.4.6.1 (a) Medium 2. The wave travels slower in denser medium or refraction is towards the normal in the denser medium.

(b) 1 : 1

(c) 3 : 2

(d) 3 : 2

(e) 1.5

2.4.6.2 Incident angle into the plastic from air = 35.1°

Incident angle into the glass from the plastic = 24.2°

2.4.6.3 37.8°

2.4.6.4 (a) 36.3° (The geometry of the figure allows you to determine this without any calculations, but it is good if you have actually used the physics to do this.)

(b) 1.4

2.4.6.5 (a) By measuring it, the angle of incidence of the light into the prism is = 48°

The angle of refraction for the red light = 42°

Therefore refractive index for red light = 1.11

(b) The angle of refraction for the blue light = 32°

Therefore refractive index for blue light = 1.40

(c) Refractive index depends on wavelength of the ray of light (remember that frequency stays constant in refraction). The shorter the wavelength of the radiation, the larger the refraction (remember 'blue bends best').

2.4.7.1 (a) A ray from the object travelling parallel to the principal axis will refract so that its extension backwards passes through the focus of the lens.

(b) A ray from the object that passes through the centre of the lens will pass straight through the lens. (Note that although its path does not change, because it is passing through the glass it will slow down – i.e. be refracted.)

2.4.7.2 Images are:

(a) Real if they occur on the opposite side of the lens as the object (R).

(b) Virtual if they occur on the same side of the lens to the object (V).

(c) Upright if they are the same way up as the object (U).

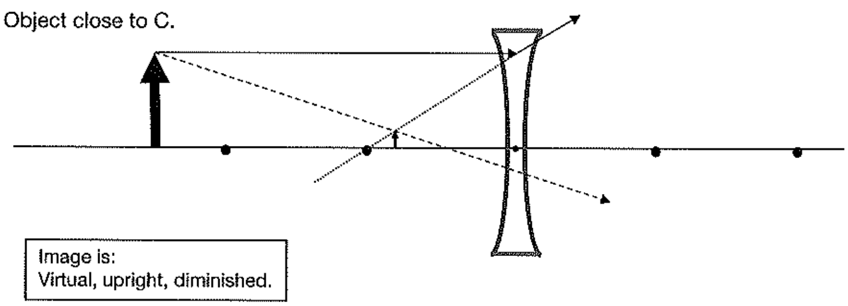
(d) Inverted if they are upside down (I).

(e) Diminished if they are smaller than the object (D).

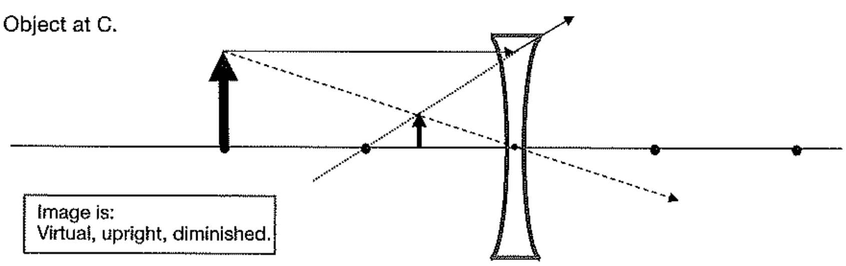
(f) Magnified if they are larger than the object (M).

(g) Same size (S).

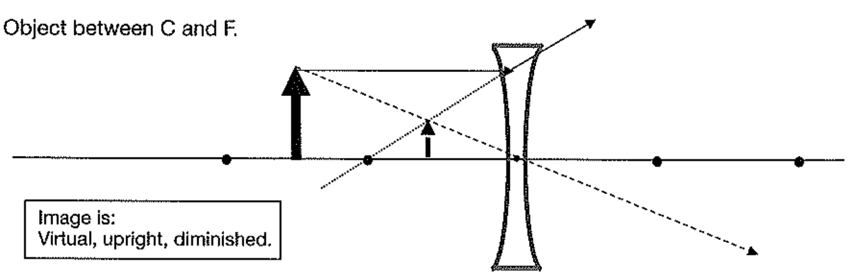
2.4.7.3 (A) Object close to C.



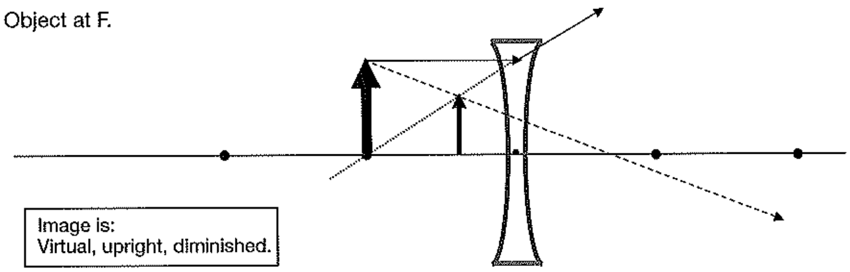
(B) Object at C.



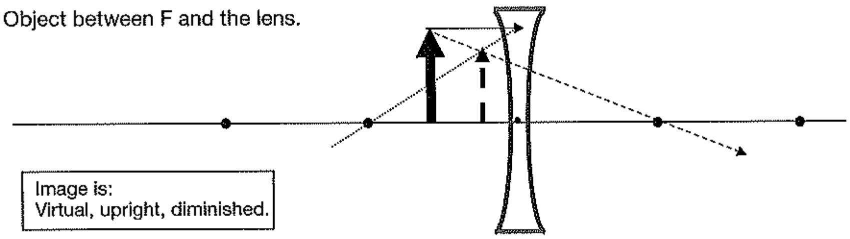
(C) Object between C and F.



(D) Object at F.



(E) Object between F and the lens.



2.4.7.4 D

2.4.7.5 D

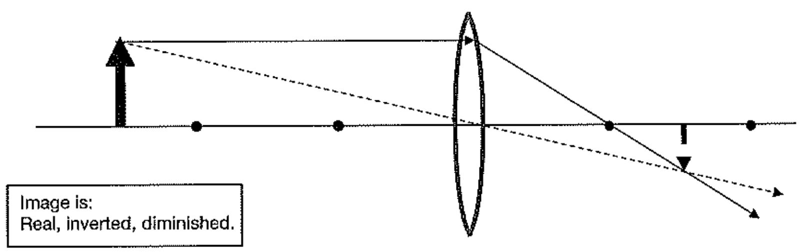
2.4.7.6 B

2.4.7.7 A

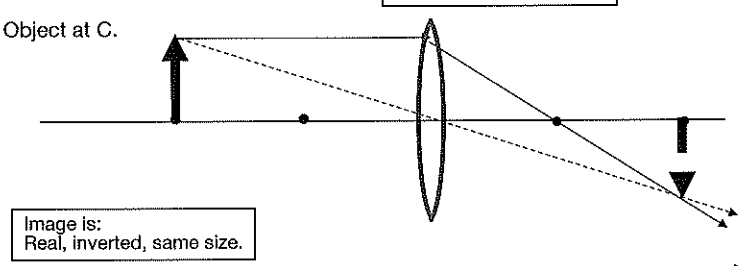
2.4.8.1
(a) A ray from the object travelling parallel to the principal axis will refract so that it passes through the focus of the lens.
(b) A ray from the object that passes through the centre of the lens will pass straight through the lens. (Note that although its path does not change, because it is passing through the glass it will slow down – i.e. be refracted.)

2.4.8.2 Images are:
(a) Real if they occur on the opposite side of the lens as the object (R).
(b) Virtual if they occur on the same side of the lens to the object (V).
(c) Upright if they are the same way up as the object (U).
(d) Inverted if they are upside down (I).
(e) Diminished if they are smaller than the object (D).
(f) Magnified if they are larger than the object (M).

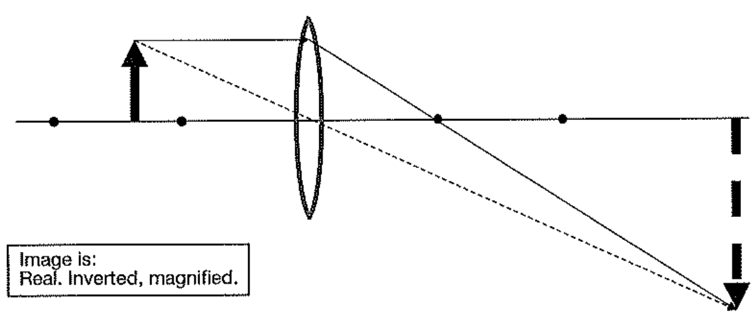
2.4.8.3
(A) Object closer to C.



(B) Object at C.



(C) Object between C and F.



(D) Object at F.

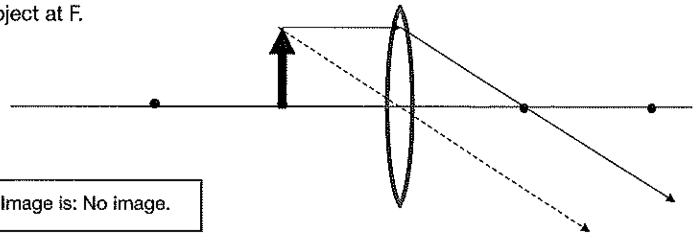


Image is: No image.

(E) Object between F and the lens.

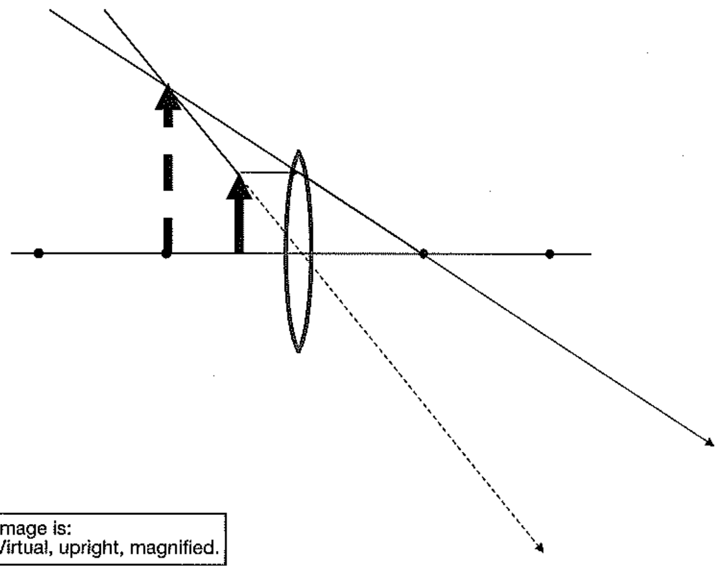
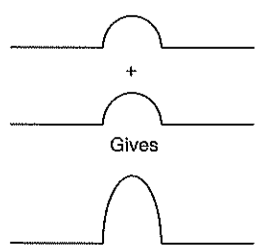
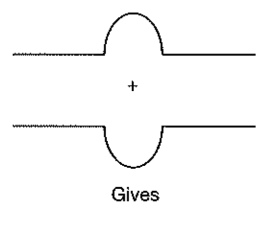


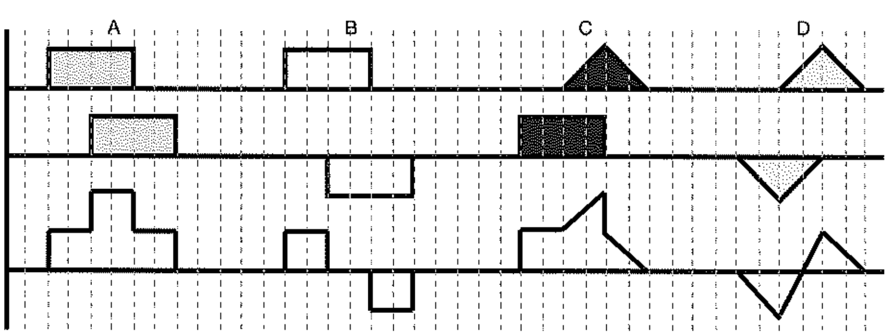
Image is:
Virtual, upright, magnified.

2.5.1.1 (a) When two (or more) sources of vibration each produce a wave in a medium, the waves interfere with each other. The amplitude of the combined wave is equal to the sum of the amplitudes of the component waves. This process is called superposition and the combination wave is the resultant wave.

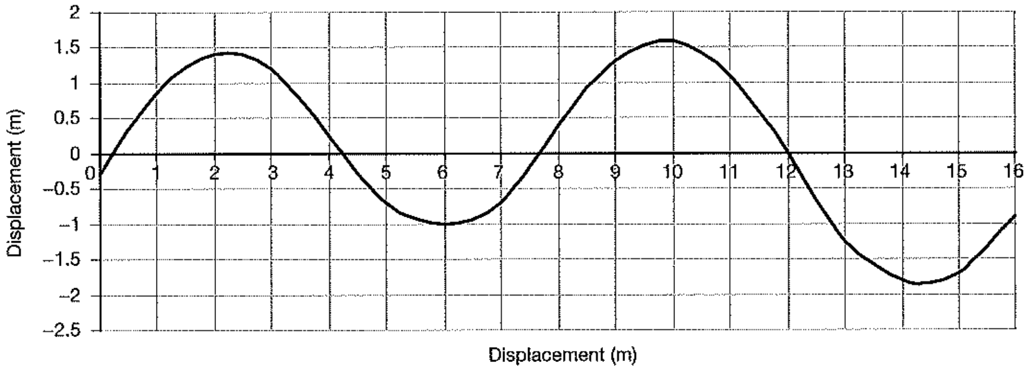
(b)

Constructive interference occurs when the superposition results in larger amplitudes in the resultant wave compared to the two components.	Destructive interference occurs when the superposition results in smaller amplitudes in the resultant wave compared to the two components.
	

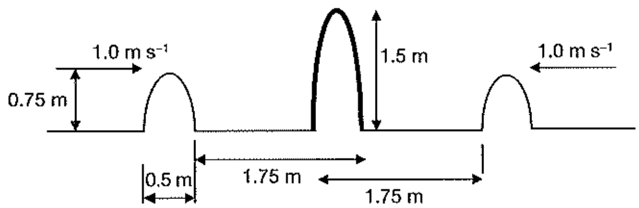
2.5.1.2



2.5.1.3



2.5.1.4



2.5.2.1 (a) A standing wave pattern is a vibration pattern that is produced in mediums when two identical waves travel in opposite directions in the medium and superimpose on each other.

(b) Node

(c) Antinode

2.5.2.2 (a) The fundamental vibration is the first harmonic.

(b) The *n*th harmonic has *n* times the frequency of the fundamental.

2.5.2.3

Wave	Harmonic	Number of nodes	Number of antinodes	Wavelength (m)	Frequency (Hz)
A	Fundamental	2	1	0.48 m	708.3
B	Second	3	2	0.24 m	1416.7
C	Third	4	3	0.16	2125
D	Fourth	5	4	0.12	2833.3

2.5.3.1 (a) There are two types of pipes in which standing waves occur – open and closed.

(b) These pipes, like stretched strings, form the basis of many musical instruments.

(c) When standing waves form in pipes or tubes, a node will always be found at a closed end.

(d) A pipe which supports a standing wave will always have an antinode at an open end.

(e) The lowest frequency standing wave that can be formed is again known as the first or fundamental harmonic.

(f) In an open pipe, the wavelength of the fundamental frequency is twice the length of the pipe.

(g) In a closed pipe, the wavelength of the fundamental frequency is four times the length of the pipe.

(h) Because of this, closed pipes only have odd numbered harmonics.

2.5.3.2 (a) 1.5 m

(b) 226.7 Hz

2.5.3.3 (a) 1.8 m

(b) 188.9 Hz

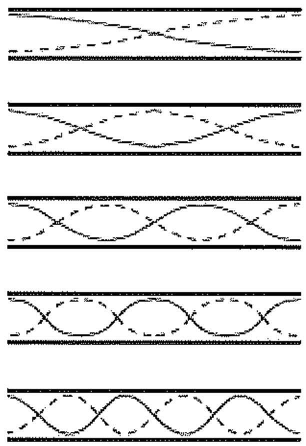
2.5.3.4 (a) 1.6 m

(b) 212.5 Hz

2.5.3.5 0.332 m

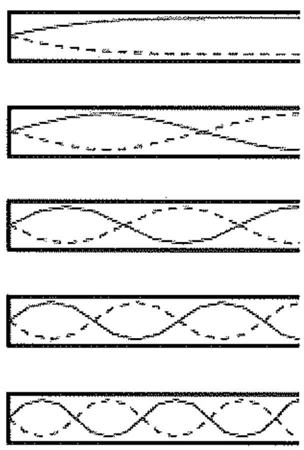
2.5.3.6 0.332 m

2.5.3.7



Harmonic	Wavelength (m)	Frequency (Hz)
Fundamental (first)	1.2 m	283.3
Second	0.6 m	566.7
Third	0.4 m	850
Fourth	0.3 m	1133.3
Fifth	0.24 m	1416.7

2.5.3.8



Harmonic	Wavelength (m)	Frequency (Hz)
Fundamental (first)	2.4	141.7
Third	0.8	425
Fifth	0.48	708.3
Seventh	0.34	1000
Ninth	0.27	1259.3

2.6.1.1 Mechanical resonance is what happens when the frequency of a sound matches with the resonant frequency of a material – the material starts to vibrate with that frequency.

2.6.1.2 The classic example is of the opera singer breaking a glass with her voice when she hits the right note.

2.6.1.3 You will feel the walls vibrating, your chest cavity may also sense the vibrations, you may feel the vibrations through the floor.

2.6.1.4 If the troops march in step they can set up a mechanical vibration in the bridge itself. If this becomes too intense, the bridge structure may be compromised.

2.6.1.5 (a) The vibrations from the strings of the guitar set up standing waves within the air inside the sound box and this amplifies the sound produced by the guitar.

(b) All musical instruments work by producing soundwaves. Soundwaves are mechanical waves (transferring energy from particle to particle) and the soundwaves are set up by resonance from vibrating reeds, or strings etc in the musical instruments.

2.7.1.1 Light emitted by luminous objects consists of tiny particles of matter.

2.7.1.2 Sir Isaac Newton, 1675.

2.7.1.3 Light travels in straight lines and light can travel through a vacuum.

2.7.1.4 Direct observations of the properties of the light, but no experimental evidence for the existence of particles.

2.7.1.5 The particles simply collided with and bounced off the surface they hit.

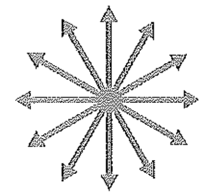
2.7.1.6 He proposed that light particles decelerated upon entering a denser material because the gravitational pull of the denser material was greater, thereby slowing the particles down.

2.7.1.7 Interference of light rays.

2.7.1.8 We cannot explain all properties with one theory, so we still have a dual model for the nature of light.

2.7.2.1 Electromagnetic radiation consists of self-propagating, alternating electric and magnetic fields which travel through space at the speed of light.

2.7.2.2 Michael Faraday in 1845.



2.7.2.3 He observed that light was affected by magnetic fields.

2.7.2.4 James Clerk Maxwell in 1873.

2.7.2.5 In 1887 Heinrich Hertz discovered what we now call radio waves.

2.7.2.6 (a) The polarisation of light.

(b) Polarisation is a property of waves that can oscillate with more than one orientation.

2.7.2.7 It was unable to explain the spectral lines of elements and the threshold frequency in the photoelectric effect.

2.7.3.1 Max Planck, 1900.

2.7.3.2 He was attempting to explain the observations that he made studying the energy emitted from hot objects – black body radiation.

2.7.3.3 The radiations given off by hot objects were due to the oscillations of the particles of the matter and that energy involved with these oscillations was quantised according to their frequency.

2.7.3.4 Einstein, in 1905, extended Planck's idea of quanta to cover all electromagnetic radiations and used it to explain the threshold frequency in the photoelectric effect.

2.7.3.5 The weakness was that in order to explain the threshold frequency in the photoelectric effect, the electromagnetic radiations involved had to be composed of particles. This is the particle side of the dual theory of light.

2.7.3.6 The frequency of the photons.

2.7.3.7 (a) The photoelectric effect is the emission of electrons from a photoemitting surface when photons with frequency above the threshold frequency are incident on the surface.

(b) The threshold frequency.

2.8.1.1 (a) Acoustics is the interdisciplinary science that deals with the study of all mechanical waves in gases, liquids, and solids including vibration, sound, ultrasound and infrasound.

(b) The purpose of acoustics is to make sound better to listen to, to make it softer, or louder, to reduce echoes, to produce echoes. It can be used to produce sound to warn us of some impending danger, or to insulate us from inside or outside noise.

2.8.1.2 Movie theatres often have fabric on the walls or panelling especially designed to scatter sound that hits it. Concert halls do this using panelling which has holes or irregularities or shapes in it to break up the soundwaves so they do not reflect or reflect in such a way to interfere with each other and cancel.

2.8.1.3 The function of sound barriers is to help insulate the area beyond the barrier from the traffic noise on the motorway by absorbing it and reflecting it back onto the motorway.

2.8.1.4 High density materials both reflect sound well and transmit it, so they are useful as sound barriers on motorways but not as flooring in multiple storey buildings.

2.8.1.5 Low density materials like air and carpet or fibreglass insulation which absorb but do not transmit or reflect sound well are used in wall cavities and as floor covering in units to reduce the transmission of sound to other units.

2.8.2.1

Type of wave	Nature of wave	Travels through	Relative speed	Relative destructive power
Primary (P)	Longitudinal	Solids, liquids, gases	Fastest	Less destructive
Secondary (S)	Transverse	Solids only	Slower	Less destructive
Surface (L)	Transverse	solids	Slowest	Most destructive

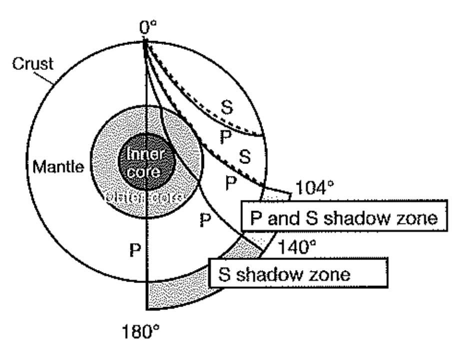
2.8.2.2 S waves are the most destructive type of seismic wave because they directly affect structures on the surface while the other types of waves travel through the Earth.

2.8.2.3 (a) The epicentre is the position on the Earth's surface directly above the focus of an earthquake. The focus is the position within the Earth's crust where the earthquake occurred.

(b) An approximate 'rule of thumb' is to determine the difference in arrival time of the P and S waves at the station and multiply it by 8. This will give the approximate distance to the epicentre in km. (Obviously computer analysis is used in practice to determine the position of the sources of earthquakes.)

2.8.2.4 (a) The shadow zone resulting from the inability of S waves to travel through the outer core provides the main evidence that this is liquid.

(b) Because S waves do not travel through liquid, 104° from the focus, there are no S waves received. Because of the refraction of the P waves in and out of the liquid core, there is also a P wave shadow zone, between 104° and 140°. This is a complete shadow zone.



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2.8.3.1 Tsunamis are large ocean waves caused by sudden movement of the ocean floor due to earthquakes, landslides on the sea floor, land slumping into the ocean, major volcanic eruptions or large meteorite impacts.

2.8.3.2 Subduction zones are regions where one oceanic tectonic plate is being forced down into the mantle underneath the adjacent plate.

2.8.3.3 The friction between the subducting plate and the overriding plate is enormous. The movement between the two plates is not always slow and steady. Sometimes they get 'stuck'. Forces between them build up and eventually they move past each other suddenly causing the overlying water to move. The resulting wave moves outwards and away from this event.

2.8.3.4 The tsunami will be a transverse matter wave because the movement of the crustal plates pushes a large section of water upwards to form the tsunami on the surface.

2.8.3.5 Underwater volcano.

Massive landslide into the ocean.

Collapse of an underwater volcano.

2.8.3.6 The subduction zone movement constitutes an earthquake, so earthquakes will accompany tsunamis formed this way.

2.8.3.7 Building a low beach wall will protect from small tsunamis but will destroy the freedom of the landscape.

2.8.4.1 Because all earthquakes and most tsunamis involve movements in the Earth's crust, early detection depends on detecting small movements in the crust before they become much larger movements. Crustal movement can be observed as relative movement of points on the Earth's surface, ground tilt, ground strain, and fault slip (creep). Factors like these are regularly monitored to detect this deformation and early warning of earthquakes can also be early warning of tsunamis.

2.8.4.2 Japan's EWS network has about 200 seismographs and 600 seismic intensity meters. It also collects data from over 3600 seismic intensity meters managed by local governments and the National Research Institute for Earth Science and Disaster Prevention (NIED). The data collected are inputted to the computers at the headquarters in Tokyo for immediate analysis and is capable of issuing a warning, if necessary, within one and a half minutes.

2.8.4.3 Earth's surface is being deformed as earthquake faults accumulate strain and slip or slowly creep over time. Using satellites fitted with GPS movements of the Earth's crust can be monitored. The GPS can measure the precise position (within 5 mm or less) of stations near active faults relative to each other. Months or years later, we observe the same stations again. By determining how far and how fast the stations move over time seismologists can make predictions about future earthquakes.

2.8.5.1
(a) Damping and isolation – These are devices in building foundations which can reduce the motion caused by ground shaking. While the earthquake waves move the ground beneath them, the inertia of the buildings lessens the vibrations they experience

(b) Reducing resonance – If the period of the earthquake wave and the natural period of the building coincide, then the building will 'resonate' and its vibration will increase the probability of damage. The height of a building is a major factor in determining its natural vibration frequency, so architects avoid heights which will resonate with the most commonly expected earthquake waves in a region.

(c) Energy dissipating devices – Energy dissipating devices are used to minimise shaking. They act like shock absorbers in a moving car by absorbing the energy before it passes through to the body of the building itself.

(d) Building configuration – Refers to the size and shape of a building as well as its internal structural elements.

(e) This determines the way seismic forces are distributed within the structure, and their relative magnitude. For example, walls that don't line up with each other could be a problem. Buildings which have complex (and therefore probably more interesting) designs may withstand earthquake waves less well.

2.8.5.5 Centre of mass – Is the point by which an object can be balanced without rotation occurring. If the mass is unevenly distributed, then the centre of mass will be outside of the geometric centre, and the more it will react to earthquake waves.

2.9.1.1 Whether light consisted of waves or particles.

2.9.1.2
(a) The aether was the hypothetical medium proposed by those who believed in the particle nature of light to carry the particles and transfer light energy.

(b) Because all other known waves at the time were matter waves.

(c) Fill space – light travelled everywhere.

Be stationary in space – light travelled in straight lines. If the aether was in motion, its movement would change the path of light travelling through it.

Be transparent – we can't see it.

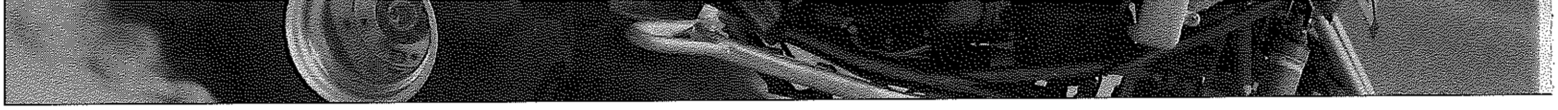
Permeate all matter – light travels everywhere.

Have an extremely low density – it cannot be detected.

Have great elasticity – transfer of energy over long distances requires the medium transmitting the wave to be Highly elastic otherwise significant amounts of energy will be 'lost' to the particles of the medium.

2.9.1.3
(a) To measure the speed of light relative to the Earth from two different directions at 90° and so gain evidence for the existence of an aether.

(b) No. It would simply have been one piece of evidence supporting the hypothesis of the existence of the aether.



2.9.1.4 (a) Null result.

(b) None (well, they observed that they did not get the interference pattern in the interferometer that they had expected).

(c) A null result means no conclusion can be drawn.

2.9.1.5 (a) Interferometer.

(b) To take advantage of the high inertia of the granite to minimise vibrations from external sources which would have made their results unreliable and invalid.

(c) They were reflecting light rays from mirrors, and superimposing the two beams entering the interferometer in order to produce an interference pattern.

2.10.1.1 The intensity of the electromagnetic radiation falling on an object is inversely proportional to the square of the distance of the object from the source of the radiation.

2.10.1.2 (a) The intensity of the electromagnetic radiation used to gather information about stars and galaxies which are a long way away is very low.

(b) By building larger ground based telescopes and by collecting data from radio waves as well as light rays, as well as by putting the Hubble telescope and others into space orbits where interference from the atmosphere allows more and more accurate data to be collected.

(c) By constructing huge arrays of ground based telescopes in several countries which are linked together electronically and using computing technology to analyse the data collected.

(d) By using satellite and computing technologies for global communications, using optical fibres locally and booster stations for medium distances.

2.10.1.3 (a) 4 units (b) 9 units

(c) 16 units (d) 81 units

(e) 144 units (f) 576 units

2.10.1.4 (a) 155 m (b) 69.3 m

(c) 28.3 m (d) 23.1 m

(e) 16.3 m (f) 14.1 m

2.10.2.1 The diagram clearly shows that while the intensity of light at distances 1, 2 and 3 from the star is the same, it is falling on larger areas, notably areas of 1, 4 and 9. This means the intensity will be reduced per unit areas by factors of 1, 4 and 9 which represent the squares of the distances from the star.

2.10.2.2

	Statement	True or false?	If false, correction
(a)	If the intensity of a light source is doubled, the intensity at each distance from the source will increase by a factor of $\sqrt{2}$.	F	Increased by factor of 2.
(b)	If the distance from a light source is tripled, then the intensity received will be reduced by a factor of 9.	T	
(c)	The intensity at a distance of 4 from a light source will be one quarter the intensity at a distance of 2.	T	
(d)	The intensity at distance 3 from a light source will be half the intensity at a distance 6 from double the light source.	F	Intensity will be double at distance 3.

2.10.2.3 (a)

Distance from light source (km)	1	1.5	2	3	4	5	6	7	8
Light intensity at distance (units)	1000	640	250	111	62.5	40	28	20	16

(b) The vertical distance on the graph between intensities 1000 and 250 resulted in the author's computer drawing an almost straight line to connect these points. The distance was too large to illustrate the curved nature of the line.

(c) Using the form of the inverse square law where Id^2 = constant, we get:

For plot $100 \times 1^2 = 1000$

For plot $250 \times 2^2 = 1000$

For plot $62.5 \times 4^2 = 1000$

So these plots support the law.

(d) Gravitational force between two masses.

Electrostatic force between two charged objects.

Nuclear radiation falling on an object.

Data Sheet

Acceleration of free fall, g	9.81 m s^{-2}
Gravitational constant, G	$6.67 \times 10^{-11} \text{ N m}^2 \text{ kg}^{-2}$
Avogadro constant, N_A	$6.02 \times 10^{23} \text{ mol}^{-1}$
Gas constant, R	$8.31 \text{ J K}^{-1} \text{ mol}^{-1}$
Boltzmann constant, k	$1.38 \times 10^{-23} \text{ J K}^{-1}$
Stefan-Boltzmann constant, σ	$5.67 \times 10^{-8} \text{ W m}^{-2} \text{ K}^{-4}$
Coulomb constant, k	$8.99 \times 10^{9} \text{ N m}^2 \text{ C}^{-2}$
Permittivity of free space, ε_0	$8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}$
Permeability of free space, μ_0	$4\pi \times 10^{-7} \text{ T m A}^{-1}$
Speed of light in a vacuum, c	$3.00 \times 10^{8} \text{ m s}^{-1}$
Planck constant, h	$6.63 \times 10^{-34} \text{ J s}$
Elementary charge, e	$1.60 \times 10^{-19} \text{ C}$
Electron rest mass, m_e	$9.110 \times 10^{-31} \text{ kg} = 0.000549 \text{ u} = 0.511 \text{ MeV } c^{-2}$
Proton rest mass, m_p	$1.673 \times 10^{-27} \text{ kg} = 1.007276 \text{ u} = 938 \text{ MeV } c^{-2}$
Neutron rest mass, m_n	$1.675 \times 10^{-27} \text{ kg} = 1.008665 \text{ u} = 940 \text{ MeV } c^{-2}$
Atomic mass unit, u	$1.661 \times 10^{-27} \text{ kg} = 931.5 \text{ MeV } c^{-2}$

1 light year (ly) = 9.46×10^{15} m

1 parsec (pc) = 3.26 ly

1 astronomical unit (AU) = 1.50×10^{11} m

1 radian (rad) = $\frac{180°}{\pi}$

1 kilowatt hour (kWh) = 3.60×10^{6} J

1 atm = $1.01 \times 10^{5} \text{ N m}^{-2}$ = 101 kPa = 760 mmHg

Equations

Heating Processes

- $Q = mc\Delta T$
  Q = heat transferred to or from the object, m = mass of object, c = specific heat capacity of the object, ΔT = temperature change
- $Q = mL$
  Q = heat transferred to or from the object, L = latent heat capacity of the material, m = mass of object
- $\eta = \dfrac{\text{energy output}}{\text{energy input}} \times \dfrac{100}{1}\%$
  η = efficiency

Ionising Radiation and Nuclear Reactions

- $N = N_o(\frac{1}{2})^n$ (for whole numbers of half-lives only)
  N = number of nuclides remaining in a sample, n = number of whole half-lives, N_o = original number of nuclides in the sample
- $\Delta E = \Delta mc^2$
  ΔE = energy change, Δm = mass change, c = speed of light (3×10^8 m s^{-1})

Electrical Circuits

- $I = \dfrac{\Delta q}{t}$
  I = current, Δq = the amount of charge that passes a point in the circuit, t = time interval
- $V = \dfrac{W}{q}$
  V = potential difference, W = work, q = charge
- $R = \dfrac{V}{I}$
  R = resistance, V = potential difference, I = current
  For ohmic resistors, resistance, R, is a constant
- $P = \dfrac{W}{t} = VI$
  P = power, W = work = energy transformed, t = time interval, V = potential difference, I = current
  - Equivalent resistance for series components, I = constant
  - $V_t = V_1 + V_2 + \dots V_n$
  - $R_t = R_1 + R_2 + \dots R_n$

  I = current, V_t = total potential difference, V_n = the potential difference across each component, R_t = equivalent resistance, R_n = resistance of each component
  - Equivalent resistance for parallel components, V = constant
  - $I_t = I_1 + I_2 + \dots I_n$
  - $\dfrac{1}{R_t} = \dfrac{1}{R_1} + \dfrac{1}{R_2} + \dots \dfrac{1}{R_n}$

  V = potential difference, I_t = total current, I_n = current in each of the components, $\dfrac{1}{R_t}$ = the reciprocal of the equivalent resistance, $\dfrac{1}{R_n}$ = the reciprocal of the resistance of each component

Linear Motion and Force

- $v = u + at$, $s = ut + \frac{1}{2}at^2$, $v^2 = u^2 + 2as$

  s = displacement, t = time interval, u = initial velocity, v = final velocity, a = acceleration
- $a = \frac{F}{m}$

  a = acceleration, F = force, m = mass
- $W = \Delta E$; where the applied force is in the same direction as the displacement, $W = Fs$,

  W = work, F = force, s = displacement, ΔE = change in energy
- $p = mv$, $\Delta p = F\Delta t$

  p = momentum, v = velocity, m = mass, F = force, Δp = change in momentum,

  Δt = time interval over which force F acts
- $E_k = \frac{1}{2}mv^2$

  E_k = kinetic energy, m = mass, v = speed
- $\Delta E_p = mg\Delta h$

  ΔE_p = change in potential energy, m = mass, g = acceleration due to gravity, Δh = change in vertical distance
- $\Sigma mv_{\text{before}} = \Sigma mv_{\text{after}}$

  $\Sigma mv_{\text{before}}$ = vector sum of the momenta of all particles before the collision, Σmv_{after} = vector sum of the momenta of all particles after the collision
- For elastic collisions: $\Sigma\frac{1}{2}mv^2_{\text{before}} = \Sigma\frac{1}{2}mv^2_{\text{after}}$

  $\Sigma\frac{1}{2}mv^2_{\text{before}}$ = sum of the kinetic energies before the collision, $\Sigma\frac{1}{2}mv^2_{\text{after}}$ = sum of the kinetic energies after the collision

Waves

- $v = f\lambda$

  v = speed, f = frequency, λ = wavelength
- angle of incidence = angle of reflection
- $l = n\frac{\lambda}{2}$ for strings attached at both ends and for pipes open at both ends
- $l = (2n - 1)\frac{\lambda}{4}$ for pipes closed at one end

  n = whole numbers 1, 2, 3 … relating to the harmonic, l = length of string or pipe,
  λ = wavelength of soundwave
- $I \propto \frac{1}{r^2}$

  I = intensity, r = distance from the source
- $\frac{\sin i}{\sin r} = \frac{v_1}{v_2} = \frac{\lambda_1}{\lambda_2}$

  i = incident angle (relative to the normal), r = angle of refraction (relative to the normal),
  v_1 = velocity in medium 1, v_2 = velocity in medium 2, λ_1 = wavelength in medium 1,
  λ_2 = wavelength in medium 2

PERIODIC TABLE OF THE ELEMENTS

KEY

Atomic Number	79
Symbol	Au
Standard Atomic Weight	197.0
Name	Gold

1 H 1.008 Hydrogen																	2 He 4.003 Helium
3 Li 6.941 Lithium	4 Be 9.012 Beryllium											5 B 10.81 Boron	6 C 12.01 Carbon	7 N 14.01 Nitrogen	8 O 16.00 Oxygen	9 F 19.00 Fluorine	10 Ne 20.18 Neon
11 Na 22.99 Sodium	12 Mg 24.31 Magnesium											13 Al 26.98 Aluminium	14 Si 28.09 Silicon	15 P 30.97 Phosphorus	16 S 32.07 Sulfur	17 Cl 35.45 Chlorine	18 Ar 39.95 Argon
19 K 39.10 Potassium	20 Ca 40.08 Calcium	21 Sc 44.96 Scandium	22 Ti 47.87 Titanium	23 V 50.94 Vanadium	24 Cr 52.00 Chromium	25 Mn 54.94 Manganese	26 Fe 55.85 Iron	27 Co 58.93 Cobalt	28 Ni 58.69 Nickel	29 Cu 63.55 Copper	30 Zn 65.38 Zinc	31 Ga 69.72 Gallium	32 Ge 72.64 Germanium	33 As 74.92 Arsenic	34 Se 78.96 Selenium	35 Br 79.90 Bromine	36 Kr 83.80 Krypton
37 Rb 85.47 Rubidium	38 Sr 87.61 Strontium	39 Y 88.91 Yttrium	40 Zr 91.22 Zirconium	41 Nb 92.91 Niobium	42 Mo 95.96 Molybdenum	43 Tc Technetium	44 Ru 101.1 Ruthenium	45 Rh 102.9 Rhodium	46 Pd 106.4 Palladium	47 Ag 107.9 Silver	48 Cd 112.4 Cadmium	49 In 114.8 Indium	50 Sn 118.7 Tin	51 Sb 121.8 Antimony	52 Te 127.6 Tellurium	53 I 126.9 Iodine	54 Xe 131.3 Xenon
55 Cs 132.9 Caesium	56 Ba 137.3 Barium	57–71 Lanthanoids	72 Hf 178.5 Hafnium	73 Ta 180.9 Tantalum	74 W 183.9 Tungsten	75 Re 186.2 Rhenium	76 Os 190.2 Osmium	77 Ir 192.2 Iridium	78 Pt 195.1 Platinum	79 Au 197.0 Gold	80 Hg 200.6 Mercury	81 Tl 204.4 Thallium	82 Pb 207.2 Lead	83 Bi 209.0 Bismuth	84 Po Polonium	85 At Astatine	86 Rn Radon
87 Fr Francium	88 Ra Radium	89–103 Actinoids	104 Rf Rutherfordium	105 Db Dubnium	106 Sg Seaborgium	107 Bh Bohrium	108 Hs Hassium	109 Mt Meitnerium	110 Ds Darmstadtium	111 Rg Roentgenium	112 Cn Copernicium						

Lanthanoids

57 La 138.9 Lanthanum	58 Ce 140.1 Cerium	59 Pr 140.9 Praseodymium	60 Nd 144.2 Neodymium	61 Pm Promethium	62 Sm 150.4 Samarium	63 Eu 152.0 Europium	64 Gd 157.3 Gadolinium	65 Tb 158.9 Terbium	66 Dy 162.5 Dysprosium	67 Ho 164.9 Holmium	68 Er 167.3 Erbium	69 Tm 168.9 Thulium	70 Yb 173.1 Ytterbium	71 Lu 175.0 Lutetium

Actinoids

89 Ac Actinium	90 Th 232.0 Thorium	91 Pa 231.0 Protactinium	92 U 238.0 Uranium	93 Np Neptunium	94 Pu Plutonium	95 Am Americium	96 Cm Curium	97 Bk Berkelium	98 Cf Californium	99 Es Einsteinium	100 Fm Fermium	101 Md Mendelevium	102 No Nobelium	103 Lr Lawrencium

Elements with atomic numbers 113 and above have been reported but not fully authenticated.
Standard atomic weights are abridged to four significant figures.
Elements with no reported values in the table have no stable nuclides.

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